Cost and Capacity Management

Use **VCF Operations** as a VI administrator to manage cost and capacity for hybrid and multi-cloud environments. Capacity management includes capacity forecasting, workload optimization, and cost transparency. Showback, chargeback, and budgeting tools helps track and predict expenses.

In This Chapter

- Configuring Cost
- Performing VCF Automation Based Multitenancy in VCF Operations
- Optimizing Capacity in VCF Operations

Getting to Know VCF

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes.

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

Configuring Cost

SDDC costing is out-of-the box with VCF Operations.

After you log in to VCF Operations, you must set the currency in **Administration** > **Global Settings** > **Cost/Price** to be able to configure cost or view the **Cost Overview** page. For more information, see Editing Cost Drivers. You can also configure other cost related settings using the **Global Settings** page. For more information see List of Global Settings.

Cost Overview

VCF Operations supports costing for VMware Cloud accounts like vCenter and VCF. You can track expenses for a single virtual machine (VM), and understand how those expenses contribute to the overall cost of your VMware Cloud accounts.

Navigate to the Cost Overview home page to view all the details about the costs associated with your VMware Cloud accounts. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Cost** > **Overview**. You can view the Cost of Ownership, Potential Savings, and Realized Savings for your VMware Cloud accounts after you configure them in VCF Operations. For more information, see Integrating Data Sources with VCF Operations.

VCF Operations also supports costing for vSphere Kubernetes Service (VKS) cluster. VCF Operations calculates the cost for VKS cluster, supervisor cluster, and Kubernetes node, by aggregating the cost of the supporting VMs.

VMware Cloud account: VCF

The cost components of a VCF account and vCenter account are Cost of Ownership, Potential Savings, and Realized Savings.

Cost of Ownership - The cost of ownership widget displays the cost expenditure by capacity, by cost drivers and by VCF domains. You can use the by capacity pie chart to view the compute, storage, and VM direct cost associated with your VMware Cloud accounts. The cost drivers bar graph provides details of the cost drivers associated with your VCF accounts and the horizontal graph for VCF domains provides the expense details of your VMware Cloud accounts for individual datacenters.

Potential Savings

The potential savings widget displays the amount of savings you can potentially achieve from your VCF accounts. The pie chart for resources displays the cost saving opportunity distributed across Idle VMs, Orphaned VMs, Oversized Hosts, Powered Off VMs, Reclaimable Hosts, and VM Snapshots. The horizontal graph for VCF domains provides the overall

potential cost savings for your VCF private cloud and potential savings for individual VCF domains. To know more about Potential Savings see, Potential Cost Savings Dashboard.

Realized Savings

The realized savings widget displays the amount of savings you have made from your VCF accounts. The pie chart for resources displays the cost savings opportunity distributed across Idle VMs, Orphaned VMs, Oversized Hosts, Powered Off VMs, Reclaimable Hosts, and VM Snapshots. The horizontal graph for VCF domains provides the realized cost saving for your overall cloud infrastructure and realized savings for individual VCF domains. To know more about Realized Savings see, Realized Cost Savings Dashboard and Realized Cost Savings Using Reclamation Suggestion.

To know the granular cost visibility and to track your expenses of virtual machines accurately in a private cloud, see Overview of Cost Drivers.

Overview of Cost Drivers

Cost Drivers are the aspect that contributes to the expense of your business operations. Cost drivers provide a link between a pool of costs. To provide a granular cost visibility and to track your expenses of virtual machines accurately in a private cloud, VCF Operations has eight key cost drivers. You can see the total projected expense on your private cloud accounts for the current month and the trend of cost over time.

You can set a cost for the server hardware: traditional, server hardware: hyper converged, storage, license, applications, maintenance, labor, network, facilities, and additional cost drivers in VCF Operations:

> **Note:**
> The total cost set by you is distributed across resources in the data center. For example, if you set the total cost for the RHEL license, the cost is divided across all the hosts that have VMs with the RHEL license.

VCF Operations maintains a reference cost for these cost drivers based on industry standards. This reference cost helps you for calculating the cost of your setup, but might not be accurate. For example, you might have received discounts during a bulk purchase or you might have an ELA with VMware with custom pricing. To get accurate values, you can modify the reference cost of cost drivers in VCF Operations, which overrides the values in the reference database. Based on your inputs, VCF Operations recalculates the total amount for the private cloud expenses. After you add a private cloud into VCF Operations, it automatically discovers one or more vCenter Servers that are part of your Private Cloud. In addition, it also retrieves the inventory details from each vCenter. The details include:

- Associated clusters: Count and names
- ESX hosts: Count, model, configuration, and so on.
- Datastores: Count, storage, type, capacity
- VMs: Count, OS type, tags, configuration, utilization

Based on the configuration and utilizations of inventory, and the available reference cost, VCF Operations calculates the estimated monthly cost of each cost driver. The total cost of your private cloud is the sum of all these cost driver expenses.

You can modify all expenses of your data center. Depending on the cost driver, expenses are calculated based on unit rate, overall cost, or a percentage of another cost. For example, hardware maintenance can be specified as a percentage of hardware purchase cost, or as a fixed monthly cost. Based on your inputs, the final amount of the expense is calculated. If you do not provide inputs regarding expenses, the default values are taken from the reference database.

> **Note:**
>
> If the vCenter has been connected for more than six months, the trend displays the total cost for the last six months only. Otherwise, the trend displays the total cost from the month the vCenter was added into VCF Operations.

> **Note:**
> You can edit the cost driver values either in All Datacenter mode or Specific Datacenter mode. If you import or export cost drivers, ensure that you download and import the cost driver configuration file in the same mode (either All Datacenter mode or Specific Datacenter mode).

For the vCenter and VCF infrastructure types the following private cloud cost drivers are applicable.

- Server Hardware : Traditional
- Server Hardware : Hyper - Converged
- Storage
- License
- Applications
- Maintenance
- Labor
- Network
- Facilities
- Additional Cost

Export and Import Cost Drivers

Except for additional cost drivers, you can export or import the remaining cost drivers that are associated with your private cloud. Using this functionality, you can also edit the cost driver values as excel sheet instead of editing them from the user interface. You can select all or individual cost drivers and export them.

Table 1738: Expense Types

Cost Drivers	Description
Select Datacenter	The Select Datacenter option allows you to choose the data center for which the cost driver changes are applicable. **Note:** You can select a specific data center and modify the cost driver values of that data center, or you can modify the cost drivers and apply the changes to all the data centers.
Export	Click export to export the cost details for all the cost drivers. Except for additional cost drivers, you can select individual cost drivers and export them also. For more information, see Import or Export Cost Drivers.
Import	Click import to browse and upload the updated cost driver configuration file (xls/csv). The import cost driver file should have the same template as that of the exported file. **Note:** If the uploaded file contains errors, error messages appear. Correct the errors and upload the file again, or ignore the errors and proceed with the upload. For more information, see Import or Export Cost Drivers.
Server Hardware : Traditional	The Server Hardware: Traditional cost driver tracks the purchase costs and details for server hardware used to calculate depreciation and operating costs. **Note:** You can select an individual server and specify the unique cost and purchase details for each server. For more information, see Editing Server Hardware: Traditional.
Server Hardware : Hyper-Converged	The Server Hardware : Hyper-Converged cost driver, tracks the expenses associated with hyper converged infrastructure components. Any servers that are being identified by the vSAN adapter are considered as a Hyper-Converged servers. The Server Hardware : Hyper-Converged cost driver includes expenses for the Hyper Converged servers like vSAN Ready Nodes. The expense provided is for both compute and storage. For more information, see Editing Server Hardware: Hyper-Converged.
Storage	You can view the storage cost at the datastore level based on the tag categories assigned to datastores in vCenter. You see the storage total distribution based on the same tag categories and the remaining fall under the uncategorised cost details. **Note:** The vSAN data stores are not displayed as part of this cost driver page. For more information, see Edit Monthly Cost of Storage.

Cost Drivers	Description
License	You see the licenses cost for VMware licenses and for other operating systems in your private cloud. **Note:** For Non-ESX physical servers, VMware license is not applicable. For more information, see Edit Monthly Cost of License.
Maintenance	You see the maintenance cost distribution for the server hardware and operating system maintenance. You can track your total expense with hardware and operating system vendors. For more information, see Edit Monthly Cost of Maintenance.
Labor	You see the labor cost distribution for the servers, virtual infrastructure, and operating systems. You can view the total administrative cost for managing physical servers, operating systems and virtual machines. You can track all expenses spent on human resources to manage the data centers. **Note:** • Labor cost includes expenses on backup appliance virtual machine (VDP virtual appliance). • For physical servers, operating system labor cost and servers labor costs are applicable, virtual infrastructure cost is not considered. For more information, see Edit Monthly Cost of Labor.
Network	You see the networks costs by NIC type. You can track a network expense based on different types of NICs attached to the ESX server. You can view the total cost of physical network infrastructure that includes the internet bandwidth, and is estimated by count and type of network ports on the ESX Servers. **Note:** For physical servers, the network details are not captured. So, the network cost is considered as zero. For more information, see Edit Monthly Cost of the Network.
Facilities	You see the cost distribution for the facilities such as real estate costs, such as rent or cost of data center buildings, power, cooling, racks, and associated facility management labor cost. You can point to the chart to see the cost details for each facility type. For more information, see Edit Monthly Cost of Facilities.
Additional Cost	You can create the additional metric based or fixed cost expenses across objects such as such as backup and restore, high availability, management, licensing, VMware software licensing. For more information, see Adding or Editing Additional Costs.
Application Cost	You can see the cost of different application services you are running in your environment compared to your overall expenses. Some examples of application cost are, cost of running SQL server cluster and cost of running Anti-virus on VMs. For more information, see Edit Application Cost.

Editing Cost Drivers

You can manually edit the monthly cost of all the eight expense types from the current month onwards.

The configuration used for cost drivers determines how VCF Operations calculates and displays the cost.

Editing Server Hardware: Traditional

You can view or edit the cost of individual servers based on their configuration, purchase cost, or purchase date. Additionally, you can specify the server cost for individual servers running in your VCF Operations environment. Once the server hardware cost is updated, cost drivers automatically update the total monthly cost for each server.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cost Drivers** tile.
2. In the Cost Drivers tab, click **Server Hardware : Traditional**.
3. Select any server from the server list and then click **Edit**.

   Note:

   You can edit the cost of one or more servers. To bulk edit the cost of multiple servers, select the servers and then click **Edit**. For example, you can sort and filter the servers and then bulk edit the desired servers.

   If the cloud account does not contain any Traditional type servers, the list appears empty.
4. Edit the server cost details and then click **Save**.

Category	Description
Purchase Type	Displays server purchase types: Owned or Leased.
Purchase Date	Displays the purchase date of the server.
Purchase Id	Displays the server purchase order ID. You can enter an unique identifier to group servers purchased together.
Purchase Cost per Server	Displays the actual cost of the server.

Editing Server Hardware: Hyper-Converged

You can view or edit the cost of Hyper converged Infrastructure (HCI) component in your individual servers. You can specify the cost per server and compute percentage exclusively for the HCI servers running in your VCF Operations environment. After you update the server hardware cost, cost drivers update the total monthly cost for each server.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cost Drivers** tile.
2. In the Cost Drivers tab, click **Server Hardware : Hyper-Converged**.
3. Select any HCI server from the server list and then click **Edit**.

   Note:

   You can edit the cost of one or more HCI servers. To bulk edit the cost of multiple HCI servers, select the servers and then click **Edit**. For example, you can sort and filter the servers and then bulk edit the desired servers.

   If the cloud account does not contain any Hyper-Converged type servers, the list appears empty.
4. Edit the server cost details and then click **Save**.

Category	Description
Purchase Type	Displays server purchase types: Owned or Leased.
Purchase Date	Displays the purchase date of the server.
Purchase Id	Displays the server purchase order ID.
Purchase Cost per Server	Displays the actual cost of the server.
Compute Percentage	Displays the compute percentage to determine the cost of vSAN servers. Edit the compute percentage value to adjust the storage rate of the vSAN data stores.

Edit Monthly Cost of Storage

The storage hardware is categorized according to the data store tag category. You can edit the monthly cost per storage GB for the data stores based on their storage category (using tags) and storage type (NAS, SAN, Fiber Channel, or Block).

To edit the cost based on the storage category, create tags and apply them to the data stores on the vCenter user interface. For more information, see the What are vSphere Tags and How to Use Them and Tags and Categories Overview.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cost Drivers** tile.
2. In the Cost Drivers tab, click **Storage**.

   Note:

   If the cloud account does not contain any external storage, the list appears empty. This is expected behavior for deployments that use an all-HCI architecture.
3. Optional: Select a tag category.

   Datastores that do not have a tag category are grouped in a catch all category called Uncategorized.

   Assume that you have two tag categories (for example, Profile and Tiers) with three tags in each category, you can select either Profile or Tiers from **Tag Category** to categorize the data stores based on tags.

Category	Description
Edit Mode	You can select the storage cost to be applicable for all the data centers or a specific data center. • **All Datacenters** mode helps you to customize a single cost driver value for all the data centers. Any customization done for the Specific data center mode are lost. • **Specific Datacenter** mode helps you to customize different cost driver values for different data centers. Any customization done for All data centers mode are lost.
Select Data center	You can select the data center for which you want to change the storage cost. This field is applicable only if Specific Datacenter is selected in Edit Mode.
Tag Category	• **Category** displays the tag categories for data stores and also the tags associated with the category.
Data stores	Displays the total number of data stores for a specific category or type.
Total Storage (GB)	Displays the total storage for a specific category or type.
Monthly Cost Per GB	Displays the monthly cost per GB for a specific category or type. You can edit this value for defining the monthly cost per GB for data stores.
Monthly Cost	Displays the total monthly cost for a specific category or type.

4. Click **Save**.

Edit Monthly Cost of License

You can edit the different license costs for your VCF Operations environment. You can now set a total fixed cost for the license in VCF Operations. The total license cost is divided across all the hosts present in the data center.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cost Drivers** tile.
2. In the Cost Drivers tab, click **License**.
3. Select the required edit mode for changing the license cost.
   - **All Datacenters** - mode helps you to customize a single cost driver value for all the data centers. Any customization done for the Specific datacenter mode are lost.
   - **Specific Datacenter** - mode helps you to customize different cost driver values for different data centers. Any customization done for All datacenters mode are lost.

Note:
When you select Edit for specific data center as the edit mode, then the select data center option is activated. Select the data center from the drop-down menu

4. Optional: Click **Customize License Assignment** to customize the licensing cost associated with your host. For more information, see Customizing License Assignment.
5. Edit the license cost in different categories and then click **Save**.

   The Cost drivers display all the licenses in your VCF Operations environment.

Category	Description
Name	Displays the different license categories: • VMware by Broadcom Licenses. – **VMware Private AI Foundation Add-on**, **VCF Edge**, **vSAN Add-on**, **VCF Edge**,**VCF**, **VMware vSphere Foundation**, **VMware Software Per Core**: VMware ESX servers 8.0 and later is charged by Per Core. Servers are classified and placed under any one of these licenses. This placement is based on the licenses applied to the vCenter systems. The license cost also accounts for instances where vCenter systems utilize multiple licenses. You can view the license information in the **Manage** > **Licenses** > **Licenses** page. If no license information is available, the default license is VMware Per Core. For example, the cost value of $9.7 has been calculated based on the three-year term license pricing value ((350/3)/12). – **VMware Software**: The VMware Software license is applicable for ESX licensing only. The monthly cost of VMware Software for VMware ESX server 7.0 or earlier is charged by Per Socket. The cost value of $51.32 has been calculated using the standard license pricing value. – **VMware vSAN**: vSAN license is applicable only to ESX servers that are classified as Server Hardware- Hyper Converged and this cost is distributed to all the datastores under these servers. There are two cost components, Monthly cost of VMware vSAN Per Socket and Monthly cost of VMware vSAN SnS have been included for the vSAN cost calculation. The default values for these components are based on the reference database values. **Note:** vSAN licenses are included in the VCF licenses. If you are using a VCF license, update the cost value as applicable to avoid adding extra cost. • **Windows Server**: The licensing cost for the Windows operating system falls under one of the following categories: Per Core License, applicable for • Windows Server 2016 • Windows Server 2019 Per Socket License, applicable for • Windows NT 4.0 • Windows Server 2003 • Windows Server 2008 • Windows Server 2012 Per Instance License, applicable for • Windows XP • Windows Vista • Windows 98 • Windows 95 • Windows 8 • Windows 7 • Windows 3.1 • Windows 2000 • Windows 10 • **SUSE** or **Red Hat**: Cost drivers categorize the Linux operating system under **SUSE** or **Red Hat**. • **Other Operating Systems**: Displays the category of the

Category	Description
Sockets	Displays the number of sockets on which the specific operating system is running.
Cores	Displays the number of VCF licenses that are core based.
TIB	Displays the add on vSAN storage.
Charged by	Displays whether a cost is charged by socket, core, TIB, or ELA. **Note:** The Charged By column can be edited to switch the display. You can either view the cost that is charged by socket, core, instance, TIB, or switch it to view the cost charged by ELA.
Total Cost	Displays the total cost of the specific operating system.

6. Click **Save**.

According to your inputs, VCF Operations calculates and displays the total cost and updates the Charged by column with the option that you have selected.

Customizing License Assignment
You can customize the licensing cost associated with your host using the custom license assignment option. Based on your requirement you can add or delete different operating system licenses on your host. With the custom license assignment option, you can increase or decrease the licensing cost associated with your host.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cost Drivers** tile.
2. In the Cost Drivers tab, click **License**.
3. Select the required **Edit Mode** for changing the monthly license cost.
   - **All Datacenters** - mode helps you to customize a single cost driver value for all the data centers. Any customization done for the Specific datacenter mode are lost.
   - **Specific Datacenter** - mode helps you to customize different cost driver values for different datacenters. Any customization done for All datacenters mode are lost.

     Note:
     When you select Edit for specific data center as the edit mode, then the select data center option is activated. Select the data center from the drop-down menu
4. To customize the license cost for a specific server, click **Customize License Assignment**.
5. Select the host for which you want to customize the license cost and click **Assign**.
6. From the menu, select one or multiple operating systems and click **OK**.

   The new operating system is listed under the Current Assignment column.
7. To remove an existing operating system from the host, under **Current Assignment** click X icon next to the operating system.

   The license cost of the removed operating system is reduced from the total cost.
8. Click **Save**.

   The license cost is updated for the host, the * sign next to the host indicates that the license cost for the host has changed.

Category	Description
Server	You can select the server for which you want to customize the license cost.
Current Assignment	Displays the current operating systems associated with the host.

Category	Description
Default Assignment	Displays the default operating systems associated with the host.
Filter	Filters the hosts based on the operating system type.
Reset	Resets the license cost of the host to the default value.

9. After you view the result, close the window.

To view report updates immediately after editing drivers, execute a manual cost calculation:

1. Navigate to **Administration** > **Control Panel** > **Cost Calculation**.
2. Click **Run** to execute the calculation.

If a manual calculation is not executed, the system will process the changes during the next automatic cost calculation, which runs by default at midnight.

Edit Application Cost

VCF Operations allows you to edit the application cost of an application present in your cloud environment. You can only modify the cost associated with the application, as all the other attributes are predefined.

Create applications in VCF Operations. For more information, see Configuring Business Applications.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cost Drivers** tile.
2. In the Cost Drivers tab, click **Applications**.
3. Click the edit icon next to the application cost you want to edit.

   Note:
   You can now specify the cost of packaged applications that are discovered by the Service Discovery Management Pack. Earlier the option to specify the application cost was available only for business applications defined by the user.
4. Modify the cost of the application.
5. Click **Save**.

Edit Monthly Cost of Maintenance

You can edit the monthly cost of maintaining your cloud environment. Maintenance cost is categorized into hardware maintenance cost and operating system maintenance cost. Hardware maintenance cost is calculated as a percentage of the purchase cost of servers. Operating system maintenance cost is calculated as a percentage of the Windows licensing costs. You can specify a total fixed cost for maintenance in VCF Operations. The total maintenance cost is divided across all the hosts present in the data center.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cost Drivers** tile.
2. In the Cost Drivers tab, click **Maintenance**.
3. Select the required edit mode for changing the monthly maintenance cost.
   - **All Datacenters** - mode helps you to customize a single cost driver value for all the data centers. Any customization done for the Specific datacenter mode are lost.
   - **Specific Datacenter** - mode helps you to customize different cost driver values for different data centers. Any customization done for All datacenters mode are lost.

     Note:
     When you select Edit for specific data center as the edit mode, then the select data center option is activated. Select the datacenter from the drop-down menu

4. To customize the maintenance cost for a specific server, click **Edit For Individual Servers**.
5. Click **+Add Cost Per Server**.
6. From the **Select Server's for customization** select the required server and click **OK**.
7. Specify the Server Hardware Percentage and OS Percentage and click **Save**.
   View the change in maintenance cost after you have run the cost calculation cycle.

Edit Monthly Cost of Labor

You can edit the monthly cost of labor for your cloud environment. You can set a total fixed cost for labor in VCF Operations. The total labor cost is divided across all the hosts present in the data center. The labor cost is combination of the total cost of the server administrator, virtual infrastructure administrator, and the operating system administrator.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cost Drivers** tile.
2. In the Cost Driver tab, click **Labor**.
3. Select the required edit mode for changing the monthly labor cost.
   - **All Datacenters** - mode helps you to customize a single cost driver value for all the data centers. Any customization done for the Specific datacenter mode are lost.
   - **Specific Datacenter** - mode helps you to customize different cost driver values for different datacenters. Any customization done for All datacenters mode are lost.

     Note:
     When you select Edit for specific data center as the edit mode, then the select datacenter option is activated. Select the datacenter from the drop-down menu
4. Edit the total monthly labor cost, click **Edit detailed cost of labor**.

Header	Description
Category	Displays the categories of labor cost, operating system, servers, and virtual infrastructure/ VM.
Calculated by	Displays whether the cost is calculated hourly or monthly.
Total Monthly Cost	Displays the total monthly cost of the particular category
Reference Cost	Displays the reference cost for the category from the cost drivers database

5. Expand the operating system and virtual infrastructure/ VM categories to customize the labor cost for virtual machines that affect the additional cost metrics of virtual machines.
6. Expand the server category to edit the labor cost for servers. You can also edit the labor cost for individual servers.
   a) Click **Edit For Individual Servers** and then click **+Add Cost Per Server**.
   b) From the **Select Server's for customization** select the required server and click **OK**.
   c) Specify the Monthly hours of labor per hour, Labor hourly rate, and click **Save**. The monthly labor cost is displayed.
7. Click **Save**.

The total monthly cost is updated. The hourly rate option or the monthly cost option that you select is updated in the **Calculated by** column.

Edit Monthly Cost of the Network

You can edit the monthly cost for each Network Interface Controller (NIC) type or can edit the total cost of all the networking expenses associated with the cloud. You can now set a total fixed cost for network resources in VCF Operations. The total network cost is divided across all the hosts present in the data center.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cost Drivers** tile.
2. In the Cost Driver tab, click **Network**.
3. Select the required edit mode for changing the monthly network cost.
   - **All Datacenters** - mode helps you to customize a single cost driver value for all the data centers. Any customizations done for the Specific data center mode are lost.
   - **Specific Datacenter** - mode helps you to customize different cost driver values for different data centers. Any customizations done for All data centers mode are lost.

     Note:
     When you select Edit for specific data center as the edit mode, then the select data center option is activated. Select the data center from the drop-down menu
4. Edit the monthly cost of network and click **Save**.
   - Modify the values for 1-Gigabit NIC, 10-Gigabit NIC, 25-Gigabit NIC, 40-Gigabit NIC, and the 100-Gigabit NIC.
     a. To customize the network cost for a specific server, click **Edit For Individual Servers**.
     b. Click **+Add Cost Per Server**.
     c. From the **Select Server's for customization** select the required server and click **OK**.
     d. Specify values for 1-Gigabit NIC, 10-Gigabit NIC, 25-Gigabit NIC, 40-Gigabit NIC, and 100-Gigabit NIC.
     e. View the change in network cost after you have run the cost calculation cycle.
   - Modify the total monthly cost of all network expenses associated with the cloud.

Edit Monthly Cost of Facilities

For your cloud environment, you can specify the total monthly cost of facilities or edit the facilities cost for real estate, power, and cooling requirements. You can now set the total fixed cost for facilities in VCF Operations. The total facilities cost is divided across all the hosts present in the data center.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cost Drivers** tile.
2. In the Cost Driver tab, click **Facilities**.
3. Select the required edit mode for changing the monthly facilities cost.
   - **All Datacenters** - mode helps you to customize a single cost driver value for all the datacenters. Any customization done for the Specific data center mode are lost.
   - **Specific Datacenter** - mode helps you to customize different cost driver values for different datacenters. Any customization done for All datacenters mode are lost. If you select specific datacenter as the edit mode, then the select dataceneter drowdown is displayed. You can select a specific datacenter from the drop-down menu to edit.
4. Edit the monthly facilities cost and click **Save**.
   - Modify the cost of rent or real estate per rack unit and modify the monthly cost of power and cooling per kilowatt-hour.
     a. To customize the facilities cost for a specific server, click **Edit For Individual Servers**.
     b. Click **+Add Cost Per Server**.
     c. From the **Select Server's for customization** select the required server and click **Ok**.
     d. Specify the Cost Per Kilowatt and Real Estate Cost Per Rack Unit and click **Save**.
     e. View the change in network cost after you have run the cost calculation cycle.
   - Modify the total monthly cost of facilities.

Adding or Editing Additional Costs

The additional cost lets you add any additional or extra expense that is not covered by other expenses categorized by VCF Operations. No reference value is present for this expense.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cost Drivers** tile.
2. In the Cost Driver tab, click **Additional Costs**, and then click **ADD**.
3. Enter a name and then select the **Object Type**.

   Note:

   Additional cost driver allows you to assign costs at vCenter, Datacenter, Cluster Compute Resource, Host System, and Virtual Machine level. For example, if you want to keep a cluster protected using the disaster recovery services, which involves an additional cost of $5000, you can do that by creating the additional cost driver for the Cluster Compute Resource where the disaster recovery services are applied, with a fixed monthly cost of $5000.
4. Select the **Scope** for the additional cost.

   You must select the scope for vCenter, Datacenter, Cluster Compute Resource, or Host System object types. You can select a particular cluster or all the clusters that are available to assign the additional cost. For Virtual Macines, all VMs across the inventory are selected as the default scope.
5. Select the **Additional cost type** and based on your selection, enter the **Rate Per Input** or the **Monthly Cost Per Entity**.
   - **Fixed**: Use the fixed cost type to set the static monthly cost for all objects.
   - **Metric/ Property Based**: Use the metric/property based cost type to set additional costs based on the Metric/ Property value available during the time of Cost Calculation. The metric/property value can be numeric/string.
     - For numeric metrics/properties, the monthly costs are calculated based on usage. For example, if you select the CPU|Demand (MHz) metric for a Virtual Machine, and set **Rate Per Input**, the the cost will be determined by its usage.
     - For string metrics/properties, the monthly costs are determined by the metric/property value and monthly cost per entity. For example, if you select Summary|Datastore Type(s) property for a Virtual Machine, you must enter the datastore type as the **Metric Value** and its corresponding monthly cost as the **Monthly Cost per entity**. The additional cost is only applicable to the specified datastore type.
6. Click **Save**.

Import or Export Cost Drivers

The cost driver editing process has been enhanced to support export and import of existing cost driver configurations. You can download (export) the existing cost driver configurations as an xls/csv file, edit the cost drivers and import the updated file back to the system. You must ensure that the import cost driver file should have the same template as that of the exported file.

Where to find the Import or Export Option

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cost Drivers** tile. In the **Cost Drivers** tab, select either **Import** or **Export**.

The import and export functionality is applicable for vCenter cost drivers.

Using the import and export option, you can perform the following actions on the cost drivers:

- Export or import the cost driver configuration file.
- Read and edit the cost driver configuration file.
- Validate the updated cost driver configuration file and report errors.
- Identify the error from the log file and correct the errors.

If the uploaded file contains errors, error messages appear. Correct the errors and upload the file again, or ignore the errors and proceed with the upload.

Cost Analysis

You can run cost, price, and bill analysis for your custom objects and groups. You can compare cost and price metrics across various custom groups. You can run an analysis and then view a list of the top and bottom 5, 10, and 20 objects based on selected metrics. Additionally you can also compare the metrics of cost, price, and VMware Cloud bills across objects.

New or Edit Cost Analysis

As part of analyzing cost, use the Analysis page to fill in the details for your cost analysis scenario. You can select a metric, an object type, and the time range for which you want to analyze cost. Cost analysis are rendered using lists or bar graphs.

Where You Can Create or Edit Cost Analysis

In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Cost** > **Analysis**. On the Analysis page, click **Add** to create an analysis.

Table 1739: New or Edit Analysis Options

Option	Description
Analysis Name	Provide a name for your cost, price, or bill analysis.
Metric	Select the metric options based on the type of analysis you want to create: Cost, Price, or Bill analysis. For metric name to metric key mapping for each adapter type, see Cost Analysis - Metrics and Metric Keys.
Object/Group Type	Select an object or group type for which you want to calculate cost. Object types are filtered and displayed based on your metric selection. The objects for which the metric is applicable are displayed. For example, you can find out the Total Cost of a VM by selecting the Total Cost as a metric and VM as an object type. You can also select a Group from the Container drop-down option.
Time Range	Select the time range for which you want to analyze and calculate costs. The time range displayed depends on the metric and object type you select. **Note:** To view the monthly cost breakup for your historical data, set the time range to **3 months**, **6 months**, or **12 months**.
Analysis Scope	If you selected an object from **vCenter** in the Object/Group Type field: • Select or enter the specific object type to narrow down your scope of analysis. You can also click the **Select** option to select a specific object type based on which the analysis is created. If you selected a group from **Container** in the Object/Group Type field: • This field is not activated. When you run the analysis, you will see a list of custom groups with the selected type. For example, Department, Function, Location, etc.
View Type	Select the view type: • By default, **Top N** is selected to display the top 10 metrics for the selected object. Use the drop down to select top 5/10/20 or bottom 5/10/20 metrics. • Select **Comparison** to run a comparison. You can run the comparison analysis for up to 20 objects, click **Add Objects** to add the objects. **Note:** You can select a second metric as **Metric 2** and run a comparison between two metrics for the selected object. The list of metrics displayed for Metric 2 is based on the first metric and object type.
Run	Click to run the analysis. You see two kinds of results: a list or a bar chart. For more information, see the **Results: List or Compare** section below this table.
Save	Click to save the analysis. You can view saved the analysis from the **Analysis** page. For more information, see Saved Analyses.

Results: List and Compare

Lists

The list results are displayed when you **Run** the Analysis. The list allows you to view the top and bottom 5, 10, and 20 results of your analysis. You can see a list or a bar graph of the selected object type defined in your analysis and the associated metrics for each object type. For example, if you selected Total Cost as a metric and VM as an object type, your list will display the all the VMs and their associated Total Cost.

Compare

You can also compare costs, prices, or expenses by selecting specific objects from your defined analysis. Click **Select Objects** to select specific objects for comparison. You can see a list or a bar graph of the selected object from your defined analysis and the associated metrics for each object.

Analysis of Cost Metrics

Cost is what you spend in currency value on managing and maintaining your infrastructure. Total Cost gives you an idea about the overall cost of ownership of your infrastructure. You can analyze cost metrics like Cost of Ownership, Potential Savings, Realized Savings and so on, for your infrastructure that includes, VMs, hosts, and clusters. You can use lists and compare between selected objects types. For example, you can view the Top 10 VMs within a datacenter based on Total Cost for the last month or you can compare Cost Center A, Cost Center B, and Cost Center C based on Total Cost for the past 6 months.

How Cost Analysis Helps

Some use cases for cost analysis are as follows:

- Financial reporting helps IT teams justify the expenses they incur to the finance team. It helps track expenses across private clouds.
- To drive accountability for resource usage, you can view the cost of infrastructure at the VM level. You can generate a bill for your usage and drive accountability within the application teams.
- Your infrastructure operations teams may want to quantify savings through reclamation or revoking of resources like idle VMs, snapshots, and so on. In VCF Operations you can see such resources and view the price associated with them. Reclaimable or right sizeable resources have a cost associated that can be seen as Potential Savings.
- Realized Savings covers the cost savings from reclamation opportunities recommended by VCF Operations.

Metrics and Objects You Can Run Cost Analysis For

Objects for Cost metrics (Total Cost, Realized Saving, Potential Saving, MTD Cost, and Monthly Projected Total Cost):

- vCenter (all objects that have Cost: VM, Host, Cluster, Data Center, Resource Pools, and Datastore)
- CloudHealth (all key objects that have Cost: Services, Accounts, Regions, and so on)
- Horizon (all key objects that have Cost or are introducing Cost in this release – User, Session, VDI Pool, and RDS Farm)
- Container (Business Apps and Custom Groups)
  Note: For Custom Groups, it is the summary of Total Cost of all VMs under that custom group.

Where You Find Cost Analysis

In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Cost** > **Analysis**. On the Analysis page, click **Add** to create an analysis. For more information about creating an analysis, see New or Edit Cost Analysis.

Analysis of Price Metrics

You can analyze price metrics for your infrastructure. The Total Price metric is based on the pricing policy defined by the user. You can use lists and compare pricing between selected objects types.

How Price Analysis Helps

Some use cases for price analysis are as follows:

- Typically, when your IT teams want the system to do the metering, but want to charge the application teams using their own Business or Pricing model, they can setup their own rate cards in VCF Operations and in turn use chargeback reports as a billing mechanism for the users to help them understand what they are being charged for.
- Service providers as well as organizations following a chargeback model to drive accountability and to help generate usage/allocation based bills for their application teams or cost centers.
- Chargeback helps IT teams to drive behavior that is better achieved when there is a cost associated with the provisioned resources.

For vCenter pricing details, see vCenter Pricing Details and for VCF Automation pricing details, see Configuring Provider Pricing Details.

Metrics and Objects You Can Run Price Analysis For

Objects for Price Metrics (Total Price):

- vCenter (all objects that have Price: VM and Cluster)
- Container (Business Apps and Custom Groups)
  Note: For Custom Groups, it is the summary of Total Cost of all VMs under that custom group.

Where You Find Price Analysis

In the left menu, click **Manage** > **Cost** > **Analysis**, and click **Add** to create an analysis. For more information about creating an analysis, see New or Edit Cost Analysis.

Analysis of Bills

Bills is the exact value you pay against the VMware Cloud infrastructure through Google Cloud VMware Engine, and VMware Cloud on AWS. Currently these bills are available if the seller on record is VMware. Using this analysis you can view, list, and compare the different bills you have received for the given cloud over a period of time

Metrics and Objects You Can Run VMware Cloud Bills Analysis For

Objects for VMware Cloud Bills (Total Component Expense, Bill Line Item Expense, Outstanding Expense, Monthly Commit Expense, Monthly OnDemand expense, and Monthly Total Expense).

Where You Find VMware Bills Analysis

In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Cost** > **Analysis**. On the Analysis page, click **Add** to create an analysis. For more information about creating an analysis, see New or Edit Cost Analysis.

Cost Analysis - Metrics and Metric Keys

As part of Cost Analysis, you can analyze a range of metrics divided into three sections — Cost, Price, and Bill.

The metric names displayed in the cost UI are generalized and do not correspond directly to the underlying metric keys. The tables below map the UI metric names to the actual metric keys for each adapter and analysis (cost, price, and bills). Use this mapping to compare data accurately across different objects within an adapter and to establish a baseline for cost analysis.

vCenter Adapter Metrics: Analyzing Cost

Metric Name per Cost Analysis UI	Metric Name	Metric Key
Object Type: Datacenter		
Total Cost	Cost|Monthly Aggregated Total Cost (Currency/Month)	cost|aggrTotalCost
Realized Saving	Cost|Realized Savings|Total Realized Cost (Currency)	cost|realized_savings|realizedTotalCost
Potential Saving (Current month) whole month	Reclaimable|Potential Savings (Currency)	reclaimable|cost
Object Type: Cluster Compute Resource		
Total Cost	Cost|Monthly Cluster Total Cost (Currency/Month)	cost|totalCost
Object Type: Host System		
Total Cost	Cost|Monthly Server Fully Loaded Cost (Currency/Month)	cost|totalLoadedCost
MTD Cost	Cost|MTD Server Total Cost (Currency)	cost|totalMTDCost
Object Type: Virtual Machine		
Total Cost	Cost|Effective MTD Cost (Currency)	cost|effectiveTotalCost
Monthly Projected Total Cost (current month) whole month	Cost|Monthly Effective Projected Total Cost (Currency/Month)	cost|effectiveProjectedTotalCost
Potential Saving	Cost|Potential Savings (US$)	cost|reclaimableCost
Object Type: Datastore		
Total Cost	Monthly Total Cost	cost|totalCost

vCenter Adapter Metrics: Analyzing Price

Metric Name per Cost Analysis UI	Metric Name	Metric Key
Object Type: Virtual Machine		
Total Price	Summary|Metering|Total price	summary|metering|value
Object Type: Cluster Compute Resource		
Total Price	Summary|Metering|Total price	summary|metering|value

Kubernetes Adapter Metrics: Analyzing Cost

Metric Name per Cost Analysis UI	Metric Name	Metric Key
Object Type: Kubernetes Node		
Total Cost	Cost|Daily Total Cost	cost|totalCost
Object Type: Kubernetes Namespace		
Total Cost	Cost|Daily Total Cost	cost|totalCost
Object Type: Kubernetes Cluster		
Total Cost	Cost|Daily Total Cost	cost|totalCost

VCF Operations for networks Adapter Metrics: Analyzing Cost

Metric Name per Cost Analysis UI	Metric Name	Metric Key
Object Type: Operations for Network Application		
Total Cost	Cost|Effective MTD Cost	cost|effective_total_cost
Monthly Projected Total Cost	Cost|Monthly Effective Projected Total Cost	cost|effective_projected_total_cost
Object Type: Operations for Network Tier		
Total Cost	Cost|Effective MTD Cost	cost|effective_total_cost
Monthly Projected Total Cost	Cost|Monthly Effective Projected Total Cost	cost|effective_projected_total_cost

VCF Automation Adapter Metrics: Analyzing Cost

Metric Name per Cost Analysis UI	Metric Name	Metric Key
Object Type: Project		
Total Cost	Cost|Effective MTD Cost	cost|effectiveTotalCost
Monthly Projected Total Cost	Cost|Monthly Effective Projected Total Cost	cost|effectiveProjectedTotalCost
Object Type: Deployment		
Total Cost	Cost|Effective MTD Cost	cost|effectiveTotalCost
Monthly Projected Total Cost	Cost|Monthly Effective Projected Total Cost	cost|effectiveProjectedTotalCost

VCF Automation Adapter Metrics: Analyzing Price

Metric Name per Cost Analysis UI	Metric Name	Metric Key
Object Type: Project		
Total Price	Summary|Metering|Total Cost	summary|metering|value
Object Type: Deployment		
Total Price	Summary|Metering|Total Cost	summary|metering|value

Saved Analyses

The Analysis page lists all the saved analyses.

Where You Find Saved Analyses

In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Cost** > **Analysis**.

Saved Analyses Options

Option	Description
Horizontal Ellipsis >Run	Run one or more analyses.
Horizontal Ellipsis > Delete	Delete all the selected analyses.
Vertical Ellipsis > Run	Run the selected analysis.
Vertical Ellipsis > Edit	Edit the selected analysis. For more information, see New or Edit Cost Analysis.
Vertical Ellipsis > Delete	Delete the selected analyses.

Cost Settings

You can configure the depreciation method and depreciation period for the Financial Accounting Model to compute the amortized cost of server hardware.

Cost Settings for Financial Accounting Model

You can set your depreciation model and years for all the vCenter instances in VCF Operations using the following steps.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cost Settings** tile.
2. On the **Cost Settings** page, select the **Depreciation Years** between two and seven.
3. Select the **Depreciation Model** as per your requirement and click **Save**. To compute the amortized cost of the Server Hardware cost driver, you can configure the depreciation method and the depreciation period. Cost Drivers supports two yearly depreciation methods and you can set the depreciation period from two to seven years.

Table 1740: Configuring Depreciation Preferences

Method	Calculation
Straight line	`Yearly straight line depreciation = [original cost/ number of depreciation years]`

Max of Double or Straight	`Yearly max of Double or Straight = Maximum (yearly depreciation of double declining balance method, yearly depreciation of straight line method)`
	`Yearly straight line depreciation = [original cost/ number of depreciation years]`
	`Yearly depreciation of double declining method= [(original cost - accumulated depreciation) * depreciation rate].`
	`Depreciation rate = 2 / number of depreciation years.`
	Note:
	`Double declining depreciation for the last year = original cost - accumulated depreciation`

Note:

Updating the depreciation years impacts the server's monthly cost. To see the updated monthly server cost in the Cost Drivers tab, you must run the cost calculation from the **Administration** > **Control Panel** > **Cost Calculation** page after making changes to the depreciation years.

Cost Drivers calculates the yearly depreciation values and then divides the value by 12 to arrive at the monthly depreciation.

Edit Cluster Cost Ratio

You can split the cluster cost for CPU and Memory and set the cluster cost ratio. You can edit a single cluster or bulk edit the cluster cost ratio for all the clusters in VCF Operations.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cost Settings** tile.
2. In **Cost Settings** page, select a cluster and click **Edit**.
3. Edit the cluster cost ratio, enter the values between one to 10 for **CPU** and **Memory**, and then click **Save**.

Example for 5-year Straight-Line Depreciation Method

Year	Original Cost	Accumulated Depreciation	Straight Line Depreciation Cost
Year 1	10000	0	`[(10000-0)/5] = 2000`
Year 2	10000	2000	`[(10000-2000)/4] =2000`
Year 3	10000	4000	`[(10000-2000)/3] =2000`
Year 4	10000	6000	`[(10000-2000)/2] =2000`
Year 5	10000	8000	`[(10000-2000)/1] =2000`

Example for Max of Double and Straight-Line Depreciation Method for a 5-year period

Year	Original Cost	Depreciation Rate	Accumulated Depreciation	Max of Double and Straight-Line Depreciation Cost
Year 1	10000	For a 5-year depreciation period, the rate is 2 / 5 = 0.4	0	`Maximum([(10000-0)*0.4], [(10000-0)/5]) = Maximum(4000, 2000) =4000` which is 333.33 per month.
Year 2	10000	For a 5-year depreciation period, the rate is 2 / 5 = 0.4	4000	`Maximum([(10000-4000)*0.4], [(10000-4000)/4]) = Maximum (2400, 1500) =2400` which is 200 per month.
Year 3	10000	For a 5-year depreciation period, the rate is 2 / 5 = 0.4	6400	`Maximum([(10000-6400)*0.4], [(10000-6400)/3]) = Maximum (1440, 1200) =1440` which is 120 per month.
Year 4	10000	For a 5-year depreciation period, the rate is 2 / 5 = 0.4	7840	`Maximum([(10000-7840)*0.4], [(10000-7840)/2]) = Maximum (864, 1080) =1080` which is 90 per month.
Year 5	10000	For a 5-year depreciation period, the rate is 2 / 5 = 0.4	8920	`Maximum([(10000-8920)*0.4], [(10000-8920)/1]) = Maximum (432, 1080) =1080` which is 90 per month.

Cloud Providers Overview

By default, you can see that common public cloud providers (Google Cloud, IBM Cloud, VMware Cloud on AWS, Azure VMware Solution, and Google Cloud VMware Engine), as well as Public Cloud-based VMware solutions are included in VCF Operations. You can also add your own cloud provider by using a standard VCF Operations template.

You can configure the new cloud provider as per the standard VCF Operations template and perform a migration scenario. The VCF Operations template contains data points for vCPU, CPU, RAM, OS, region, plan term, location, and built-in instance storage, you must provide these values when you add cloud providers. The result of the migration scenario helps you assess the cost savings achieved using your cloud provider against the default cloud providers.

You can edit the rate card for new cloud providers and default cloud providers. However, you cannot delete the default cloud providers.

Add or Edit Cloud Provider

You can use the Add Cloud Provider workspace to add or edit a public cloud provider. You can edit the cloud provider rate card for default cloud providers and add the new cloud provider.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cloud Providers** tile.
2. To add or edit the cloud provider, click **Add** or select the **Edit** option from the vertical ellipsis next to the clour provider you want to edit.
3. Enter or edit the **Cloud Provider Name**.

   Note:

   The name and logo for the predefined system cloud providers are fixed and cannot be modified.
4. Select the cloud provider logo and click **Upload Logo**.
5. Click **Next**.
6. Click **Download Template** to download the rate card Excel format template. Edit the **Instance Configuration** and **Pricing Configuration** values.

   Note:
   When you edit a cloud provider the Download Template link is replaced with Download Existing Rate Card. You can add, edit, or delete the rows with the instance names. You can fill in the instance settings values and the mandatory price configuration values as mentioned in the template.
7. Select the updated template and click **Upload Rate Card**.
8. Click **Validate**.

   Note:

   VCF Operations validates the rate card and reports success or failure. If errors are reported, correct the errors in the file and re-upload the revised rate card.
9. Click **Finish**.

The new cloud provider is now part of the VCF Operations cloud provider list.

Cluster Cost Overview

VCF Operations calculates the base rates of CPU and memory so that they can be used for the virtual machine cost computation. Base rates are determined for each cluster, which are homogeneous provisioning groups. As a result, base rates might change across clusters, but are the same within a cluster. For more information, see Editing Cluster Cost Calculation Methods.

1. VCF Operations first arrives at the fully loaded cost of the cluster from the cost drivers. After the cost of a cluster is determined, this cost is split into CPU and memory costs based on the industry standard cost ratios for the different models of the server.
2. The CPU base rate is first computed by dividing the CPU cost of the cluster by the CPU capacity of the cluster. This base rate is then prorated by dividing the CPU base rate by the expected CPU use percentage to arrive at a true base rate used to calculate virtual machines charges.
3. The memory base rate is first computed by dividing the memory cost of the cluster by the memory capacity of the cluster. This base rate is then prorated by dividing the memory base rate by the expected memory use percentage to arrive at a true base rate used to calculate virtual machine charges.

4. The base rates are monthly rates.

Cluster Cost Elements	Calculation
Total Compute Cost	Total Compute Cost = (Total Infrastructure cost, which is a sum of all cost drivers) – (Storage cost) – (Direct VM cost, which is sum of OS labor, VM labor and any Windows Desktop licenses).
Per GHz CPU base rate	Per GHz CPU base rate = (Cluster Total CPU Cost) / (Expected CPU Utilization * Cluster CPU Capacity in gHZ).
Per GB RAM base rate	Per GB RAM base rate = (Cluster Total Memory Cost) / (Expected Memory Utilization * Cluster RAM Capacity in GB).
Average CPU Utilization	Average CPU Utilization = (Cost attributed to CPU utilization of VMs in a cluster, out of Total compute cost) / (Total number of VMs in the cluster).
Average Memory Utilization	Average Memory Utilization = (Cost attributed to Memory utilization of VMs in a cluster, out of Total compute cost) / (Total number of VMs in the cluster).
Expected CPU Utilization	The utilization percentage level of CPU that the cluster is expected to operate. **Note:** When you select actual utilization as the cost calculation mode, the cost engine rounds actual utilization up to the next multiple based on the ceiling cost factor, which defaults to 5.
Expected Memory Utilization	The utilization percentage level of Memory that the cluster is expected to operate. **Note:** When you select actual utilization as the cost calculation mode, the cost engine rounds actual utilization up to the next multiple based on the ceiling cost factor, which defaults to 5.

Cluster Cost Computation with Allocation Model

You can use the allocation model to compute the cost of clusters in VCF Operations. When you perform cost computation using the allocation model, you can set the over commit ratio for CPU, RAM, and storage.

Note:
The allocation ratio can be set at both cluster level and data store cluster level. You can also mention the storage base rate, which is displayed at the data store level.

Table 1741: Cluster Base Rate Computation with Allocation Model

Base Rate	Formula
vCPU Base Rate	vCPU base rate = B1 = (Cost attributed to CPU) / (Number of vCPUs in a cluster)
RAM Base Rate	RAM base rate = B2 = (Cost attributed to RAM) / Number of vRAMs in a cluster) **Note:** The cost computation is based on Over Commit ratio. If the Over Commit ratio is 1:4, and total cores in cluster are 6, then vCPU count = 24, in case if the allocated vCPU exceeds this targeted number, then the maximum value is selected.

Table 1742: Virtual Machine Cost Computation with Allocation Model

Cost	Formula
Virtual Machine Cost	Virtual machine cost = (Number of vCPU allocated x B1 of cluster it belongs to) + Number of vRAMs allocated x B2 of cluster it belongs to) + storage cost + direct cost. **Note:** Storage allocated represents the Storage Base Rate based on allocation.

Editing Cluster Cost Calculation Methods

You can edit the cluster cost calculation method based on your business requirement. The cost of a cluster is derived from cost drivers. Virtual machine cost is calculated by multiplying base rates with the utilization of the VMs.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Cluster Cost** tile.
2. Click **CHANGE**.
   The Cluster Cost Calculation Methods dialog box is displayed.
3. Select any one of the Cluster Cost Calculation methods.
   - **Cluster Usable Capacity After HA and Buffer**: The cluster cost calculated total capacity minus resources needed for High Availability (HA) and the capacity buffer setting.

     Base rates are calculated based on the total cost of the cluster and Usable Capacity after HA and Buffer. Virtual machine costs are calculated from these base rates. Things to note:

     - A lower buffer reduces the base rates and causes the virtual machines to become cheaper.
     - A higher buffer increases base rates and causes the virtual machines to become more expensive.
     - Base rates and virtual machine costs do not change with the utilization of the cluster.
     - The difference between Usable Capacity after HA and Buffer and actual utilization is used to compute unallocated costs.
   - **Cluster Actual Utilization**: To calculate the base rates using the month to date average utilization of the cluster resources, select this option.

     Base rates are calculated based on the total cost of the cluster and average utilization. Virtual machine costs are calculated from these base rates. Things to note:

     - Lower utilization level causes base rates to be high and virtual machines also become more expensive.
     - Higher utilization level causes base rates to be lower and virtual machines to become cheaper.
     - Base rates and virtual machine costs can change frequently based on the utilization of the cluster.
     - Unallocated cost of the cluster is near to zero.
     - The costs for unused resources are distributed across all virtual machines based on their actual utilization within the cluster.

     Note:

     You can set the Cluster Utilization Ceiling Factor to influence the expected CPU and Memory utilization calculation when the cluster cost calculation method is set to actual utilization. The default ceiling value is 5. To update the value, navigate to **Administration** > **Global Settings**, and then click **Cost/Price** > **Cluster Utilization Ceiling Factor**. Enter the ceiling value between zero and 20 and click **Save**.

     To view the change in cost metrics, run the Cost Calculation Status and select a cluster.

     For example, if the Actual Utilization of the cluster for CPU is 32 % and Memory is 45%, and the ceiling value specified is 10, then

     - Cluster Expected CPU Utilization (%) = 40
     - Cluster Memory Expected Utilization (%) = 50

Actual Cluster Utilization is rounded off to the ceiling value.

If you set the Cluster Utilization Ceiling Factor to 0, the Expected Utilization is equal to the Actual Utilization. If you set the Cluster Utilization Ceiling Factor to 20, the Expected Utilization is rounded off to the next multiple of 20. The Expected Utilization value is always a multiple of the configured Cluster Utilization Ceiling Factor.

4. Click **SAVE**.

Publish Tag Based Cost as Individual Metrics

You can publish additional cost metrics per tag using VCF Operations. Activate Tag based Costing Metrics at the Global Settings level. You can publish additional cost metrics per tag using VCF Operations. Activate Tag based Costing Metrics at the Global Settings level. When you apply tags that are mapped to additional cost drivers to virtual machines, each tag is considered an independent instance metric of the virtual machine.

How to Activate Tag Based Costing Metrics

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Global Settings**, and then click **Cost/Price** and activate the Tag based costing metrics. Add a VM tag or custom properties based entries in Additional Cost Drivers. To do so, add an additional cost driver for the VM object type, then select metric based, and choose **Properties** > **Summary** > **vSphere Tag**. For the Metric Value, enter the vSphere tag in the format `TagCategory-TagValue`. You can include multiple tags separated by commas. To know how to set the Additional Cost at a VM level, see Adding or Editing Additional Costs.

After you assign the cost for VMs with tags or custom properties, run the Cost Calculation for the changes to take effect. The additional cost metrics can be viewed from the following location, only for virtual machines which have the tags mentioned in the Additional Cost Driver: **Inventory** > **VM** > **Cost** > **Daily Tags and Custom Properties Cost**.

Using Pricing Cards in VCF Operations

You can create pricing cards in VCF Operations to calculate the price associated with your virtual infrastructure. You can assign pricing cards to vCenter instances or Clusters, depending on the pricing strategy. The pricing cards allows you to set the price for each resource present in your virtual environment.

VCF Operations has two types of pricing cards, rate-based pricing card and cost-based pricing card. After configuring a pricing card, you can assign it to one or more vCenters or Clusters as determined by the pricing strategy.

Note:

You can create pricing cards or you can migrate vCenter pricing cards to vCenter pricing policies. For details about creating a new pricing card, see Add New Pricing Card.

How Is Price Calculated

In rate-based pricing policy VCF Operations calculates the virtual infrastructure price based on the rate card. For rate-based pricing policy VCF Operations lets you define cost elements.

VCF Operations recalculates the price every 24-hours, the price calculation for the new pricing cards is done in the next price calculation cycle.

Hierarchy of Pricing Policy

In VCF Operations, policy is assigned to Clusters and/or vCenter instances. The price is calculated for virtual machines, then aggregated and rolled up to vCenter. If there are two policies, a default policy for vCenter and another policy for Cluster, then the price calculation is based on the cluster policy for all the resources under the cluster. After that the cluster cost is rolled up to vCenter.

When a virtual machine is under VCF Automation hierarchy and vCenter hierarchy, then the pricing is calculated based on the VCF Automation hierarchy and the virtual machine is removed from the vCenter resources and included under VCF Automation resources.

Add New Pricing Card

You can add and assign a new pricing card to vCenter and Clusters in VCF Operations. The pricing card can be cost-based or rate-based, both of which are customizable. After configuring the pricing card, you can assign it to one more vCenter or Clusters based on your pricing strategy.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Policy Definitions** tile.
2. Select a policy and in the right pane, click **Edit Policy**.
3. In the <policy name> [Edit] workspace, click the **VC Pricing** card.
4. Click **VC Pricing** and then click **New Pricing Card** and configure the details of the pricing card.

Table 1743: Pricing Card Configuration

Parameter	Description
Name and Description	1. Enter a name and description for your pricing card. 2. Optional: Select Default for Unassigned Workloads. 3. Click **Next**. Default pricing card applies to all vCenter resources which do not have a direct cost policy assigned to them.
Basic Charges	Pricing is **Based on Cost** or **Based on Rate**. Select the type of pricing card. For **Cost-based** pricing: 1. Enter the Cost Factor for the following. a. CPU Cost b. Memory Cost c. Storage Cost d. Additional Cost 2. Select the charging period as per your requirement, the options are **Hourly**, **Daily**, **Weekly**, and **Monthly**. 3. Select how to charge for the resources, the options are **Always** or **Only When Powered On**. 4. Click **Next**. The base cost is defined in the Cost Drivers section of VCF Operations. If selected, a multiplication factor is required. For example, if you select 1.1 as a factor, the cost is multiplied by 1.1 resulting in a 10% increase to the calculated cost. The price equation using cost is: <cost> x <multiplication factor> = Price **Note:** For accurate private cloud VM prices, you must edit the cost drivers, see Editing Cost Drivers and for accurate hybrid cloud provider VM prices, you must edit the cloud provider rate card, see Add or Edit Cloud Provider. For **Rate-based** pricing: 1. Enter the CPU Rate per vCPU. 2. Enter the Memory Rate per GB. 3. Enter Storage Rate per GB. 4. Select the ChargingPeriod for all the values. 5. Select the Charge On Power State for all the values.

Parameter	Description
Guest OSes	1. Click **Create Guest OS Rate**. 2. Enter the Guest OS Name. 3. Select the charging period as per your requirement, the options are **Hourly**, **Daily**, **Weekly**, and **Monthly**. 4. Enter the base rate.
Tags	Enter the Tag name and Tag Value. Define the charging method and base rate. • Recurring - enter a base rate and define recurring interval as the charge period. The absolute rate value is required and it is added to the overall price. • One time - define the one-time base rate charge. The absolute value is required and it is added as a one time price. • Rate Factor - A multiplication factor is required that is applied to the select charge category. Select how to charge the Tag based on powered on state.
Overall Charges	You can define overall charges to VMs that match this policy. 1. Enter the VM setup charges. 2. Enter the Recurring charge period. 3. Select the time period from the drop-down menu.
Assignments	You can assign the new pricing card to vCenters and Clusters. 1. Select the vCenter or Cluster to which you want to apply the pricing card. 2. Click **Add** and Click **Finish**.

The new pricing card details are displayed in the Pricing tab.

Cost Calculation Status Overview

You can check the ongoing status of manually triggered cost calculation process.

Cost calculation by default, occurs daily and whenever there is a change in the inventory or cost drivers values. You can trigger the cost calculation manually so that changes in the inventory and cost driver values reflect accordingly on the VM cost without having to wait there for any failures in the cost calculation process. It also shows default schedules time for next cost calculation process.

Note:

To run the manual cost calculation, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Cost Calculation** tile. On the Cost Calculation page, click **Run**.

Other Costs

VCF Operations publishes other cost metrics as part of the cost calculation job, such as reclaimable host costs, realized cost savings from reclamation suggestions, and costing for oversized and undersized VMs.

Reclaimable Hosts Cost Metric

You can use the cost metrics at cluster level to identify the clusters with reclaimable hosts and the potential cost savings from reclaiming these hosts. To know the cost associated with all the reclaimable hosts in a cluster, refer to the value of Total Host Reclaimable Host Cost metric.

How to View Reclaimable Host Cost

To view the reclaimable host cost, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory** > **vCenter** > **Cluster Compute Resource**, and then go to the **Metrics** tab. Expand **Metric** > **Cost** > **Potential Savings** and select **Reclaimable Host Cost ($/Month)**.

You can also view the total host reclaimable cost using **Inventory** > **vCenter**, click **vCenter** > **vSphere World**, and then click **Metrics** > **Cost**.

> **Note:**
> If the cluster does not have reclaimable hosts, then the cost metric associated with the reclaimable host is not displayed.

Realized Cost Savings Using Reclamation Suggestion

In VCF Operations you can track the cost savings using reclamation suggestions. Using the reclamation option, you can view the cost, capacity, and allocation metrics related to individual data centers. For more information, see Reclamation Settings. The metrics provides an estimate of the potential savings achieved through VCF Operations.

You can track the realized cost savings and actual capacity reclaimed for data centers, in the following scenarios. For more information see, Using Reclaim to Free Up Resources

- Reclaim the cost for Idle VMs by deleting the VM.
- Reclaim the cost for Powered off VMs by deleting the VM.
- Reclaim the cost for Idle VMs by powering off the VM.
- Reclaim the cost for snapshots VMs by deleting the snapshot.
- Reclaim the cost for orphaned disks by deleting the orphaned disk space.
- Reclaim the cost by removing vCPU and Memory from an oversized VM.
- Reclaim the cost by removing a host from the vCenter.

Costing for Oversized VM and Undersized VM

Rightsizing is defined as changing the amount of resources allocated to a VM based on the Recommended Size for a VM. Recommended Size is the maximum projected utilization for the period from the current time to 30 days after the warning threshold value for time remaining. The warning threshold is the period during which the time remaining is green. If the warning threshold value for time remaining is 120 days, which is the default value, the recommended size is the maximum projected utilization 150 days into the future. While rightsizing a VM can reclaim capacity.

Quantifying the Effect on Capacity Due to Rightsizing

Demand Model

- **Reclaimable CPU Usage (GHz)**: If an oversized VM's CPU usage is 100 MHz before rightsizing, removing vCPU's will not change its CPU usage and it should still be at 100 MHz. This means there is no reclaimable capacity associated with overallocation of vCPUs. Reclaimable CPU Usage for oversized VM's will always be 0 MHz.
- **Reclaimable Memory Consumed (GB)**: An oversized VM can have reclaimable memory only if consumed memory is greater than the new recommended size of the VM. The reclaimable memory capacity is the difference between consumed memory and recommended size.
- **Increased CPU Usage (GHz)**: CPU usage of an undersized VM is expected to be the current CPU Demand. The difference between CPU Demand and CPU Usage is the expected increase in capacity utilized after rightsizing.
- **Increased Memory Consumed (GB)**: It can be expected for consumed memory to increase by the same amount of memory recommended to add to an undersized VM.

Allocation Model

In the allocation model, you can directly pick the recommendation provided which is given as a part of the metric groups **Summary|Oversized** and **Summary|Undersized**.

Potential Cost Savings Calculation Detail

- **Oversized CPU Utilization**: $0 since Reclaimable CPU Usage (GHz) is always 0.
- **Oversized Memory Utilization**: Reclaimable Memory Consumed (GB) * Cluster Memory Base Rate.
- **Oversized CPU Allocation**: vCPU(s) to Remove * Allocation Cluster CPU Base Rate.
- **Oversized Memory Allocation**: Memory to Remove * Allocation Cluster Memory Base Rate.

Potential Cost Increase Calculation Detail

- **Undersized CPU Utilization**: Increased CPU Usage (GHz) * Cluster CPU Base Rate.
- **Undersized Memory Utilization**: Increased Memory Consumed (GB) * Cluster Memory Base Rate.
- **Undersized CPU Allocation**: vCPU(s) to Add * Allocation Cluster CPU Base Rate.
- **Undersized Memory Allocation**: Memory to Add * Allocation Cluster Memory Base Rate.

The rightsizing value calculated here is available as part of

- **Potential Savings** metric (For VM) for Oversized VMs.
- **Potential Increase** metric (For VM) for Undersized VMs.

> **Note:**
> Reclaimable memory consumed, Increased CPU Usage, and Increased memory consumed are metrics that are available for reference under **Summary|is Oversized** metrics and **Summary|is Undersized** metrics respectively.

Performing VCF Automation Based Multitenancy in VCF Operations

Partners and enterprises can host and sell cloud services that are built on VMware software technology using VCF Automation. VCF Automation provides constructs to segment the virtual infrastructure and offers it as a service to tenants of these partners.

Activating Chargeback Capabilities in VCF Operations

To activate Chargeback capabilities in VCF Operations, deploy VCF Automation.

Complete the following step to activate the capabilities that are related to Chargeback in VCF Operations.

Procedure

- **Integrating Chargeback capabilities by deploying VCF Automation using the VCF Installer Deployment Wizard**. For more information about deploying VCF Automation using the VCF Installer Deployment Wizard, see Start a New VCF Fleet or a New VCF Instance Deployment by Using the VCF Installer Deployment Wizard.

  If you have not installed VCF Automation using the VCF Installer, deploy VCF Automation from VCF Operations as a Day-N operation. Navigate to **Build** > **Lifecycle** > **VCF Management** > **Components** > **Add Component** > **VCF Automation**. For more information, see Deploying VCF Automation as a Day-N Operation.

Managing Historical Data

You can view and manage historical data retention that is related to system settings. You can tune these settings to manage storage space consumed by VCF Operations.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Global Settings**, and then click **Data Retention**.

   > **Note:**
   >
   > The **Generated Bills** and **Generated Tenants Reports Retention** options are visible only if you have deployed VCF Automation.

   Generated Bills

2. In the **Generated Bills** field, enter the number of days for which you want to retain the generated bills.

   Note:

   You can retain the generated bills to a maximum of 1440 days.

   Generated Tenant Report Retention

3. In the **Generated Tenants Reports Retention** field, enter the number of days to retain generated tenant reports. The minimum number of days you can set is 30 and the maximum is 90. If deactivated, the generated tenant reports are retained as per the setting in the **Generated Reports Retention** field. This option is visible only if you have deployed VCF Automation.

Configuring Provider Pricing Details

You can edit the parent policy settings and can configure your Region Quota settings using the **Provider Pricing** tab in VCF Operations.

If you want to copy the Provider pricing settings from the policy currently being edited to another policy, click **Copy local changes to other policy** and select the policy to which you want to copy the settings. The copied pricing configuration overrides any existing local pricing configuration in the target policy.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Policy Definition** tile.
2. Select the required policy or click **Add** to add a new policy.
3. Select the required policy, click the **horizontal ellipsis > Edit** or in the right pane, click **Edit Policy**.
4. In the <policy name> [Edit] workspace, click the **Provider Pricing** card.

   Note:

   This policy workspace space is not available if VCF Automation is not connected to VCF Operations.

5. Click the Lock icon next to the **Price Base Settings** to override parent policy settings.
6. Select if you want to activate or deactivate the pricing engine.
7. Configure the Region Quota settings.
   - **Compute Rate**: Click the Lock icon to edit the parent policy settings.
     The following options appear for **CPU Rate**.
     - CPU GHz Based/vCPU Count Based: Select if you want to charge the CPU rate based on GHz or vCPU count.
     - Charge Period: The Charge Period indicates the frequency of charging.
     - Charge Based On: The Charge Based On indicates the pricing model based on which the charge is applied.

     Note:

     This field appears when you charge the CPU based on GHz.
     - Default Base Rate (per GHz): Enter a valid number for Default Base Rate.

     Note:

     This field appears when you charge the CPU based on GHz.
     - Default Base Rate (per vCPU): Enter a valid number for Default Base Rate.

     Note:

     This field appears when you charge the CPU based on vCPU.
     - Base Rate Slabs: Using slabs, you can optionally charge different rates depending on the number of CPUs used or the amount of GHz that is determined by the value specified in the **Charge Based On** field. Click **Add Slab** and enter the following details:
       – Greater than or equal: Enter a valid number .
       – Base Rate: Enter a base rate.
       – Explanation: Displays an explanation of the slab based on the values entered in the above fields.

The base rate slab allows you to change the rate of charging based on the resources used. For example, if you enter 5 in Greater than or equal and 5 as Base Rate, it means if the usage is 5 vCPU (or GHz) and above, then the Base Rate of 5 will be applied for the whole usage.

The slabs that you add appear in the table below. To edit or delete an entry, click the vertical ellipses next to it and select the **Edit** or **Delete**. To delete multiple entries at once, select the slabs to delete and click the horizontal ellipses at the top and select **Delete**.

- Fixed Cost (per charge period per VM): Enter a valid number. fixed costs are not dependant upon the Charge Based on selection or measured usage. Fixed costs do not depend on the units of charging.

Enter details for **Memory Rate**:

- Charge Period: The Charge Period indicates the frequency of charging.
- Charge Based On: The Charge Based On indicates the pricing model based on which the charge is applied.
- Default Base Rate (per GB): Enter a valid number for Default Base Rate.
- Add Slab: Using slabs, you can optionally charge different rates depending on the value specified in the **Charge Based On** field. Click **Add Slab** and enter the following details:
  - Greater than or equal: Enter a valid number.
  - Base Rate: Enter a base rate.
  - Explanation: Displays an explanation of the slab based on the values entered in the above fields.

  The base rate slab allows you to change the rate of charging based on the resources used. For example, if you enter 50 in Greater than or equal and 10 as the Base Rate, it means if the usage is 50 GB and above, then the Base Rate of 10 will be applied for the whole usage.

  The slabs that you add appear in the table below. To edit or delete an entery, click the vertical ellipsis next to it and select the **Edit** or **Delete**. To delete multiple entries at once, select the slabs to delete and click the horizontal ellipses at the top and select **Delete**.
- Fixed Cost (per charge period per VM): Enter a valid number. Fixed costs are not dependant upon the Charge Based on selection or measured usage

- **Storage Rate**: Click **Storage Rate** in the left pane and then click the Lock icon to edit the parent policy settings.
  - Click **Create Storage Class Rate** and enter the following details:
    - Storage Class Name: Select a storage class name from the drop-down list.
    - Charge Period: The Charge Period indicates the frequency of charging.
    - Charge Based On: The Charge Based On indicates the pricing model based on which the charge is applied.
    - Default Base Rate (per GB): Enter a default base rate.

    The storage policy that you add appears in the table below. To edit, delete, or add new slabs, click the vertical ellipsis next to the storage class rate and select the desired option. Using slabs, you can optionally charge different rates depending on the value specified in the **Charge Based On** field.
- **Network Rate**: Click **Network Rate** in the left pane and then click the Lock icon to edit the parent policy settings.
  - **Network Data**
    - External Network Transmit (per MB): Enter the rates for external network transmit.
    - External Network Receive (per MB): Enter the rates for external network receive.
  - **Network Transmit Rate (Bandwidth)**
    - Charge Period: The Charge Period indicates the frequency of charging.
    - Usage: Select the usage based on which you want to charge.
    - Default Base Rate (per MBps): Enter a default base rate.
    - Add Slab: Using slabs, you can optionally charge different rates depending on the network data consumed. Click **Add Slab** and enter the following details:
      - Greater than or equal: Enter a valid number.
      - Base Rate: Enter a base rate.
      - Explanation: Displays an explanation of the slab based on the values entered in the above fields.

      The base rate slab allows you to change the rate of charging based on the resources used. For example, if you enter 50 in Greater than or equal and 10 as the Base Rate, it means if the usage is 50 Mbps and above, then the Base Rate of 10 will be applied for the whole usage.

The slabs that you add appear in the table below. To edit or delete the entries, click the vertical ellipsis next to the slab and select the desired option.

  - **Network Receive Rate (Bandwidth)**
    - Charge Period: The Charge Period indicates the frequency of charging.
    - Usage: Select the Usage based on which you want to charge.
    - Default Base Rate (per MBps): Enter a default base rate.
    - Add Slab: Using slabs, you can optionally charge different rates depending on the network data consumed. Click **Add Slab** and enter the following details:
      - Greater than or equal: Enter a valid number.
      - Base Rate: Enter a base rate.
      - Explanation: Displays an explanation of the slab based on the values entered in the above fields.

      The base rate slab allows you to change the rate of charging based on the resources used. For example, if you enter 50 in Greater than or equal and 10 as the Base Rate, it means if the usage is 50 Mbps and above, then the Base Rate of 10 will be applied for the whole usage.

      The slabs that you add appear in the table below. To edit or delete the entries, click the vertical ellipsis and select the desired option.
- **Recurring Charges**: Click **Recurring Charges** in the left pane and then click the Lock icon to edit the parent policy settings. Use this section to charge recurring costs for for each additional regular and VKS VM based on the frequency of charging and the power state.
  - Regular VMs
    - Additional Cost per VM: Enter the cost to be charged for each VM based on the settings in the Charge Period and Charge Based on Pwer State.
    - Charge Period: The Charge Period indicates the frequency of charging.
    - Charge Based on Power State: This determines how the power state of the VM affects the charges.
  - VKS VMs
    - Additional Cost per VKS VM: Enter the cost to be charged for each VKS VM based on the settings in the Charge Period and Charge Based on Pwer State.
    - Charge Period: The Charge Period indicates the frequency of charging.
    - Charge Based on Power State: This determines how the power state of the VM affects the charges.
- **Guest OS Rate**: Click **Guest OS Rate** in the left pane and then click the Lock icon to edit the parent policy settings. This section is used to charge for any additional charges that are to be applied on Virtual machines based on their Discovered Guest Operating System by vCenter.
  - Click **Create Guest OS Rate** and enter the following details:
    - Guest OS Name: Enter a guest OS name.
    - Charge Period: The Charge Period indicates the frequency of charging.
    - Charge Based on Power State: This determines how the power state of the VM affects the charges.
    - Base Rate: Enter a base rate.

    The guest OS rates that you add appear in the table below. To edit or delete the entries, click the vertical ellipsis and select the desired option.
- **vCenter Tag Rate**: Click **vCenter Tag Rate** in the left pane and then click the Lock icon to edit the parent policy settings. This section is used to charge for any additional amounts applied to the VMs based on their discovered vCenter Tags.
  - **Fixed Rate** (fixed rate charges a fixed amount for each VM with a matching tag): Click **Create Base Rate** and enter the following details:
    - vCenter Tag Key: Enter a tag key.
    - vCenter Tag Value: Enter a tag value.
    - Charge Period: The Charge Period indicates the frequency of charging.
    - Charge Based on Power State: This determines how the power state of the VM affects the charges.
    - Base Rate: Enter a base rate.

    The base rates that you add appear in the table below. To edit or delete the entries, click the vertical ellipsis next to the base rate and select the desired option.
  - **Alternate Pricing Policy**: Click **Create Base Rate** and enter the following details:
    - vCenter Tag Key: Enter a tag key.

    - vCenter Tag Value: Enter a tag value.
    - Policy: Select the alternate pricing policy name.
    - Priority: Select the priority for the alternate pricing policy. When the metadata or tag-based charges overlap, setting a priority allows you to define which policy should be processed first.

    The base rates that you add appear in the table below. To edit or delete the entries, click the vertical ellipses next to the base rate and select the desired option.
- **One Time Fixed Cost**: Click **One Time Fixed Cost** in the left pane and then click the Lock icon to edit the parent policy settings. This section is used to charge for one time incidental charges on Virtual machines, such as creation/setup charges, or charges for one-off incidents like installation of a patch. These costs do not repeat on a recurring basis.
  - VM Creation:
    - One Time Fixed Cost per Regular VM: Enter the one time fixed cost.
    - One Time Fixed Cost per VKS VM: Enter the one time fixed cost for a VKS VM.
  - vCenter Tag: Click **Create vCenter Tag** and enter the following details:
    - vCenter Tag Key: Enter a tag key.
    - vCenter Tag Value: Enter a tag value.
    - One Time Fixed Cost: Enter the one time fixed cost.

    The details that you add appear in the table below. To edit or delete the entries, click the vertical ellipsis next to the tag entrey and select the desired option.
- **Rate Factors**: Click **Rate Factors** in the left pane and then click the Lock icon to edit the parent policy settings. Use Rate Factors to either increase or discount the prices against individual resources consumed by the Virtual Machines or by whole charges against the Virtual Machine.
  - vCenter Tag: Click **Create vCenter Tag** and enter the following details:

    The details that you add appear in the table below. To edit or delete the entries, click the vertical ellipsis next to the tag entry and select the desired option.
    - vCenter Tag Key: Enter a tag key.
    - vCenter Tag Value: Enter a tag value.
    - Change the price of: Select the required option from the drop-down list.
    - By applying a factor of: Enter a valid number. For example, if you want to increase the price of the CPU which has a tag 'Tag1-Value1' by 20% then select CPU from the **Change the price of** drop-down menu and enter 1.2 in **By applying a factor of**.

8. Click **Preview** to review your changes and then, click **Save**.

You can assign policies to the required VCF Automation/Region Quota under **Policy Assignment**. Navigate to **Administration** > **Configurations**, and then click the **Policy Assignment** tile. For details, see Assigning Policies.

Creating Notification Rules for Tenant in VCF Operations

Cloud service providers can add, manage, and edit notification rules for tenants in VCF Operations. Notifications for Region Quotas are sent to VCF Automation and if the tenant notification settings are properly configured within VCF Automation, VCF Automation will automatically forward the notifications to the tenant.

1. To manage your notifications, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, select **Administration** > **Configurations** and then click the **Tenant Notifications** tile. On the toolbar, click **Add** to add a rule, or click the vertical ellipsis and select **Edit** to edit the selected rule.
2. Enter the following notification details.

Option	Description
Name	Provide a name for the notification rule.
Description	Describe the notification rule.
Notification Status	Either activate or deactivate a notification setting. Deactivating a notification stops the alert notification for that setting and activating it activates the alert notification.

3. Click **Next**.
4. Select the **Region Quota/Organization** toggle option if you want to send notifications at the organization level. The list of Region Quotas/Organizations is updated based on your selection.
5. From the list of Region Quotas/Organizations, drag and drop the Region Quota or Organization into the left pane.
6. Click **Next**.
7. Define criteria for the notification rule. From the list of alert definitions, drag and drop the alert definition(s) for which you want to trigger a notification, into the left pane.

   The notification will be sent to VCF Automation when ANY of the selected alert definitions triggers an alert.
8. Click **Create**.

You can view the created notification from **Administration** > **Configurations** and then click the **Tenant Notifications** tile. You can also delete, activate, deactivate, import, or export tenant notifications by clicking the horizontal ellipsis next to the **Add** button and selecting the relevant option.

Managing Chargeback Reports

Cloud service providers can view, schedule, and generate reports in VCF Operations. You can create report templates for your tenants.

A report template contains views and dashboards. Views present collected information for an object. Dashboards give a visual overview of the performance and state of objects in your environment.

As a cloud service provider, you can:

- View and access all report templates. For information on accessing reports, see Accessing Report Templates .
- Create report templates for tenants by selecting the **For Tenants** option on the **Name and Description** page while creating or editing a report template. For more information, see Create a Report Template.
- Generate a report for a specific Region Quota. You can order the reports by the date and time that they were created, the report name, the object on which the report is generated, the owner, or the status. You can view the generated reports on the **Generated reports** tab in the **Reports** page (navigate to **Operate** > **Dashboards** > **Reports**). For more information, see Generate and Regenerate a Report.
- Schedule a report for a specific Region Quota. For more information, see Schedule a Report. Cloud service providers can also generate tenant reports on demand.
- Publish a generated report to the tenant. For more information, see Publish a Report to Tenants.
- Configure the retention period of reports for a specific number of days after the removal of an Organization. The default is 10 days. Maximum is 1440 days. Navigate to **Administration** > **Global Settings** > **Data Retention** > **Organization Reports**. You can set a maximum of 1440 days.

Generating Bills in VCF Operations

Chargeback allows you to generate monthly bills to provide an account of the overall expenses used for resources in an organization.

As a cloud service provider, you can generate bills at the Region Quota level. You can provide a bill summary at organization level by separating the costs that are incurred by the different regions. At the Organization level, the charges for resources are calculated at the Region Quota level and below and then aggregated.

1. To generate a bill, in the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, select **Cost** > **Bills**.
2. From the **Bills** page, click the **Generate Bills** button.
3. Enter the following details.

Option	Description
Name	Enter a name for the bill.
Description	Describe the bill.
Start Date	Select a start date for the bill.
End Date	Select an end date for the bill.
Publish to Tenant	Select this option to publish the bill to tenants, which can then be viewed in VCF Automation.
Advanced Settings	
Pricing Enabled Policy	If you want to generate a bill using a specific policy, then select a policy from the list. Only the policies that have Provider pricing configured will appear in the list. It is not mandatory to select a policy from the list. If you do not select any policy, then the bills will be calculated based on Provider pricing configurations present in the default policy.

4. Click **Next** to select resources.
5. Select if you want to generate a bill at the organization level or at a region quota level. The list of resources is updated based on your selection.
6. From the list of resources, drag and drop the Region Quota or Organization into the left pane.

   Note:
   You can use filters to filter resources by Region Quota and Organization.
7. Click **Create**.

If the bill was published to the tenant, the tenant administrator can view the bill on their tenant portal under **Administer** > **Bills**.

For bills generated at the Organization level, you can view the bill summary of the Organization and associated Region Quotas.

Scheduling Bill Generation Using Automation Central

You can create a schedule for bill generation using the Automation Central feature of VCF Operations.

To activate Chargeback capabilities in VCF Operations, the cloud service providers must deploy VCF Automation.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Automation Central**.

   Alternatively, in the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, select **Cost** > **Bills**. In the **Bills** page, click the three horizontal dots next to the **GENERATE BILL** button on the **Bills** tab and click **Schedule Bill**.
2. In the **Automation Central** page, click **Add Job**.

   The **Create New Job** page opens. This page displays the following cards:

   - Reclaim
   - Rightsize
   - Billing
   - Actions
   - Configuration Drift

3. Click the **Billing** card.
4. In the **Billing Information** step of the wizard, provide the following inputs and then click **Next**:

Property	Description
Name	Specify a name for the job. This is displayed in the calendar.
Description	Provide a description for the job.

5. In the **Select an Object** step of the wizard, use the toggle button to display either Region Quota or Organization objects in the right-hand pane. Drag the Region Quota or Organization objects to the left-hand pane to generate bills for those objects. You can select multiple objects. Click **Next**.
6. In the **Schedule** step of the wizard, provide the following inputs:

   Scheduled bills are always published to tenants.

Property	Description
Start Date	Set a start date which is on, or after the current date, for the job to start. Select a date from the calendar displayed on the page. **Note:** There will be a delay of up to five minutes for a job to run. The actual start time for an action also depends on the action itself and the number of target objects involved.
Time zone	For the start time and date to be calculated for the job, select a time zone. Choices are: • Browser - based on the current location reported by the browser. • Host - based on the current location of the host machine. • GMT based time zone - Based on a location calculated as per UTC+0.
Start Time	Select a start time for the job. The drop-down menu provides options in five-minute intervals, starting closest to your current browser reported time.
Recurrence	Set a recurrence for the job. The choices are: • **Weekly**. Select the days of the week when you want the job to run, by clicking the day of the week abbreviation. • **Monthly**. Select the months when you want the job to run, by clicking the month abbreviation. By default, all the months are selected. For the months that you select, you can configure the job to run: – On specific days of the month, by clicking the number. Or, the last day of the month, without specifying the exact date, by clicking **Last**. – On the specific number of the week (first, second, third, fourth or the last) in the month, combined with the specific day of the week. You can make multiple selections for each of the options in the drop down menu. For the **Daily**, **Weekly**, **Monthly** options, you can set when the job must end based on a date, or the number of occurrences.
End	Provide a date when the schedule ends.
Bill Start Day	From the drop-down options, provide the start date for the billing period.
Bill End Day	From the drop-down options, provide the end date for the billing period.
Notifications	In the **Notifications** section, select the **Receive Updates on Job via Email** check box to receive notifications two hours

Property	Description
	before the job is set to run and after it has been executed. For the email to be sent, you must also select the email outbound plugin from the drop down menu, and enter the email address to which the email must be sent. If you have not created an email outbound plugin, see the topic, Outbound Settings in VCF Operations. **Note:** You can send updates via email to only one email ID.

7. Click **Create** to complete the steps in the wizard and create the job.

The job to create a bill as per the schedule is created and added to the Automation Central calendar. In the VCF Operations navigation bar at the top, when you click **Manage** and then in the left navigation pane, select **Automation Central**, you see the calendar of events. You can also switch the tabs to **History**, **Jobs** to see more information about the jobs.

Note:
The **Report** tab does not display any information related to billing.

Download and Publish Bills

After you generate bills to provide an account of the overall expenses used for resources in an organization, you can download the bills and publish the bills to tenants.

Prerequisite

- Generate a bill. For more information, see Generating Bills in VCF Operations.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, select **Cost** > **Bills**.

Download a PDF

2. From the **Bills** page, select the bill and then click the **vertical ellipsis** > **Download PDF**.

Publish a bill to the tenant

3. From the **Bills** page, select the bill and then click the **vertical ellipsis** > **Publish**.

View Chargeback Summary, Organizations, and Projects

You can view Chargeback summary, organization, and project details.

Overview

From the **Cost** > **Chargeback** > **Overview** tab you can view a summary of the resources in an Organization, an overview of CPU, memory, and storage use of an Organization, and detailed information about each Organization, such as the number of Region Quotas, VMs, running VMs, cost, and VKS clusters and Pods in VCF Operations.

Organizations

From the **Cost** > **Chargeback** > **Organization** tab you can see a list of Organizations, the associated Region Quotas, and namespaces. You can also view Cost and Price comparison trends for each Organization, Region Quota, and namespace.

Projects

From the **Cost** > **Chargeback** > **Projects** tab you can view a list of Projects and namespaces. A chart comparing the organization cost, project cost, and namespace cost vs price trend for each is displayed.

Optimizing Capacity in VCF Operations

In VCF Operations, capacity optimization focuses on efficiently using and managing resources within a virtualized infrastructure, such as a VMware vSphere environment. These features help organizations maximize the utilization of their resources while ensuring that capacity remains within acceptable limits.

Capacity Optimization

This aspect of VCF Operations involves forecasting and planning for future resource needs based on historical data and current usage patterns. The goal is to ensure that the virtual environment has the necessary resources to accommodate workload growth. Capacity optimization involves activities like identifying underutilized or overprovisioned resources, predicting when additional resources will be needed, and suggesting right-sizing of virtual machines and ESX hosts.

Capacity Optimization Concepts

Capacity optimization in VCF Operations involves various key concepts and practices to ensure optimal resource utilization, plan for growth, and maintain performance within a virtualized environment.

How Does Capacity Optimization Work in VCF Operations

Capacity optimization in VCF Operations is achieved using powerful integrated functions - capacity overview, workload balancing and optimization, repurposing of underutilized resources, and what-if predictive scenarios - to reach optimal system performance.

Capacity planners must assess whether physical capacity is sufficient to meet current or forecasted demand. With robust capacity planning and optimization, you can manage your production capacity effectively as your organization addresses changing requirements. The objective of strategic capacity optimization is to reach an optimal level where production capabilities meet ongoing demand.

VCF Operations analytics provide precise tracking, measuring and forecasting of data center capacity, usage, and trends to help manage and optimize resource use, system tuning, and cost recovery. The system monitors stress thresholds and alerts you before potential issues can affect performance. Multiple pre-set reports are available. You can plan capacity based on historical usage, and run what-if scenarios as your requirements expand.

Capacity Optimization Using Overview, Reclaim, Workload Optimization and What-If

Capacity management is organized into three primary functions, Assess, Optimize, and Plan. Together, they provide a comprehensive view of data center activity and resource trending.

Use the Assess page to conduct on-the-spot analysis of your environment's capacity health. This dashboard highlights Time Remaining until resources (CPU, Memory, or Disk Space) run out and provides high-level Optimization Recommendations to reduce waste. The Cluster Utilization graph shows the time remaining until CPU, Memory and Disk Space run out based on demand and allocation models.

The Optimize page allows you to rebalance and tune compute resources through dedicated tabs - Reclaim, Rightsize, Workload Placement, and Resource Efficiency. Specifically, the Reclaim tab identifies underutilized workloads and calculates the potential cost savings and reclaimable capacity (CPU, Memory, Disk Space) available for redeployment. Workload Placement provides for moving virtual workloads and their file systems dynamically across datastore clusters within a data center or custom data center. From the Reclaim and Rightsize tabs, you can also schedule automation jobs that act on these recommendations, so reclaim and rightsize actions run on your own schedule and conditions instead of requiring manual follow-up each time.

Plan enables forward-looking capacity planning through What-If Analysis and Committed Scenarios, allowing you to interact with and manipulate data outcomes based on your future requirements. The What-If analysis function- can run scenarios that help determine where additional system resources can be accommodated.

> **Note:**
> You may see a data center or cluster labeled as optimized when it has few or no days remaining before CPU, memory, or storage is predicted to run out. That is because these are two different measures of data center and cluster health. A data center can be running at optimum levels based on policy settings for balance and

consolidation, yet be almost out of resources. It is important to consider both measures when managing your environment.

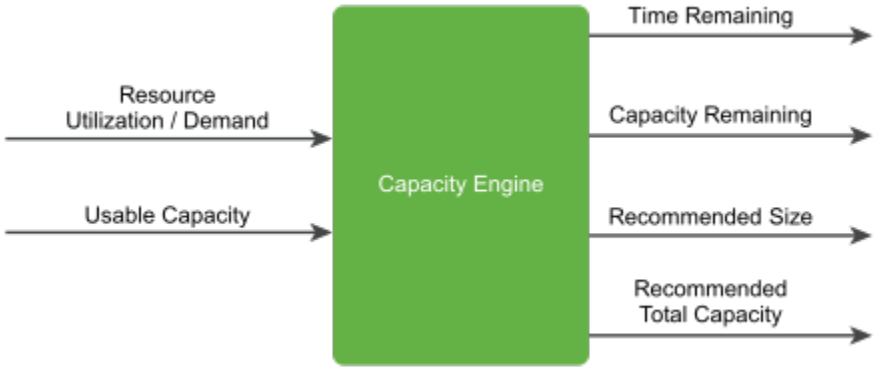
How Does VCF Operations Calculate and Forecast Capacity

Capacity analytics helps you assess the utilization and capacity remaining in objects across your environment. An evaluation of the historical utilization of resources generates a projection of the future workload. You can plan for infrastructure procurement or migrations based on the projection and avoid the risk of capacity shortage and high infrastructure costs.

Capacity analytics uses the capacity engine to assess historical trends, including utilization peaks. The engine selects an appropriate projection model to predict future workload. The amount of historical utilization data that the engine considers depends on how much data has been collected..

Capacity Engine and Calculations

The capacity engine analyzes historical utilization and projects future workload by using real-time predictive capacity analytics, which is based on an industry-standard statistical analysis model of demand behavior. The engine takes the Demand and Usable Capacity metrics as input and generates the output metrics, which are Time Remaining, Capacity Remaining, Recommended Size, and Recommended Total Capacity, as shown in the following figure.



The projection window for the capacity engine is 1 year into the future. The engine collects data points every five minutes and processes them once in 24 hours to calculate output metrics. Any changes made to the policy applied to the objects and object groups, or total capacity, will trigger an immediate recalculation.

The capacity engine projects the future workload in a projected utilization range. The range includes an upper bound projection and a lower bound projection. Capacity calculations are based on the time remaining and risk level. The engine considers the upper bound projection for a conservative risk level and the mean of the upper bound projection and lower bound projection for an aggressive risk level. For more information about setting risk levels, see, Capacity Details.

The capacity engine calculates the Time Remaining, Capacity Remaining, Recommended Size, and Recommended Total Capacity.

You can define custom metrics to compute capacity. The custom metric configuration is available in the policy settings. The calculation algorithm itself does not change. The custom metrics that you configure in the policy is used by VCF Operations in the capacity calculations. You can select the default metrics shipped with VCF Operations, or create super metrics and select them for custom capacity calculations. Changes that you make take effect after the next collection cycle.

Note:

Enabling custom metrics in the capacity calculations is an advanced configuration. Custom metrics alter the way VCF Operations calculates capacity across your enivronment. Use this setting only when needed.

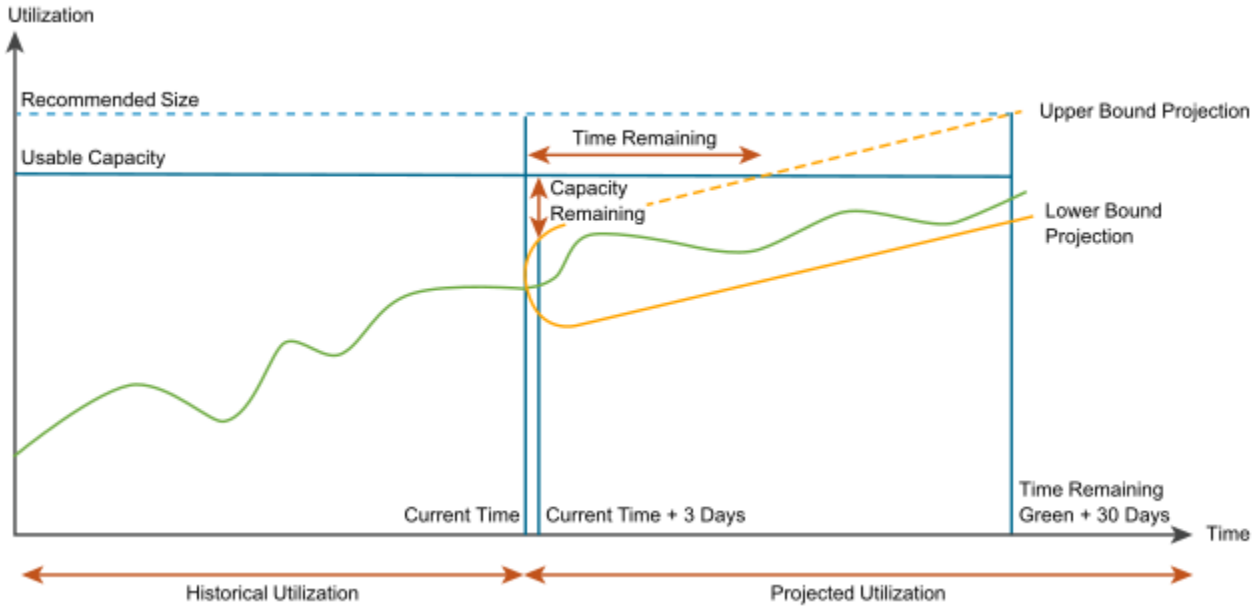
For more details on how to enable custom metrics, see see, Capacity Details.

Time Remaining	The number of days remaining till the projected utilization crosses the threshold for the usable capacity. The usable capacity is the total capacity minus the resources that vSphere HA admission control reserves for failover.

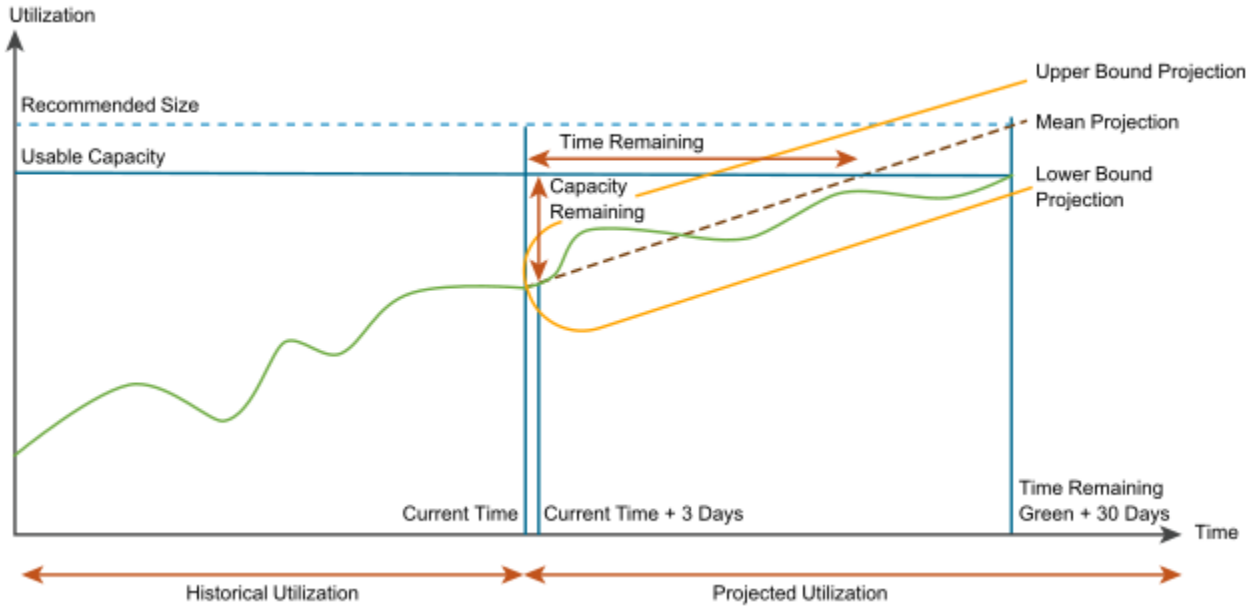
Capacity Remaining	The largest difference between the usable capacity and the projected utilization between now and 3 days into the future. If the projected utilization is above 100% of the usable capacity, the capacity remaining is 0.
Recommended Size	The maximum projected utilization for the period from now to 30 days beyond the time remaining warning threshold. The warning threshold is the period during which the time remaining is green. You set this value in the Time Remaining policy element. See Policy Time Remaining Element. For example, if the warning threshold for time remaining is 120 days, which is the default value, the recommended size is the maximum projected utilization 150 days into the future. The recommended size does not include the capacity that vSphere HA reserves for failover. To obtain your actual sizing requirement, add your vSphere HA spare capacity requirement on top of the recommended size. VCF Operations caps the recommended size that is generated by the capacity engine to keep the recommendations conservative. • VCF Operations caps an oversized recommended size at 50% of the currently allocated resources. For example: a virtual machine is configured with 8 vCPUs but has never used more than 10% of that. Instead of recommending a reclaim of 7 vCPUs, the recommendation is capped to reclaiming 4 vCPUs. • VCF Operations caps an undersized recommended size at 100% of the currently allocated resources. For example: a virtual machine is configured with 4 vCPUs, and has been constantly running very hot. Instead of recommending the addition of 8 vCPUs, the recommendation is capped at adding 4 additional vCPUs.
Recommended Total Capacity	The maximum projected utilization for the projection period from the current time to 30 days after the warning threshold value for time remaining. The recommended total capacity includes vSphere HA settings. For example, if the warning threshold value for time remaining is 120 days, which is the default value, the recommended size is the maximum projected utilization including vSphere HA values, 150 days into the future.

Note:
Recommended total capacity is not available for objects.

The following figure shows the capacity calculations for a conservative risk level.



The following figure shows the capacity calculations for an aggressive risk level.



Note:

- If vSphere HA is not enabled in vCenter then Usable Capacity = Total Capacity. In this case, the Usable Capacity value can be 0 only if there are no ESX hosts in the vSphere cluster.
- If vSphere HA is enabled, then Usable Capacity can be 0 in the following cases:
  - There are no ESX hosts in the vSphere cluster.
  - vSphere HA is configured incorrectly. For example: Admission Control is configured with a cluster resource Percentage policy set to 100% of CPU or Memory. Please check the HA configuration in vCenter.
  - vSphere HA Active host count is less than 2.
  - The ESX host is not vSphere HA Active if:
    - ESX host is in maintenance mode.
    - ESX host is powered off.
    - Value of “runtime.dasHostState” property is not equal to “connectedToMaster” or “master”. This can be because of some network issues between the ESX hosts.

Utilization Peaks

The historical utilization of resources can have peaks, which are periods of maximum utilization. The projection of future workload depends on the types of peaks. According to the frequency of peaks, they can be momentary, sustained, or periodic.

Momentary Peaks	Short-lived peaks that are a one-time occurrence. The peaks are not significant enough to require additional capacity, so they do not impact capacity planning and projection.
Sustained Peaks	Peaks that last for a longer time and impact projections. If a sustained peak is not periodic, the impact on the projection lessens over time because of exponential decay.
Periodic Peaks	Peaks that exhibit cyclical patterns or waves. The peaks can be hourly, daily, weekly, monthly, during the last day of the month, and so on. The capacity engine also detects multiple overlapping cyclical patterns. Because these peaks recur, their impact does not decay over time. The capacity engine projects the pattern forward, so periodic peaks have the greatest influence on capacity planning and projection.

Projection Models

The capacity engine uses models to generate projections. The engine constantly modifies projections and chooses the model that best fits the pattern of historical data. The projection range predicts the general usage pattern that covers 90% of the future data points. Projection models can be linear or periodic.

Linear Models	Models that have a steadily increasing or decreasing trend. Multiple linear models run in parallel and the capacity engine chooses the best model.

	Examples of linear models are linear regression and autoregressive moving average (ARMA).
Periodic Models	Models that discover periodicity of various lengths, such as hours, days, weeks, months, or the last day of the week or month. Periodic models detect square waves that represent batch jobs and handle data streams that contain multiple overlapping periodic patterns. These models ignore random noise. Examples of periodic models are Fast Fourier Transforms (FFTs), pulses (edge detection), and wavelets.

Forecast In Trend Views

Forecasts are generated based on the time range specified in the view settings and are forecasted for the number of days specified in the forecast setting. The forecast is generated based on 3 main algorithms - Change-point detection to find sections of the history with significant changes, linear regression to find linear trends, and cyclical analysis to identify periodic patterns.

Historical Data Window

The capacity engine captures historical data over a period of time depending on the historical data window. The historical data window that the engine uses is an exponential decay window.

The exponential decay window is a window of unlimited size in which the capacity engine gives more importance to the most recent data points. Beginning from the projection calculation start point, the engine consumes all the historical data points and decreases the weight attributed to each one exponentially based on how far back in time they are.

Altering Capacity Calculation

Capacity calculations in VCF Operations can be managed by changing the date from when the historical utilization is calculated or excluding up to 6 time periods from the capacity calculations. Storage considerations can also be excluded from capacity calculations to exclude storage utilization from influencing the overall capacity calculation. This can be done either by editing the policy settings (Capacity Details) or adjusting capacity configuration settings (Using the Capacity Page to Asses and Optimize Capacity). These changes influence future trend calculations.

Allocation and Demand Model in Workload Optimization

VCF Operations uses the demand model and allocation model for Workload Optimization. By default, only the demand model is used. You can turn the allocation model on in the active policy.

What is Allocation Model?

The allocation model determines how much compute, memory, and storage resources are allocated to object types. You define the allocation values by modifying the policy which is applied to the objects. The allocation values, also known as overcommit ratios, affect performance and cost.

The allocation model works alongside the demand model. Unlike the demand model which always affects the capacity calculations, the allocation model can be turned on or off in the policy setting. You can control the ratio by which VCF Operations overcommits either the CPU, memory, or disk space. By specifying the allocation values in the policy, you can choose whether you want to overcommit your resources or not. Overcommitting helps you measure utilization of resources in a pay-as-you-go model. When you do not overcommit, the utilization of your cluster will never exceed 100%. If your resource utilization is over the allocation ratio that you set, Capacity Remaining becomes zero.

To modify a policy and configure overcommit ratios, after you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administrations** > **Configurations**. Click the **Policy Definition** tile.

What is Demand Model?

The demand model is a concept used to predict future resource requirements based on historical data and usage patterns. The demand model is a key component of capacity planning and optimization, allowing administrators to anticipate the growth of workloads and infrastructure resources needed to support those workloads over time.

The demand model takes into account various factors, such as historical resource consumption, trends, seasonality, and potential changes in workloads. By analyzing this information, VCF Operations can provide insights into when additional resources (such as CPU, memory, storage, and network capacity) will be needed to meet the expected growth in demand.

Difference between Allocation Model and Demand Model in Workload Optimization

When clusters in a data center are not optimized, VCF Operations uses the demand model by default to optimize the VMs placed in the cluster. While VCF Operations moves VMs to optimize the cluster, it uses the capacity demand to calculate the best way to optimize the cluster.

When you turn on the Allocation Model in the active policy, VCF Operations optimizes based on the Demand Model, and takes into consideration the overcommit ratios defined by the Allocation Model. As the VCF Operations WLP engine takes into the consideration the overcommit ratios based on the allocation model for CPU, memory, and disk space, these resources will not be under stress after optimization due to the over allocation. In any optimization model which is based on allocation the target is always to reduce the stress on all clusters and ensure that after the VM migration the target clusters are not over-allocated.

How to Enable Allocation Model

After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administrations** > **Configurations**. Click the **Policy Definition** tile. Select the active policy and click the **Capacity** card. In the Capacity Setting section, click the lock icon next to the Allocation Model for your vCenter object type. The available object types are:

- Cluster compute resource. Set the overcommit ratios for CPU, memory and disk space. You can also consider powered off VMs by selecting the **Activate** checkbox.
- Datastore. Set the overcommit ratio for the vSAN datastore disk space.
- Datastore cluster. Set the overcommit ratio for cluster disk space.

After you have set the overcommit ratios, click **Save**. After setting the overcommit ratios for the Allocation Model in the active policy, go to the Workload Placement page to see the Allocation Model displayed under the **Are your clusters meeting your utilization objective?** section.

Policy Settings for Capacity

VCF Operations lets you define capacity and workload automation policies on objects and object groups. These policies help you define custom metrics, and set symptom thresholds, alerts, and recommendations for capacity planning.

To set the capacity settings for your policy, from the left menu, after you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administrations** > **Configurations**. Click the **Policy Definition** tile. Click **Add**. Alternatively, click the **Policy Definition** , select a policy from the list, and click **Edit** from the horizontal ellipsis.

For more information on policy settings for capacity, see the topics, Capacity Details and Workload Automation Details.

How to View and Assess Capacity

VCF Operations helps you assess capacity by predicting future resource needs based on historical data and usage patterns. VCF Operations helps you analyze trends, forecast demand, and allocate resources accordingly to prevent capacity shortages and overprovisioning. To view capacity, you can open the Capacity page or view the Capacity Tab in the Inventory page.

Assessing Capacity in the Capacity Page

The Capacity page in VCF Operations helps you assess workload status and how much capacity is remaining in data centers across your environment.

Where You Find the Capacity Page

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Assess** under **Capacity**.

> **Note:**
> Click on a data center graphic to display the object details screen for the data center.

For more details on how to use the capacity page to optimize capacity, see the topic, Using the Capacity Page to Asses and Optimize Capacity.

For more details on viewing objects on the capacity page, see the topic, Viewing Object Capacity in the Capacity Tab.

Viewing Object Capacity in the Capacity Tab

The **Capacity** tab provides Time Remaining and Capacity Remaining data for the selected object. Virtual Machine Remaining data is the sum of Virtual Machine remaining from each Cluster Compute Resource and is available for datastores, datastore clusters, clusters, data centers, CDC, and VC based on the average profile, or when you activate one or more custom profiles in the policy.

Where You Find the Capacity Tab at an Object Level

After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory**. Then select a group, custom data center, application, or inventory object. Click the **Capacity** tab.

Understanding the Capacity Tab

For the selected object, the **Capacity** tab displays the Time Remaining, Capacity, and Virtual Machine Remaining information. These panes display the value of the resources remaining till they run out.

Below the **Time Remaining**, **Capacity** and **Virtual Machine Remaining** panes, the time and capacity utilization metric for CPU, memory, and disk space are displayed in three panes. By default, the most constrained resource is selected. Click **CPU**, **Memory**, or **Disk Space** to change the views to these resources. These panes display the resource information based on the Demand model (default) or Allocation model (if configured).

For datastores, if you have activated Allocation Model and Capacity buffer in the assigned profile, you see the **Disk Space** information based on Allocation and Usage.

Time Remaining Pane	When you select the **Time Remaining** pane and click one of the resource types, the utilization graph displays the historical value of the utilization metric and its forecast plotted against time, projecting how swiftly resource utilization is approaching the usable capacity. For example, if the capacity calculation is using historical data based on the duration when you were provisioning a large number of VMs, but in the recent past, there was no provisioning done. Then, VCF Operations may not project a trend that is based on the recent data when there was no provisioning, but still will consider historical data when there were a large number of VMs provisioned. This historical data may show an increasing trend. In this case, you can change the capacity computation to start from a date after the VM provisioning is complete.
Capacity Pane	The **Capacity Remaining** pane indicates the unused capacity of your virtual environment to accommodate new virtual machines. VCF Operations calculates the Capacity Remaining as a percentage of the remaining capacity, compared to the total. Capacity Remaining is calculated as the utilization metric forecast 3 days from now subtracted from the Usable Capacity. VCF Operations calculates the average profile and always computes the virtual machine remaining number based on the average profile. You can change the profile by clicking the **Go to Custom Profiles** link. VCF Operations calculates virtual machine remaining numbers when you activate one or more custom profiles from the policy. The overall virtual machine remaining is based on the most constrained profile. The Average Profile size is derived by calculating the average utilization of consumers and overcommit ratios (if enabled) against the usable capacity of this object. When you select Capacity and click one of the resource types, a bar chart and a table of values based on the Demand and Allocation model (if configured) appears. The bar chart displays total usable resource, the percentage used, the percentage allocated for high availability and buffer, and the percentage remaining based on the Demand and Allocation models (if configured).

	The table displays the following information for each resource type: • **Total**: The total usable capacity for each resource type based on the Demand model or Allocation model (if configured). The difference in Total capacity and Usable capacity is set in the HA (admission control) that is set in the clusters in vSphere. • **Usable**: The total usable capacity for each resource type based on the Demand model or Allocation model (if configured). • **Used**: Approximate value how much utilization do you have now. Shows the forecast value of utilization metric in 3 days from now. • **Recommended Size**: The Total Capacity that must be available for a green level of Time Remaining. The slider in the policy controls the Time Remaining green zone, and the default value is 150 days. • **Remaining**: The Capacity Remaining metric value and also the percentage. The value of Capacity Remaining metric is calculated by forecasting the utilization metric 3 days from now and subtracting it from Usable capacity. • **Buffer**: The percentage of the capacity buffer based on the buffer value that you set in the policy. The Capacity Buffer element determines how much extra headroom you have and ensures that you have extra space for growth inside the cluster when required. • **High Availability**: The percentage of the high availability based on the high availability buffer.
Virtual Machine Remaining	The virtual machine remaining number is based on the average profile. The virtual machine remaining numbers are calculated when you activate one or more custom profiles from the policy. The overall virtual machine remaining is based on the most constrained profile. When you click on Virtual Machine Remaining, you see the number of virtual machines based on the Average Profile. Select a different policy by clicking **Policy** from the drop-down list on the top right-hand side of the page. To activate custom profiles in that policy, or other policies, click the **Go to Custom Profiles** link.
Utilization graph	Only vSAN datastore objects, and not other datastore objects, display disk space based on Buffer and HA. Data shows current and trending resource usage and pinpoints when a given cluster is projected to run out of CPU, memory, or disk space based on the allocation or demand model (default). VCF Operations takes into account the business hours which you set in the current policy. View the legend of the colors in the Utilization graph. Click on any one of the legends to toggle its display in the chart.

The **Capacity** tab is a subset of the Capacity optimization capability. For additional details, refer to Assessing Capacity in the Capacity Page.

Managing the Capacity Calculations

Click the **RESET** button on the top right of the page if you want to change the date from when the historical utilization is calculated. By default, it is calculated from the object creation or VCF Operations installation date. The Utilization graph displays the reset calculation start point. Click **RESET** if you want to change projected utilization, for example, drop the irrelevant historical data from the calculation. Using the reset button doesn't impact actual data, just the calculations being shown on the page.

When you click the RESET button, you can also exclude up to 6 time periods from the capacity calculations. This is useful if you want to exclude anomalous data patterns from impacting capacity projections. Click **Advanced Settings** to bring up the option to exclude dates. For the time intervals you select, VCF Operations will consider linearly projected values of the metrics. You can delete time intervals if you want to add more than six exclusions.

You can also exclude storage consideration from influencing the overall capacity calculation for a cluster compute resource. This option will help you deliberately maximize your storage utilization. To exclude storage from influencing the capacity calculations, use one of these methods:

- By editing the policy. Click **Edit Capacity** under the **Policy** drop-down list on the top right-hand side of the Capacity Tab page. Click the lock icon and select the option, **Exclude storage capacity**. For more information on policy settings for capacity, see, Capacity Details.
- By editing the capacity configuration settings. From the Capacity configuration settings in the Capacity page. For more details, see, Using the Capacity Page to Asses and Optimize Capacity.

These changes affect the capacity calculation for the future trend, which will impact Capacity Remaining, Time Remaining, Recommended Size, and VM Remaining (if it is available for the particular resource, namely, Datastore, Cluster, Datastore Cluster). It will also impact all the resource containers (CPU, Memory, Disk Space, etc.)

Note:

If you are unable to see the **RESET** button, make sure that you have the **Manage Capacity Calculation** permission. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel** > **Access Control** > **Roles**. Capacity permissions are available under **Manage** > **Capacity** > **Assess**.

Capacity Shortfall

Capacity Shortfall is a metric that reports capacity overcommitment as a positive number. When an object is overcommitted, Capacity Remaining cannot go below zero, so VCF Operations instead reports the size of the overcommitment as Capacity Shortfall. In the Metrics tree, this appears as **Capacity Shortfall (Profile)**, alongside **Capacity Remaining (Profile)**, under Capacity Analytics Generated.

Capacity Shortfall is available for Cluster, Datastore, and Datastore Cluster objects only. It appears in the Capacity Summary pages next to VM Remaining, and only when VM Remaining is zero. It does not appear while VM Remaining is still positive.

For all these objects you can create custom profiles directly from the capacity page. In the **Capacity** tab, click the **Go to Custom Profiles** link under Virtual Machine remaining. From the **Custom Profiles** page, click **Add** to add a new profile. You can either add new profile or import from existing object. You can modify the required metric value and click **OK**.

How Does Capacity Shortfall Work

When an object has more VMs than its capacity can support, Virtual Machine Remaining shows zero rather than a negative number, and Capacity Shortfall reports how far over capacity the object is. VCF Operations evaluates every profile configured on the object and bases Capacity Shortfall on the profile with the largest number of overcommitted VMs.

Whenever there is no capacity in that case, Capacity Shortfall detects the VM profiles in the setup and finds the profile for which we have the maximum number of VMs that are overcommitted.

Capacity Shortfall Formula

It is possible that when one datastore is connected with multiple clusters, each cluster thinks that the whole diskspace available in the datastore is for itself. We can calculate the Capacity Shortfall as:

```
Capacity Shortfall = (Used - Usable Capacity) / Profile Sizes
```

This calculation is done per profile because a datastore shared across multiple clusters can otherwise be overcounted. Each cluster's Usable Capacity may include the full capacity of the shared datastore, as if it were the only consumer. Calculating shortfall against a specific profile size avoids that overcounting.

Note:
In the above formula the Used Capacity represents the specific value that is displayed in the **Capacity** > **Capacity Remaining** tab.

If you have two clusters, one cluster has 20 VMs remaining and the other cluster has shortfall of 20 VMs, then the number of VMs remaining for the datacenter will be 20 VMs, however, if you have a shortfall of VMs in the datastore connected to that cluster, then we will see on the datacenter the remaining VMs as zero.

Metrics

The Capacity Shortfall displays metrics under Capacity Analytics Generated and under Profiles. The metrics are:

- Capacity Remaining Profile (Average)
- Capacity Shortfall Profile (Average)

vSAN HCI Mesh

The vSAN HCI Mesh is applicable if you have a vSAN OSA HCI cluster. The vSAN HCI Mesh allows vSAN clusters to remotely mount the datastore of another (remote) vSAN cluster, hence sharing the storage capacity and span its usage to a wider pool of compute resources. HCI Mesh allows multiple vSAN clusters to share their datastores remotely. If the vCenter version is 7.0U1 and greater you can share other cluster datastores and provision VMs to the remotely shared data stores.

A new card is created under **Capacity** > **Assess** > **vSAN HCI Mesh**. The vSAN HCI Mesh card is displayed only when there are eligible clusters which can be part of the vSAN HCI mesh. If the version of the vCenter is less than 7.0U1 for the selected datacenter, this card is not displayed. If the mesh is not configured, click the link, which lets you configure the clusters from vCenter.

The vSAN HCI Mesh displays the following information:

- Total Capacity - Displays the total capacity of all the vSAN datastores connected to the mesh.
- vSAN Mesh - Displays how the datastores are interconnected in the HCI mesh.
- Inner Circle - Represents the local datastore and displays capacity details like, free capacity, used capacity, and total capacity.
- Outer Circle - Represents the remote datastore and displays capacity details like, total mounted capacity, free capacity, and total capacity usage percentage.
- Arrow Heads - The arrow heads points to the clients connected to the server vSAN cluster.

Datastore View	Displays the capacity consumption details for the selected datastore. You can also identify which are the client and server clusters.
Cluster View	Displays the capacity details of the remote datastores. You can also view the capacity details for the mounted datastores. The Total Usage % displays the total capacity consumed by all the client clusters connected to these mounted datastores including the selected cluster.
Health Alerts Related to HCI Mesh	Displays the health alerts related to the HCI Mesh. You can also view alerts for individual datastores, which is part of the HCI mesh.

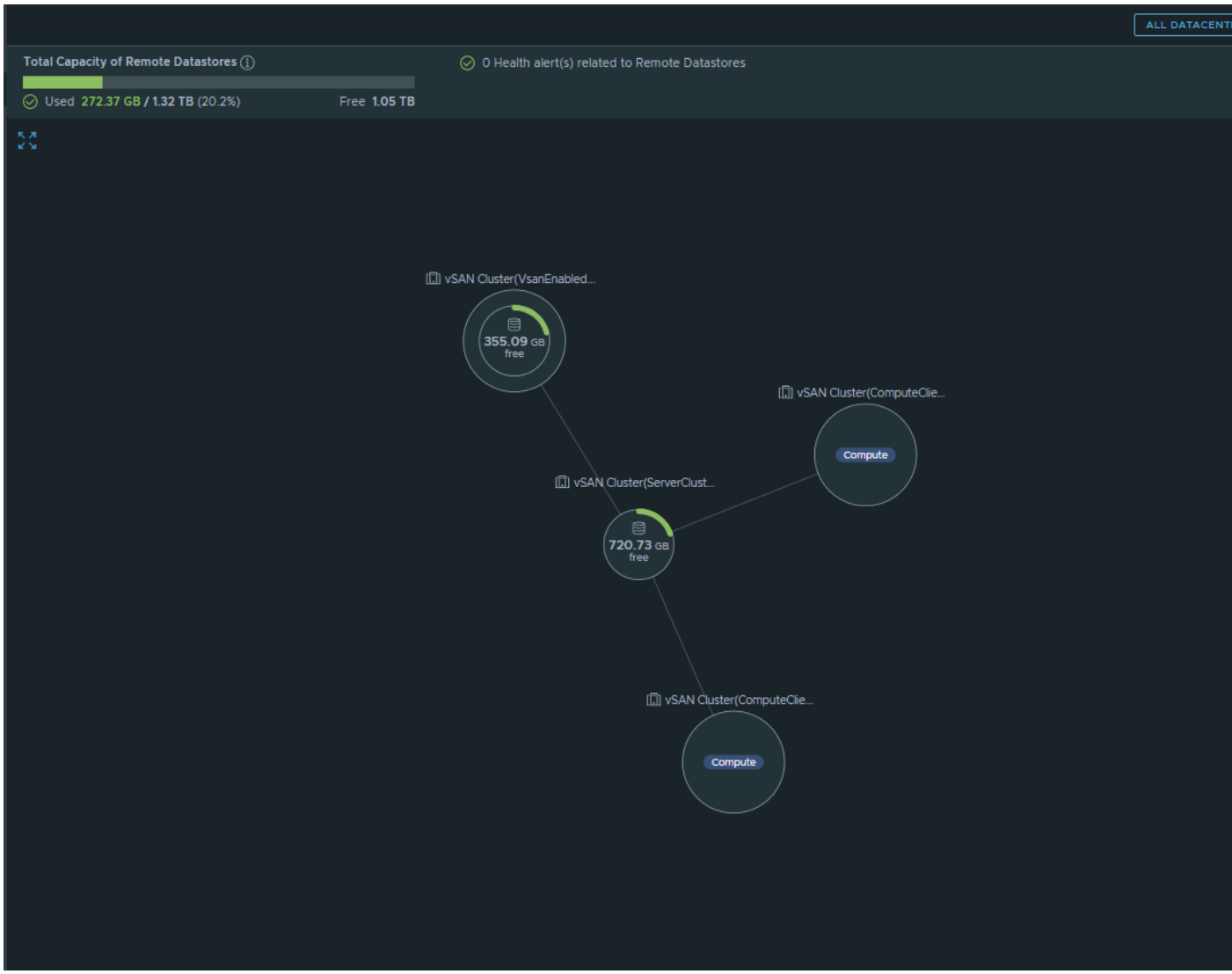
vSAN Remote Datastores

The vSAN Remote Datastore card is displayed if you have a vSAN Max ESA cluster and a vSAN Compute cluster mounting the vSAN Max cluster. The vSAN Compute cluster does not have any storage of its own and uses the storage provided by the vSAN Max cluster. vSAN Max cluster allows vSAN compute to remotely mount it's datastore, hence sharing the storage capacity and span its usage to a wider pool of compute resources. vSAN Max cluster allows multiple vSAN Compute clusters to share their datastores remotely.

The vSAN Remote Datastores card displays the following information:

- Total number of clusters mounted: Displays the total number of the vSAN Compute clusters mounted on the vSAN Max cluster.
- Total Capacity: Displays the total capacity of the vSAN datastores connected to the vSAN Max cluster.
- Free Capacity: Displays the remaining capacity of the vSAN datastores connected to the vSAN Max cluster.

Click **View Remote Datastores** for a more detailed view of the



information.

The total capacity details of the remote datastore and the health alerts (if any) of the remote datastore are displayed along with the relationship between the vSAN Compute clusters and the remote datastore they use.

You can view which vSAN Compute cluster is mounting the vSAN Max cluster and their usage trends. Hover over each cluster type to view the details related to it. For example, if you hover over a vSAN Compute cluster, you can see the number of datastore it is using along with the total amount of mounted and free capacity in gigabytes and the total usage percentage of the cluster. If you hover over the vSAN Max cluster, you can view the total, free, and used capacity in gigabytes, and the usage percentage of the datastore.

Custom Profiles in VCF Operations

A custom profile defines a hypothetical configuration of a virtual machine. Custom profiles help you determine how many instances of that virtual machine can fit in your environment, depending on the capacity remaining and the configuration of the parent object. You activate the custom profile in policies for specific object types. The custom profiles are used to calculate VM remaining when the policy is selected for those object types in the **Capacity** tab of the VCF Operations object browser.

How Custom Profiles Work

As with default profiles, custom profiles are based on metric values for a virtual machine. You can add as many custom profiles as you need for an object. For example, you might create one custom profile for a virtual machine that has a memory demand model of 2 GB. You create another custom profile that has a memory demand model of 4 GB.

VCF Operations uses custom profiles to calculate the number of instances of those virtual machines that can fit in your environment for specific objects. The number of virtual machines is based on the capacity allocation and demand defined for the object.

After you define the metric values for a custom profile, you configure for which policies and object types the custom profiles are activated.

How Custom Profiles are Related to Policies

After you create a custom profile, you activate it in a policy for one or more of the following object types:

- Datastores
- Datastore clusters
- Cluster compute resource

The custom profile is then available for all objects of that object type in the **Capacity** tab of the VCF Operations inventory browser.

To determine how many instances of the virtual machine with a specific configuration as defined by the custom profile you can include in the parent object, go to the **Capacity** tab of the parent object. Select the **Capacity Remaining | Virtual Machine Remaining** pane. Select the policy in which the custom profile is activated from the top right hand side drop-down list. The custom profiles appear in the Virtual Machine remaining section and indicate how many instances of the virtual machine can fit in your environment. The overall virtual machine remaining is based on the most constrained profile.

Where You Find Custom Profiles

After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administrations** > **Configurations**. Click the **Custom Profiles** tile under the **Other** section. A list of all the custom profiles available in VCF Operations is displayed in a table. Click the name of a custom profile from the **Name** column to view the metric values and policies of that custom profile on the right pane.

Table 1744: Options in the Custom Profiles Page

Option	Description
Toolbar options	In the toolbar click **Add** to add a custom profile for a specific object type. Click the **Vertical Ellipses** against a profile to perform the following actions: • **Delete**. Delete the selected profile. • **Export**. Export the selected profile. • **Import**. Import selected profiles. • **Clone**. Clone selected profiles.
Sorting options	Filter the list to display profiles that match the filter you create. You can sort by name, description, object type, or adapter type.
Filter and Search	Use the advanced filter and search box to look for custom profiles. Click the drop-down icon to search by name, description, object type, adapter type.

Custom Profiles Add and Edit Workspace

You can create a custom profile that models a virtual machine's resource requirements, then apply the profile to a cluster compute resource, datastore cluster, or datastore to determine how many VMs of that profile can fit on the object. In the Custom Profiles workspace, you define the metrics for the virtual machine and select the object types the profile applies to.

Where You Create or Edit a Custom Profile

After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administrations** > **Configurations**. Click the **Custom Profiles** tile. The Custom Profiles tile displays all the available custom profiles in a table. You have the option to add custom profiles via the **ADD** button, or filter and search for profiles using the search box. To see the properties of a custom profile, click the custom profile name in the table. The custom profile details are displayed on the right-hand side pane.

Click the ellipsis next to a profile name to edit, delete, clone, or export the profile. Alternatively, to delete or export the more than one profile, select the checkbox next to the profile name and click the horizontal ellipsis next to the **ADD** button, and perform an action. You can also import custom profiles after clicking the horizontal ellipsis next to the **ADD** button.

Add or Edit Custom Profiles Wizard

When you click the **ADD** button, or **Edit** from the horizontal ellipsis, you get a three step wizard. Click the **Next** button after every step to proceed with the wizard.

Table 1745: Profile Step

Option	Description
Name	Descriptive name of the custom profile. This field is mandatory.
Description	Meaningful description for the custom profile. Provide specific information that other users must know about this profile.
Profile Type	The profile type for which the custom profile is created. Currently, you can create a custom profile only for a virtual machine. The Capacity Remaining is calculated for this object type.

Table 1746: Metrics Step

Option	Description
Define Values Custom Profile	Populate the value and unit for the capacity metrics. You can also copy the metric values for the custom profile from an existing object. When you click the **Copy from Existing Object** button, the **Select an Object** pane opens on the right-hand side. Browse to the VM whose metric values you want to copy, or use the advanced search and filter to search by VM Name, vCenter, VM Tag, or Custom Group. Click **Copy** when done.

Table 1747: Policies Step

Option	Description
Checkbox	Select the policies to enable the custom profile in. Objects with those policies applied will use this custom profile to calculate the VM Remaining value: • Cluster Compute Resource • Datastore • Datastore Cluster

Click **CREATE** to create the custom profile or **UPDATE** if you are updating an existing custom profile.

Clone a Custom Profile

Click the ellipsis next to a profile name to clone an existing profile. When you click **Clone**, you get a three step wizard. In the **Profile** step of the wizard, you can provide the name, description, and select the profile type. In the **Metrics** and **Policies** steps of the wizard, the details from the custom profile that you are cloning from are already populated. You can

skip these steps and click the **CREATE** button to create a cloned custom profile with the same values and policies as the original custom profile.

Custom Profiles Import and Export

Export a custom profile as an XML file and import it again in a different VCF Operations instance to easily transfer custom profile configurations.

Where you Import or Export Custom Profile

After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administrations** > **Configurations**. Navigate to the **Custom Profiles** tile. Do one of the following:

- To import a custom profile, click the horizontal ellipsis next to the **ADD** button and select Import. The **Import Custom Profile** dialog box opens. Browse to the XML configuration file and select what VCF Operations should do in case of a conflict. Click **IMPORT** to complete the process.
- To export an existing custom profile, select the box next to the custom profile name. Click the vertical ellipsis next to the name and select **EXPORT**. The XML configuration file is downloaded to your web browser.

Predefined Dashboards for Capacity

VCF Operations provides predefined dashboards to help you monitor and manage your virtualized environments. Capacity dashboards help you analyze the capacity of your environment by providing insights into resource consumption trends, utilization patterns, and potential bottlenecks. They can help you plan for future growth and optimize resource allocation.

For more information on the predefined dashboards available for capacity optimization in VCF Operations, see the topic, Capacity Dashboards.

How to Optimize Capacity in VCF Operations

VCF Operations helps you optimize capacity by identifying and reclaiming unused or underutilized resources, such as reclaiming idle CPU or memory from VMs. This practice can help free up resources for other workloads. Rightsizing helps you adjust the resources allocated to virtual machines based on their actual usage. VCF Operations identifies overprovisioned or underutilized VMs and provides recommendations to optimize their resource allocations.

Excluding VMs from Reclaim and Rightsizing

You can exclude VMs by tags and age, giving you more control over resource optimization by allowing specific virtual machines (VMs) to be filtered out from reclamation and right-sizing recommendations. This ensures that essential system VMs or those with expected resource consumption patterns are not subject to unexpected modifications or reclamation.

The two exclusion mechanisms are:

- Exclusion by tags - You can exclude VMs from optimization recommendations using tags created in the vCenter.
- Exclusion by age - You can exclude VMs by based on a minimum number of days of data collection before the VM can be considered for right-sizing optimization.

Considerations for exclusion based on tags:

Container-Level Setting	Tag-based exclusion settings are applied at the container level, such as a data center, custom data center, or cloud zone. If a data center has multiple containers, the exclusion setting must be designed and applied separately for each container.
Reclamation and Right-Sizing	Tag exclusions can be applied to both reclaim and right-size recommendations.
Intersection of Containers	If a VM belongs to multiple intersecting containers, for example, a VM is in both a Data Center and a Custom Data Center, an exclusion setting defined in one container's exclusion settings will apply and be visible in all other intersecting containers to which the VM belongs.
Reversing Exclusion	To reverse a tag exclusion, you must clean up the tag exclusion setting within the container configuration.

Exclusion by age allows you to set a minimum number of days of historical data collection required before a VM is considered for right-sizing optimization. Exclusion by age helps you ensure that have collected enough historical data to accurately determine if a VM is oversized or undersized based on its memory and CPU usage behavior.

Considerations for exclusion by age:

Right-Sizing Only	This exclusion setting is specifically for right-sizing optimization recommendations.
Disabling the Feature	If this setting is disabled (the default state), VCF Operations uses all the lifetime data available. VMs receive recommendations even if only a minimal number of historical data points are available and the capacity has been calculated.
Capacity Calculation Requirement	Only VMs that have the required minimum number of days of historical data will have their capacity calculated and appear in the right-sizing list.

For more information, see Exclusion Settings in Using Reclaim to Free Up Resources and Using Rightsize to Adjust Resource Allocation.

Using the Capacity Page to Asses and Optimize Capacity

The capacity page in VCF Operations offers a comprehensive view of resource utilization, allocation, trends, and forecasts, allowing administrators to make informed decisions about resource provisioning, performance optimization, and future growth.

Where you find Capacity

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Optimize**.

How Optimize Capacity Works

The Capacity Optimization and Reclaim features are tightly integrated functions that allows you to assess workload status in data centers across your environment. You can determine time remaining until CPU, memory, or disk space resources run out and realize cost savings when underutilized VMs can be reclaimed and deployed where needed.

When you open the Capacity page, graphical representations of all the data centers and custom data centers in your environment appear. VMware Cloud on AWS data centers has a unique icon to differentiate it from the other data centers.

By default, they are shown in order of time remaining, beginning from the upper left, where the most constrained data centers appear. To review the status of a data center, click the graphic. The page refreshes to display the following data:

Time Remaining	Time Remaining specifies which clusters are most constrained and displays the criticality of the cluster.
Optimization Recommendations	VCF Operations shows you the number of reclaimable VMs and the associated cost savings. Click **View Reclaimable VMs** to navigate to the **Reclaim** page.
Cluster Utilization	Cluster Utilization displays an interactive graph that shows time remaining by component. You can explore the demand percentage over time by CPU, memory, and disk space or by the most constrained component. By default, the data displayed is for the Demand model. If you have configured the Allocation model, then you can also see the CPU, memory, and disk space time remaining model based on the overcommit ratios that you have set in the policy. Click **CAPACITY CONFIGURATION DETAILS** to exclude, storage consideration for capacity calculation and to modify the criticality threshold, risk level, and allocation model. These changes affect the selected cluster's policy. Hence, any change that you make here, affects all the clusters under the same policy. Set the **Show History** and **Show Forecast** variables to create the slice of time in which you want to see time-remaining data. The vertical axis of the graph shows the total capacity being used by the current amount of CPU, memory, or disk space respectively. The bold, black line across the top of the graph depicts the historical value of usable capacity. The horizontal axis is the timeline. Vertical lines in the graph are labeled at the bottom of each line. The first vertical dotted line on the left marks the projection calculation start point. The next line is the current date - now. The third vertical marks the date the resource runs out. If a resource has little time remaining, the current date and the date that time runs out may be the same.

VCF Operations can make recommendations for increasing time remaining based on the data it receives and these recommendations appear at the bottom of the screen. You might see two options: Option 1 shows what you can achieve by reclaiming resources. Option 2 shows the results of adding capacity.

If you choose to reclaim resources, you can run that process immediately by clicking **RECLAIM RESOURCES**. To see the details or choose additional options before running a reclaim action, review the information provided in the **Optimization Recommendations** pane and then click **VIEW RECLAIMABLE VMS** to go to the **Reclaim** page.

Table 1748: Capacity Optimization Options

Option	Description
Select a datacenter	Select a data center from the carousel across the top of the page. Information about the datacenter is displayed below.
ALL DATACENTERS | X	Toggle: click **ALL DATACENTERS** on the upper right when you want to switch the view to a filtered list of all data centers. Click **X** to return to a carousel view of data centers.
View:	Filter results to include data centers, custom data centers, or both. This option appears if you select **ALL DATACENTERS** on the upper right.
Group BY:	Filter results by criticality (least time remaining data centers/custom data centers listed first) or by the vCenter to which each data center belongs. This option appears if you select **ALL DATACENTERS** on the upper right.
Sort by:	Options (Options appear if you select **ALL DATACENTERS** on the upper right): • Alarm clock graphic - lists data centers/custom data centers by time remaining. • Dollar sign - lists data centers/custom data centers by potential cost savings. • Scales graphic - lists data centers/custom data centers by level of optimization.
Select datacenter or ADD NEW CUSTOM DATACENTER	Options (options appear if you select **ALL DATACENTERS** on the upper right): • Select a data center from the carousel across the top of the page. All data following refreshes with information for the selected object. • Select **ADD NEW CUSTOM DATACENTER** to display a dialog box that allows you to define a custom data center.
Time Remaining	Appears when you select a data center or custom data center from the top of the screen. Gives overview of cluster status, including how many are at: • Critical • Medium • Normal • Unknown "Critical" can indicate a resource contention, imbalance, or other stress condition. Thresholds you set in the policies define what is critical.
Optimization Recommendations	Lists potential cost savings by reclaiming unused resources. Indicates if workloads can be optimized across clusters. **VIEW RECLAIMABLE VMS** - displays the **Reclaim** screen, where you can research and run potential VM reclamation actions. **VIEW OPTIMIZATION** - displays the **Workload Optimization** screen, where you can optimize workloads based on your policy settings.

Option	Description
Cluster Utilization and Time Remaining	Overall view of cluster health in the selected data center. You can select a cluster from the list to display information about that cluster, or use the options to sort and filter results. The options you select dictate the data displayed in the graph. Sort by: • Most Constrained: most constrained element • CPU (allocation or demand) • Memory (allocation or demand) • Disk Space (allocation or demand) **Note:** Demand model is always on and is the default. **Filter**: search field. **Show History for**: The period before forecasting begins (does not impact the forecast calculation). **Show Forecast For**: The forecast period. **How is the criticality determined?** Displays the criticality threshold you set for this type of object in the Policies Library. **Cluster Time Remaining Settings**: Click the **Edit** icon to edit the default policy for the selected cluster. Change the criticality threshold, risk level, allocation model and capacity buffer. Applying these changes affects all objects in the policy. For more information, see Configuring Policies in the VCF Operations Configuration Guide.
Time Remaining graph	Data shows current and trending resource usage and pinpoints when a given cluster is projected to run out of CPU, memory, or disk space based on the allocation or demand model (default). VCF Operations takes into account the business hours which you set in the current policy.
Legend	Displays a legend of the colours in the time remaining graph. Click on any one of the legends to toggle its display in the chart.
Recommendations	Option 1: Reclaim Resources. Shows resources that can be reclaimed to increase time remaining for the selected cluster. **RECLAIM RESOURCES** - displays the **Reclaim** screen, where you can research and run potential VM reclamation actions. Option 2: Add Capacity. Shows resources that can be added to increase time remaining.

Note:
You might see that a data center or cluster is labeled optimized when it has few or no days remaining before CPU, memory, or disk space is predicted to run out. The seemingly odd assessment is due to optimization and time remaining being two different measures of data center and cluster health. A data center can be running at optimum based on policy settings for balance and consolidation, yet be almost out of resources. It is important to consider both measures when managing your environment.

Using Reclaim to Free Up Resources

Use the **Reclaim** feature in VCF Operations to identify underutilized workloads and reclaim resources from across your environment.

Where You Find Reclaim

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Optimize**. Click on a data center card to display the object details screen for the data center.

How Reclaim Works

The Capacity Optimization and Reclaim features are tightly integrated functions that allows you to assess workload status and resource contention in data centers across your environment. You can determine time remaining until CPU, memory,

or storage resources run out, and realize cost savings when underutilized VMs can be reclaimed and deployed where needed.

When you open the **Reclaim** page, graphical representations of all the data centers and custom data centers in your environment appear. By default, they are shown in order of time remaining, beginning from the upper left, where the most constrained data centers appear. To review the status of a data center, click the graphic. The area following refreshes to display details about the selected data center. The **How much you can potentially save** pane reflects potential capacity savings and indicates a possible cost savings once you have reclaimed underused or powered off VMs. The **Total Reclaimable Capacity** pane gives details of the reclaimable percentages for CPU, memory, and disk space.

The table at the bottom of the page provides important information about the VMs that offer the most cost savings. The VMs are listed by **Powered Off VMs**, **Idle VMs**, **Snapshots**, and **Orphaned Disks**. The highest priority heading is at the far left. You can specify what information is included in your reclaim action. For example, when you click a column heading, the table lists, by data center and then by VM, the allocated and reclaimable CPUs and memory, respectively. Then, for example, you can select the box next to one or more VM names and click the **EXCLUDE VM(S)** button to keep those VMs from being included in any reclaim action. You can also select VMs to resize.

Note:

Ensure that you enable permissions for the users accessing the Reclaim page. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile. Edit or add a role with the following permissions enabled:

- To browse to the Orphaned Disks tab in Relcaim, you must select the checkbox next to the **Reclaim** permission.
- To perform the Delete Orphaned Disks action, you must select the checkbox next to the **Delete Orphaned Disks** permission.

You can use the search function in the **Edit Role Information** page to find and enable the above permissions. For more information, see Add or Edit Roles and Assign Permissions.

VMs in the Snapshot List

In earlier versions, a VM that qualified for both snapshot reclamation and a higher-priority reclamation factor, such as idle or power-off, would not appear in the snapshot list. VCF Operations recognized the higher priority of the power-off or idle reclamation factor, resulting in the snapshot list being empty for such objects. In previous versions, you could only choose to power off or delay an idle VM. Snapshot reclamation was not an option for idle VMs.

Starting in VCF 9.1, VCF Operations displays a VM qualifying for multiple factors such as snapshot reclamation and power-off or idle reclamation, in all the respective sections, including the **Snapshot** list, the **Idle** section, and the **Power Off** section. It offers visibility and control, allowing you to decide which type of reclamation to perform.

You can easily perform snapshot reclamation on an idle VM. Performing a snapshot reclamation will only delete the snapshots. The VM itself will not be powered off or deleted as an idle VM. The VM will be kept as an idle VM and will qualify for idle reclamation again. For more information on how to automate deleting idle and powered off VMs for reclamation, see Automation Central.

Reclamation Settings

Select the gear icon next to the page heading to customize Reclamation Settings. This affects all data centers. Using the Reclamation Settings, you can exclude, for example all snapshots from being included in the reclaim action - by deselecting the Snapshots check box. Similarly, you can include or exclude powered-off VMs, idle VMs, and orphaned disks. For more information, see Reclamation Settings.

Note:

To provide read-only access to the Reclamation Settings page for a user, configure the user role in the Access Control page (Roles tab). In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel** > **Access Control** > **Roles**. After you click **Add** or after you edit an existing role, select the **Manage Global Settings** permissions under **Operate** > **Administration** > **Global Settings**.

Run a Reclaim Action

Run a reclaim action as follows:

1. In the table headings, **Select** the types of VMs to reclaim.
2. **Click** the name of a listed cluster to show its VM list.
3. **Select** each VM or snapshot you want to reclaim.
4. Click **Delete VM(s)** to reclaim their resources.

Table 1749: Reclaim Options

Option	Description
Select a data center.	Select a data center from the carousel across the top of the page. All data refreshes with information for the selected object.
ALL DATACENTERS | X	Toggle: click **ALL DATACENTERS** on the upper right when you want to switch the view to a filtered list of all data centers. Click **X** to return to a carousel view of data centers.
View:	Filter results to include data centers, custom data centers, or both. Option appears when you select **ALL DATACENTERS** on the upper right.
Group BY:	Filter results by criticality (least time remaining data centers/custom data centers listed first) or by the vCenter to which each data center belongs. Option appears when you select **ALL DATACENTERS** on the upper right.
Sort by:	Options (Options appear when you select **ALL DATACENTERS** on the upper right): • Alarm clock graphic - list data centers/custom data centers by time remaining. • Dollar sign - list data centers/custom data centers by potential cost savings. • Scales graphic - list data centers/custom data centers by level of optimization.
Select data center or ADD NEW CUSTOM DATACENTER.	Options (Options appear when you select **ALL DATACENTERS** on the upper right): • Select a data center from the carousel across the top of the page. All data refreshes with information for the selected object. • Select **ADD NEW CUSTOM DATACENTER** to display a dialog box that allows you to define a custom data center.
How much you can potentially save.	Appears when you select a data center or custom data center from the top of the screen. Shows the total calculated potential cost savings when you accept system reclamation recommendations.
Total Reclaimable Capacity	Lists potential cost savings for the selected data center when you reclaim unused resources. Resource: CPU, memory, or disk space Reclaimable Capacity: how much capacity is available to reclaim from idle resources % Reclaimable: percentage of total CPU, memory, or storage you can reclaim.
Exclusion Settings	Apply tags at the datacenter level, to exclude multiple VMs and workloads at once, instead of selecting them individually. By applying tags, you can define exclusion rules that apply globally to specific groups of VMs. For exclusions based on the age of the VM, go to the global settings page. Click **Operate** in the top navigation pane and then go to **Administration** > **Global Settings** in the left navigation pane.
Duration older than:	Shows idle or powered off VMs that have been idle or powered off for at least the selected time period: one week, two weeks, or a month.

Option	Description
Table of Potential Cost Savings	Tabular representation of the VMs, Idle VMs, Snapshots, and Orphaned disks in the selected data center from which resources can be reclaimed. Click one of the elements - powered off VMs, idle VMs, and so on - to refresh the table with data for that element. The table lists the relevant clusters. To see the VMs hosted in a given cluster, click the chevron to the left of the cluster name. Click the check box next to the VMs you want to act on, or click the check box next to the column heading VM Name to act on all the VMs. Once you select a VM or VMs, the dimmed options above the table become visible, as follows. **Exclude VM(s)**: The selected VMs are excluded from your subsequent action. Excluding VMs from a reclamation action can reduce the potential cost savings. For powered Off VMs: • **SCHEDULE ACTION**: Displays a dialog box enabling you to schedule one or more reclaim action for powered off VMs. Expand the cluster name displayed in the table and select one or more VMs. Then, from the SCHEDULE ACTION drop down menu, select an action to be performed later. In the dialog box, you configure the schedule for the job. Scheduled jobs can be managed in Automation Central. • **DELETE VM(s)**: Deletes the selected VMs. • **EXCLUDE VM(s)**: Excludes the selected VMs. • **EXPORT ALL**: Exports the list of powered off VMs into a CSV file. For idle VMs: • **SCHEDULE ACTION**: Displays a dialog box enabling you to schedule one or more reclaim action for idle VMs. Expand the cluster name displayed in the table and select one or more VMs. Then, from the SCHEDULE ACTION drop down menu, select an action to be performed later. In the dialog box, you configure the schedule for the job. Scheduled jobs can be managed in Automation Central. • **DELETE VM(s)**: Deletes the selected VMs. • **POWER OFF**: Powers off the selected VMs. • **EXCLUDE VM(s)**: Excludes the selected VMs. • **EXPORT ALL**: Exports the list of idle VMs into a CSV file. **Note:** The list of VMs includes idle and powered off VMs. For Snapshots: • **SCHEDULE ACTION**: Displays a dialog box enabling you to schedule one or more reclaim action for snapshots. Expand the cluster name displayed in the table and select one or more VMs. Then, from the SCHEDULE ACTION drop down menu, select an action to be performed later. In the dialog box, you configure the schedule for the job. Scheduled jobs can be managed in Automation Central. • **DELETE SNAPSHOT(s)**: Deletes the selected snapshots. • **EXCLUDE VM(s)**: Excludes the selected snapshot. • **EXPORT ALL**: Exports the list of snapshots into a CSV file. **Note:** The list of VMs includes idle, powered off and active VMs. **SHOW|HIDE EXCLUDED VMS**: Toggle displays or hides the list of VMs you previously excluded. **Note:** By default, calculations for reclaimable resources are based on the demand model. But if you turn on the allocation model in the policy settings, the calculations are based on the allocation model. For Orphaned Disks: • **EXCLUDE DISK(S)**: Exclude the selected disks in the actionable list. • **DELETE**: Delete the selected disks. A confirmation dialog box is displayed, which shows the savings and disk space you can reclaim by deleting the disk. • **EXPORT ALL**: Exports the list of orphaned disks into a CSV file. You cannot reclaim orphaned disks from the UI. Instead, export the list into a CSV file and then reclaim the orphaned disks manually.

Reclamation Settings

Displays information about powered off VMs, idle VMs, snapshots and orphaned disks. This information helps to identify the amount of resources that can be reclaimed and provisioned to other objects in your environment or amount of potential savings that can be done in each month.

The types of VMs are ranked in the order of their importance in a reclamation action. A VM whose attributes match more than one VM type is included with the higher-ranking VM type. Grouping the VMs this way eliminates duplicates during calculations. As an example, powered-off VMs are ranked higher than snapshots, so that a powered-off VM that also has a snapshot appears only in the powered-off VM group.

If you exclude a given type of VM, all VMs matching this type are included with the next lower-ranked group they match. For example, to list all snapshots regardless of whether their corresponding VMs are powered-off or idle, deselect the Powered-off VMs and Idle VMs check boxes.

Further, you can configure how long a given class of VMs must be in the designated state - powered-off, for example, or idle - to be included in the reclamation exercise. You can also choose to hide the cost savings calculation.

Property	Description
Show Cost Savings	Controls whether to show Cost savings in the **Capacity** page.
VM Idleness Criteria	Defines the criteria to identify VMs that remain idle during the defined period of time. You can set the following values: • MHz of CPU consumption of VMs, and percentage of the time VMs have less than defined MHz of CPU within each day.
Show the following types of VMs for reclamation	Configure the types of VMs that are shown for reclamation. You can select the type of VMs which are shown for reclamation, and then configure parameters which those VMs have to meet to be eligible for reclamation.
Powered Off VMs	VMs that have been continuously powered off during the defined period of time. The total storage capacity used is reclaimable. Total storage reclaimable cost is computed by multiplying storage rate with storage utilization. The direct cost of VM is also attributed.
Idle VMs	You can configure the following parameters based on which VCF Operations calculates idle VMs: • How many days the VM has been idle for. • Number of days before which the VMs were provisioned based on which they are excluded.
Snapshots	VM snapshots that have existed for the entire defined period of time. Snapshots of a VM use storage space and such storage is reclaimable. The reclaimable cost is computed by multiplying storage rate with reclaimable storage value.

Property	Description
Orphaned Disks	VMDKs on datastores that are not connected to any registered VMs and have not been modified during the defined period of time. Orphaned disks are VMDKs which are associated with a VM which is not in inventory, but still available in a datastore. You can configure the minimum number of days for which VMDKs not related to any existing VM will be reported as orphaned and appear under Orphaned Disks in Reclaim page. **Note:** You can navigate to**Cost/Price** section after you click **Operate** in the top navigation bare, and then click **Ad ministration** > **Global Settings** in the left navigation. Change the value of the **Orphaned Disks Collection** time. At this time that you set, VCF Operations checks for orphaned VMDKs in vSphere Client instances. The settings for **Cost Calculation** and **Orphaned Disks Collection** are interrelated. The default value for **Cost Calculation** is 9:00 PM, and the default for **Orphaned Disks Collection** is 8:00 PM. It is recommended to schedule **Cost Calculation** after **Orphaned Disks Collection**.
Exclusion settings based on time	Number of days of data collection after which VMs can be included for Rightsizing Optimization.
Exclude Broadcom appliances from Optimization	Exclude VCF appliances from Rightsize and Reclaim. Activating this setting does not hide snapshots which can be deleted. Activating this setting shows appliances which are undersized, if applicable, but you cannot resize them.

Note:

If you cannot make changes in the Reclamation Settings page, your user role in the Access Control page must be modified by an administrator. This setting is available when you click **Operate** in the top navigation bar, and then click **Administration** > **Control Panel** > **Access Control**in the left navigation.

Idle VM Calculation and Criteria

The Reclaim Settings is a global configuration that impacts the operation of the entire system. Use these settings to configure idleness criteria and define reclaimable idle conditions for Virtual Machines (VMs). Through these configurations, VCF Operations ensures that VM idleness and Reclaim status is accurately tracked and updated according to real-time data and defined metrics.

VM Idleness Criteria and Analysis Period

The VM idleness criteria defines the specific threshold of CPU utilization that a VM must maintain within a day to be classified as idle. When a VM's CPU demand falls below the utilization percentage set in the reclaim settings for the required portion of the day, VCF Operations, sets the idleness indicator property to 1. Conversely, if the CPU demand exceeds this threshold, VCF Operations, sets the property to 0. The default criteria for this assessment are set to a CPU demand of ≤100 MHz for 100% of the time.

The identification process begins with a 24-hour analysis period for the idleness indicator. Upon the initial installation of VCF Operations, it takes exactly 24 hours on the first day to perform the initial determination of whether a VM qualifies as idle. Following this initial window, VCF Operations conducts hourly evaluations based on the preceding 24 hours of data. It is possible for a VM to be excluded from the idle classification after the first 24 hours but qualify later. For example, during a check three hours later, if its CPU demand satisfies the idleness criteria for the entire preceding 24-hour period.

Reclaimable Idle Conditions

Reclaimable idle status is determined by the number of consecutive days a VM must meet the idleness criteria. If a VM consistently maintains an idleness criteria value of 1 for the specified number of consecutive days, it is marked as reclaimable idle with a value of 1. This reclaimable idle metric is published daily, with its value adjusting dynamically based on whether the condition remains satisfied.

Reset and Evaluation Logic	VCF Operations employs a multi-tiered evaluation logic involving hourly and daily checks: • Hourly Check: The idleness status is evaluated based on data values received within the last hour. If a drop in idleness occurs during this window, the count resets to zero and the process begins again. • Daily Evaluation: The system records the total number of non-idle periods over the last day. If these non-idle periods exceed the set threshold, the property is set to 0; otherwise, it is maintained at 1. For example, if a VM becomes non-idle for three periods within a single hour, it will not be considered idle. • 24-Hour Reset: Every 24 hours, VCF Operations performs a reset of all records to re-evaluate the VM's idleness based on the specific periods within that timeframe that fell below the set limit.
Impact of Modifying the Configuration	When you make modifications to the idleness or reclaimable criteria, the following global changes occur: • Less Strict Conditions: VMs previously classified as idle will retain their status. Within one day of the change, VMs meeting the new condition will have their idleness property updated, though this update can occur earlier within the same day if the condition is met. • More Strict Conditions: Any VM that was considered idle but fails to meet a new, stricter condition will have its idleness indicator changed to 0 within one hour of the system receiving a non-compliant data point. • Decreased Consecutive Days: Reducing the required number of days may lead to an increase in the number of VMs identified as reclaimable. • Increased Consecutive Days: The system evaluates the date when the idleness indicator last transitioned to 1. If the new duration requirement is shorter than the time elapsed since the last recorded idle state, the VM remains reclaimable idle. If the requirement is not met, the VM becomes non-reclaimable until it satisfies the new requirement without violating any conditions.

Using Rightsize to Adjust Resource Allocation

Rightsizing in VCF Operations refers to the process of adjusting the resource allocations of virtual machines (VMs) to match their actual workload requirements. It involves optimizing the allocation of CPU, memory, storage, and other resources to ensure that VMs are neither overprovisioned (allocating more resources than necessary) nor underutilized (allocating fewer resources than needed). The goal of rightsizing is to achieve efficient resource utilization, improve performance, and potentially reduce costs. Use the Rightsize page to alter the number of CPUs and amount of memory in oversized and undersized virtual machines.

Where You Find Rightsizing

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Optimize**. Click on a data center and then click the **Rightsize** tab.

How Rightsizing Works

The Capacity Optimization, Reclaim, and Rightsizing features are tightly integrated functions that allows you to assess workload status and resource usage in data centers across your environment. You can determine time remaining until CPU, memory, or storage resources run out, and realize cost savings when underutilized VMs can be reclaimed and deployed where needed. With this function, you can change CPU size and memory values for oversized and undersized virtual machines to achieve optimum system performance.

When you open the page, graphical representations of all the data centers and custom data centers in your environment appear. By default, they are shown in order of time remaining, beginning from the upper left, where the most constrained data centers appear. To identify possible oversized and undersized VMs in a data center, click its graphic. The area following refreshes to display details about the selected data center.

"Oversized VMs" displays the number of VMs determined to be oversized based on policies previously set. A chart details suggested reductions in the overall number of CPUs and GBs of memory and shows the percentage of total resources the reductions represent. Similarly, "Undersized VMs" indicates the number of VMs considered to be undersized, with a chart listing suggested increases in CPU and memory.

The table at the bottom of the page provides important information about the VMs. Table headings are Oversized VMs and Undersized VMs. VMs under each heading are grouped by cluster. Click the chevron to the left of a cluster name to list all the oversized or undersized VMs, respectively, in that cluster. You can check the box next to one or more VM names

and click the **EXCLUDE VM(S)** button to prevent those VMs from being included in a resizing action. You can also select individual VMs to resize before clicking the **RESIZE VM(S)** button.

Run a Rightsize Action on Oversized VMs

Run the action as follows:

1. Click the Oversized VM tabs.
2. Select the VMs you want to exclude from the action, if any and click **EXCLUDE VM(S)**. In the confirmation dialog box, click **EXCLUDE VM(S)**.
3. Select the VMs you want to include in the resizing action, or click the checkbox next to VM Name to include all VMs.
4. Click **RESIZE VM(S)**. The Resize VM(S) workspace appears. The table displays suggested reductions for vCPU and memory. **Click** the edit icons to accomplish to changes you wish.

   Note:

   Operational Actions must be activated in the vCenter cloud adapter instance.
5. Select the checkbox at the bottom of the screen to indicate your understanding that, because workloads must restart to accommodate resizing, some work may be interrupted.

Run a Rightsize Action on Undersized VMs

Run the action as follows:

1. In the table headings, **Select** Undersized VMs.
2. Select the VMs you want to exclude from the action, if any and click **EXCLUDE VM(S)**. In the confirmation dialog box, click **EXCLUDE VM(S)**.
3. Select the VMs you want to include in the resizing action, or click the checkbox next to VM Name to include all VMs.
4. Click **RESIZE VM(S)**. The Resize VM(S) workspace appears. The table displays suggested increases for vCPU and memory. **Click** the edit icons to accomplish to changes you wish.
5. Select the checkbox at the bottom of the screen to indicate your understanding that, because workloads must restart to accommodate resizing, some work may be interrupted.

Table 1750: Rightsize Options

Option	Description
Select a data center.	Select a data center from the carousel across the top of the page. All data refreshes with information for the selected object.
ALL DATACENTERS | X	Toggle: click **ALL DATACENTERS** on the upper right when you want to switch the view to a filtered list of all data centers. Click **X** to return to a carousel view of data centers.
View:	Filter results to include data centers, custom data centers, or both. Option appears when you select **ALL DATACENTERS** on the upper right.
Group BY:	Filter results by criticality (least time remaining data centers/custom data centers listed first) or by thevCenter to which each data center belongs. Option appears when you select **ALL DATACENTERS** on the upper right.
Sort by:	Options (Options appear when you select **ALL DATACENTERS** on the upper right): • Alarm clock graphic - list data centers/custom data centers by time remaining. • Dollar sign - list data centers/custom data centers by potential cost savings. • Scales graphic - list data centers/custom data centers by level of optimization.
Select data center or ADD NEW CUSTOM DATACENTER.	Options (Options appear when you select **ALL DATACENTERS** on the upper right): • Select a data center from the carousel across the top of the page. All data refreshes with information for the selected object. • Select **ADD NEW CUSTOM DATACENTER** to display a dialog box that allows you to define a custom data center.

Option	Description
Oversized VMs display	Displays the number of VMs identified as oversized, with suggested reductions for vCPU and memory size. **Note:** The list includes idle VMs and VMs with snapshots.
Undersized VMs display	Displays the number of VMs identified as undersized, with suggested increases for vCPU and memory size. **Note:** The list includes idle VMs and VMs with snapshots.
Exclusion Settings	Apply tags to exclude multiple VMs and workloads at once, instead of selecting them individually. By applying tags, you can define exclusion rules that apply globally to specific groups of VMs. For exclusions based on the age of the VM, go to the global settings page. Click **Operate** in the top navigation pane and then go to **Administration** > **Global Settings** in the left pane.

Option	Description
Table of Oversized and Undersized VMs	Tabular representation of the Oversized and Undersized VMs in the selected data center. Click one of the headings - Oversized VMs or Undersized VMs - to refresh the table with data for that heading. The table lists the relevant VMs. To see the VMs hosted in a given cluster, click the chevron to the left of the cluster name. Click the check box next to the VMs you want to act on, or click the check box next to the column heading VM Name to act on all the VMs. Click the expand icon next to a VM to see an explanation of how VCF Operations computes the recommended size. The visual representation shows you the historical utilization pattern, the projection and the recommendation for rightsizing. You can change the graphical representation to show the history from one month to 12 months. **Note:** The rightsizing recommendation is influenced by the risk level that you set in the policy. You can modify the risk level in the policy to change the recommendation. Once you select a VM or VMs, the dimmed options above the table become visible, as follows. **Exclude VM(s)**: the selected VMs are excluded from your subsequent action. Excluding VMs from a reclamation action can reduce the potential cost savings. For Oversized VMs: • **SCHEDULE ACTION**: Displays a dialog box enabling you to schedule one or more resize actions for oversized VMs. Expand the cluster name displayed in the table and select one or more VMs. Then, from the SCHEDULE ACTION drop down menu, select an action to be performed later. In the dialog box, you configure the schedule for the job. Scheduled jobs can be managed in Automation Central. • **RESIZE VM(s)**: The system displays a dialog box with suggestions for reducing vCPUs and memory. Click the edit icons to change resource size. • **EXCLUDE VM(s)**: Excludes the selected VMs. • **EXPORT ALL**: Exports the list of powered off VMs into a CSV file. For Undersized VMs: • **SCHEDULE ACTION**: Displays a dialog box enabling you to resize VM actions for undersized VMs. Expand the cluster name displayed in the table and select one or more VMs. Then, from the SCHEDULE ACTION drop down menu, select an action to be performed later. In the dialog box, you configure the schedule for the job. Scheduled jobs can be managed in Automation Central. • **RESIZE VM(s)**: The system displays a dialog box with suggestions for increasing vCPUs and memory. Click the edit icons to change resource size. • **EXCLUDE VM(s)**: Excludes the selected VMs. • **EXPORT ALL**: Exports the list of powered off VMs into a CSV file. **SHOW|HIDE EXCLUDED VMS**: toggle displays or hides the list of VMs you previously excluded. **INCLUDE VM(s)**: include the selected VMs in the actionable list.

Using Workload Optimization

Workload Optimization in VCF Operations focuses on intelligently balancing and optimizing workloads across your virtualized infrastructure. The goal is to ensure that resources are distributed efficiently to maintain optimal performance, prevent resource contention, and make the best use of available hardware resources.

Using Workload Optimization, you can rebalance virtual machines and storage across clusters, relieving demand on an overloaded individual cluster and maintaining or improving cluster performance. You can also set your automated rebalancing policies to emphasize VM consolidation, which potentially frees up hosts and reduces resource demand.

Workload Optimization further allows you to potentially automate a significant portion of your data center compute and storage optimization efforts. With properly defined policies determining the threshold at which resource contention automatically runs an action, a data center performs at optimum.

In previous versions, you could run workload optimization on a custom data center which has clusters across multiple data centers within a single vCenter instance. The prerequisite for this was that the hosts in the different clusters were under the same network. This means that the port groups had to be the same across the data centers. You could activate cluster level optimization across data center boundaries by enabling the setting in the business intent workspace.

Starting with version 9.1, VCF Operations, Workload Placement is available for both VCF Automation managed VMs and non-VCF Automation managed VMs. Migration is supported across different physical network segments for VMs deployed using VCF Automation. Traditionally Automation Cloud Zones were not aware of network segments but now that limitation is not present.

VCF Automation Integration

When you add an instance to a VCF Automation adapter or solution pack as well as to a vCenter adapter instance that is connected to the VCF Automation server, using VCF Automation-managed resources, VCF Operations automatically adds a custom data center for the vCenter, using VCF Automation-managed resources.

On the VCF Operations side, to get the day2 chain configured, you must make the following initial configurations:

1. In vCenter, **Administration -> Solutions** and then add the VMware vSphere adapter instance for the vCenter that is configured as an endpoint in VCF Automation Server.
2. In vCenter, **Administration -> Solutions** and then add the VCF Automation adapter instance for the server that will appear in the VCF Operations and VCF Automation integration day2 chain.

VCF Operations can manage workload placement and optimization for the custom data centers that reside in VCF Automation-managed clusters.

However, VCF Operations is not permitted to set tag policies for the custom data center. (At the Workload Optimization screen, the Business Intent window is not operational for VCF Automation custom data centers.) When rebalancing a VCF Automation custom data center, VCF Operations uses all applicable policies and placement principles from both systems: VCF Automation and VCF Operations For complete information on creating and managing VCF Automation custom data centers that are managed by VCF Operations see the VCF Automation documentation.

Configuring Workload Optimization

Workload Optimization offers you the potential to automate fully a significant portion of your cluster workload rebalancing tasks. The tasks to accomplish workload automation are as follows:

1. Configure the Workload Automation Details. See Workload Automation Details.
2. If you do not use the AUTOMATE function in the Optimization Recommendation pane at the Workload Automation screen, configure the two Workload Optimization alerts to be triggered when cluster CPU/memory limits are breached, and configure them as automated. When the alerts are automated, the actions calculated by Workload Optimization are run automatically. See Configuring Workload Optimization Alerts

Prerequisites

Workload Optimization acts on objects associated with the VMware vSphere Solution that connects VCF Operations to one or more vCenter instances. The virtual objects in this environment include a vCenter data centers and custom data centers, cluster compute and storage resources, host systems, and virtual machines. Specific requirements:

- Workload Optimization is supported when the optimization candidate clusters are in the same network or port group.
- Optimization candidate clusters should have one of the following datastore configurations:
  - Shared Datastore
  - Datastore Cluster
  - vSAN Datastore
- A vCenter Adapter configured with the actions activated for each vCenter instance.
- A vCenter instance with at least two datastore clusters. sDRS must be activated and fully automated for datastore clusters configuration only.

- Any non-datastore clusters must have DRS activated and fully automated
- Storage vMotion must be set to ON at Workload Automation Details. The default is On.
- You must have permission to access all objects in the environment.

Design Considerations

The following rules constrain the possible computer and storage resource moves that can be performed.

> **Note:**
>
> When VCF Operations suggests that you optimize clusters in a data center, the system does not guarantee it can run an optimization action. VCF Operations analytics can determine that optimization is desirable and can create a rebalancing plan. However, the system cannot automatically identify all the architectural constraints that may be present. Such constraints may prevent an optimization action, or cause an action in progress to fail.

- Moving compute and storage resources is allowed only within, not across data centers or custom data centers.
- Storage resources cannot be moved across non-datastore clusters.
- Compute-resource-only moves are permitted through shared storage.
- Virtual machines defined with affinity rules or anti-affinity rules are not to be moved.
- Virtual machines cannot be moved when residing on a local datastore, unless a storage swap exists on the local datastore.
- Virtual machines cannot be moved if they have data residing across multiple datastore clusters. Compute-only moves with similar shared storage are not permitted.
- A virtual machine cannot have data that resides across different storage types. For example, if a virtual machine has a VM disk on a datastore and a second VM disk on a datastore cluster, the virtual machine does not move, even when the datastore is shared with the destination or has swap on it.
- A virtual machine can use RDM so long as the destination datastore cluster can access the RDM LUN.
- A virtual machine can implement VM disks on multiple datastores inside a single datastore cluster.
- Workload Optimization may suggest moving virtual machines that are protected by vSphere Replication or Array Based Replication. You must ensure that all the clusters within a selected data center or custom data center have replication available. You can set up DRS affinity rules on virtual machines that you do not want moving across clusters.

Workload Placement Page

Workload Placement lets you optimize virtual machines and storage across clusters to reduce resource contention and maintain optimum system performance.

Where You Find Workload Placement

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Optimize**. Click on a data center and you see four tabs - **Reclaim**, **Rightsize**, **Workload Placement** and **Resource Effeciency**.

Workload Placement Page Options

In the Workload Optimization page, you see a list of data centers in a carousel, listed under three categories:

- Critical
- Normal
- Unknown

After you select a data center, you see the **ALL DATACENTERS** button on the upper right. Click **ALL DATACENTERS** when you want to switch the view to a filtered list of all data centers. Click **X** to return to a carousel view of data centers.

Note:

The workload placement automation operates by checking the optimization status of workloads every 5 minutes. When the system identifies a workload that is not optimized, it begins the optimization procedure. After this process, the next optimization status verification is scheduled for 6 hours later. If workload placement automation is disabled, manual optimization is accessible through the user interface at any time. While the workload placement engine attempts to resolve tag violations, it is restricted from migrating tag violating VMs to clusters that are incompatible or unable to support the current load.

Table 1751: Workload Placement Page Options

Option	Description
View:	Filter results to include data centers, custom data centers, vRA-managed custom data centers, or all three. (Option appears if you select **ALL DATACENTERS** on the upper right.)
Group By:	Filter results by criticality (most out of balance data centers/ custom data centers listed first) or by the vCenter to which each data center belongs. (Option appears if you select **ALL DATACENTERS** on the upper right.)
Sort By:	Options (Options appear if you select **ALL DATACENTERS** on the upper right): • Alarm clock graphic - list data centers/custom data centers by time remaining. • Dollar sign - list data centers/custom data centers by potential cost savings with capacity optimization. • Scales graphic - Optimized.
Select data center or **ADD NEW CUSTOM DATACENTER**	Options (Options appear if you select **ALL DATACENTERS** on the upper right): • Select a data center from the carousel across the top of the page. All data following refreshes with information for the selected object. • Select **ADD NEW CUSTOM DATACENTER** to display a screen that enables you to define a custom data center.

Data Center Options

After you select a data center from the carousel, you see the following information and options.

Note:
If you point your cursor to the lower right of a data center graphic, a tooltip may appear to let you know that the data center is using automated optimization.

Optimization Status Tab

Appears when you select a data center or custom data center from the top of the screen.

Table 1752: The Optimization Recommendation Card

Option	Description
Status	• Optimized - indicates that workloads are optimized based on the settings you entered in the neighboring Operational Intent window, with no tag violations based on the settings you entered in the Business Intent window. • Not Optimized - indicates that one of the following conditions is true: workloads are not optimized based on the settings you entered in the neighboring Operational Intent window AND/OR there are tag violations based on the settings you entered in the Business Intent window. In the event of tag violations, the offending tags are listed.
OPTIMIZE NOW	Runs optimizing actions based on the settings you entered in your Operational and Business Intent settings.
SCHEDULE	Displays a dialog box enabling you to schedule one or more optimization actions. If schedules are currently set for data center or custom data center optimization, a check mark appears next to the data center or custom data center name.
AUTOMATE	Continually seeks optimizing opportunities for data center or custom data center, based on the settings in the neighboring Operational Intent window or Business Intent windows. Scheduled optimizations are turned off while automatic optimization is on. Also, automated alerts are not operational when automatic optimization is on. Once you confirm automation, the system displays message, for example, 1) "Workload Optimization is looking for opportunities to automate," 2) "Your workloads are optimized according to your settings." or 3) "No eligible moves were found within the max number of compatibility checks allowed." **Note:** To initiate Automation, you must have privileges for Environment -> Action -> Schedule Optimize Container.
TURN OFF AUTOMATION	Stops automatic optimization. Any scheduled optimizations come back online.

Note:
Sometimes an optimizing action may be recommended, for example, to consolidate two hosts, but when you run the optimization, the generated placement plan does not show any potential consolidation. The seeming inconsistency results from the fact that recommended optimization actions are based on current conditions, whereas the placement plan logic includes forecasting. If forecasting predicts that consolidation can incur stress in the future, then consolidation is not recommended.

Note:
Scheduled jobs created from the Workload Placement page will be shown in the Automation Central page. However, these jobs cannot be edited, cloned or previewed, but can only be activated or deactivated.

Table 1753: The Operational Intent Card

Option	Description
Utilization Objective	Indicates the main attribute of your current automation policy settings. Values are moderate, consolidate, or balance.
Edit	Displays the Workload Automation Policy Settings, where you can adjust settings for optimization and cluster headroom. For more information, see the topic, Workload Automation Details.

Table 1754: The Business Intent Card

Option	Description
Intent	Allows you to define zones of infrastructure within cluster boundaries.
Edit	Displays a workspace where you can select criteria for placement of VMs.

Table 1755: Details for Are your clusters meeting your utilization objective?

Option	Description
Are your clusters meeting your utilization objective?	Displays a table which presents data in the following columns based on the Demand model or Allocation model: • Name • CPU Workload • Memory Workload • Max Storage Workload • DRS Settings • Migration Threshold • Violated Tags • VM Name Max Storage Workload is the disk workload % on the most stressed vSAN datastore or storage pod. This is not a cumulative value across all pods or datastores. Enable storage based eviction details in the policy. For more information, see Policy Allocation Model Element To activate the Allocation model, set the overcommit ratios in the active policy. For more information, see the topic, Allocation and Demand Model in Workload Optimization. Migration thresholds are based on DRS priority levels, and are computed based on the workload imbalance metric for the cluster. The violated tags shows which clusters or host groups are breaching the business intent. The VM Name column shows the name of the VMs and tag value due to which tag violation is happening. Provides the option to set the DRS automation level for individual objects.
VIEW DRS SUMMARY	Select a cluster in the list, then click this link to display a page containing metrics for DRS performance and cluster balance in the selected data center.
SET DRS AUTOMATION	Select a cluster in the list, then click this link to set the level of the DRS automation for the cluster. Note that clusters must be fully automated in order for workload optimization alerts to run actions set in the policies.

History Tab

Displays a graphical depiction of executed manual and automated optimizations for clusters in the selected data center or custom data center, based on parameters you provide.

Table 1756: Details for History

Option	Description
Selected WLP process drop-down	The optimization action whose details you want to display.
Time duration drop-down	Last *n* hours - select the time parameter: last 6, 12, 24 hours or last 7 days.
Quick filter	choose a cluster name to search on.
Squares graphic	toggle between viewing processes in icon or circle form.
Circle	Toggle between viewing processes presented in a circle or on a straight line.
Back arrow - reset action.	Reset action.

If you point your cursor to a specific cluster as displayed on the screen, the details of the cluster appear in a tool tip. Click the note card icon on the lower right of the tool tip to go to the Details screen for the cluster. When displayed in the circle format, rings in the circle indicate how much CPU and how much memory was used at any given time. For example, if memory usage was higher than recommended based on your policy settings, the memory circle appears red.

Note the timeline across the bottom of the screen. When you choose parameters, for example, WLP process name, time parameter and cluster name, indicators appear along the timeline, showing when processes were initiated.

To zero in on a specific event, choose a process from the drop-down menu. You can also click points on the marker floating above the timeline, which causes a descriptive tool tip to appear, then double-click the 'Double-click to zoom' icon on the lower right.

If the event you choose includes an actual movement of VMs, you see a blue ball containing the number of VMs moved and showing the direction of the move and starting and ending clusters.

Optimization Potential Tab

When you run Workload Optimization, VCF Operations runs compatibility checks and excludes those VMs which have constraints, and only optimizes resources of those VMs which can be moved. If you want to see the total potential of your workload optimization, assuming that all VMs can be moved, click the **CALCULATE OPTIMIZATION POTENTIAL** button in the Optimization Potential tab. Optimization Potential disregards the underlying constraints and recommends moves before the compatibility checks. You can download the report to see more details. Storage DRS must be set to fully automated.

If you want to see what can be realistically optimized, click **OPTIMIZE NOW** in the **Operation Status** tab. After you click **OPTIMIZE NOW**, you can download a report to review incompatibilities.

The optimization potential report helps you understand the difference between the optimization achievable when you run **OPTIMIZE NOW** and the total optimization potential.

See also Example: Run Workload Optimization

Configuring Workload Optimization Alerts

VCF Operations provides two preconfigured alerts designed to work with the Workload Optimization feature. You must take additional action in the Policies area to turn on the alerts and automate them so that predetermined actions are run when the alerts fire.

Ensure that you have all required permissions to access the Workload Optimization UI pages and manage vCenter objects.

The following preconfigured alerts are designed to work with the Workload Optimization feature:

- Data center performance can potentially be optimized in one or more clusters.
- Custom data center performance can potentially be optimized in one or more clusters.

The preconfigured alerts fire only if the AUTOMATE function is not turned on at the Workload Optimization screen which can be accessed by clicking **Workload Placement** under **Capacity** in the left menu.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**. Click the **Policy Defintion** tile.
2. Select the policy that includes settings for the relevant data centers and custom data centers, for example, **vSphere Solution's Default Policy**.
3. Click the **Vertical Ellipses** next to **Add** and then **Edit**.
4. Click the **Alerts and Symptoms** tile.
5. Search on "can potentially be optimized" to locate the two alerts you want.
6. The alerts are activated by default/inheritance (see the **State** column).
7. The alerts are not automated by default/inheritance (see the **Automate** column). To automate the alerts, click the menu symbol to the right of the inherited value and select the green check mark.

Workload Optimization is fully automated for your environment.

To confirm that actions are taken automatically, monitor rebalance activity at the Workload Optimization screen.

Custom Datacenters in VCF Operations

A custom data center is a user-defined container for a group of objects that includes clusters, hosts, and virtual machines. Custom data centers provide capacity analytics and capacity badge computations based on the objects it contains. You can use custom data centers to forecast and analyze the capacity needs for your environment.

When you create a custom data center, you can include multiple cluster objects that span multiple vCenter instances. For example, you might have a production environment that spans multiple clusters, and you must monitor and manage the performance and capacity of the entire production environment.

After you create your custom datacenter, you can select it in the list of custom data centers to display a summary of its health, risk, and efficiency. To access the list of custom data centers, click the **Global Inventory** in the left menu.

This view displays the top alerts for the data center. To examine the capacity remaining for the custom data center, click the **Capacity** tab under**Details** after you select an object.

How Custom Datacenters Work

In vSphere, a data center serves as a container for objects that a vCenter instance manages. A custom data center is a container that can include objects from multiple vCenter instances.

Custom data centers can contain vCenter instances, data centers, clusters, hosts, virtual machines, and datastores. You can add vSphere object types to a custom data center.

When you add an object, the hierarchical children of that object become part of the custom data center. An object can belong to multiple custom data centers.

When you create custom data centers, the system runs capacity analytics on the objects in the custom data center, even if those objects span multiple vCenter instances. For example, you might need to examine the capacity analytics data across multiple clusters, and the multiple vCenter instances that manage those clusters. You do not have to analyze the capacity of one cluster or one vCenter instance at a time. You can create a custom data center, add all the clusters to it, and see the capacity analysis in a single location.

Custom Datacenters List

You can view the list of custom data centers that exist in your environment, and a summary view of its health, risk, and efficiency. In this view, you can click a custom data center to display the top alerts that the objects in the custom data center triggered.

Where You Find Custom Datacenters

After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory**. Select a vCenter and then select the **Custom Group and Datacenters** tab.

Table 1757: Custom Datacenters Toolbar and Grid Options

Option	Description
Toolbar options	In the toolbar click **Add** to add a new custom data center. Click the **Vertical Ellipses** against a custom data center name to perform the following actions: • **Edit**. Modify the custom data center. • **Delete** . Remove the custom data center. • **Clone** . Clone the custom data center. • **Go to Details**. Opens the custom data center page in the Object browser and displays the custom data center summary page
Filter	Limit the list of custom data centers to those data centers that match the text that you enter in the **Filter** text box.
Data grid	Lists the custom data centers in your environment. Click on the name of a custom data center to view the details. To view a summary of the custom data center health, risk, and efficiency, click the **Vertical Ellipses** and then **Go to Details**. To edit, delete, or clone a custom data center, **Vertical Ellipses**, and then the corresponding option.

Custom Datacenters Add and Edit Workspace

A custom data center is an object type that provides capacity analytics and capacity badge computations based on the objects it contains. You create a custom data center object and add inventory objects to it.

Where You Create or Edit a Custom Datacenter

After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**. Click the **Custom Datacenters** tile, and then click the **ADD** button.

To edit a selected custom data center, click the **Vertical Ellipses** to edit, remove, or clone.

Table 1758: Add and Edit Custom Datacenters Configuration Options

Step	Option	Description
1 - Custom Data Center	Name	Descriptive name of the custom data center.
	Description	Meaningful description for the custom data center. Provide specific information that other users must know about this custom data center. Click the **NEXT** button when you are done.
2 - Objects	Objects	This screen in this step is divided into two sections. On the right section, you can select an object from the tree view of your inventory. Drag an object to the left pane to add it to the custom data center. You can search for objects on the right-hand side of the screen after selecting an object type. Once you have found the object, drag the object from the right section of the screen to the left section of the screen under **vSphere Hosts and Clusters**. You can add vCenter instances, vSphere data centers, vSphere clusters, and ESX hosts. When you add an object, the hierarchical children of that object become part of the custom data center. An object can belong to multiple custom data centers. To remove an object when editing, click the object in the vSphere Hosts and Clusters panel and click **REMOVE**. Click the **CREATE** when you are done.

Workload Automation Policy Settings

Provides options for refining policy settings specifically for Workload Optimization.

Where You Find Workload Automation Settings

Access this screen through the Policies pages:

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**. Click the **Policy Defintion** tile.
2. Select a policy that you want to modify. Ideally, this should be an active policy. Or, click the **ADD** button to add a new policy.
3. Select the **Workload Automation** card to review the changes, or click **EDIT POLICY** to make changes.

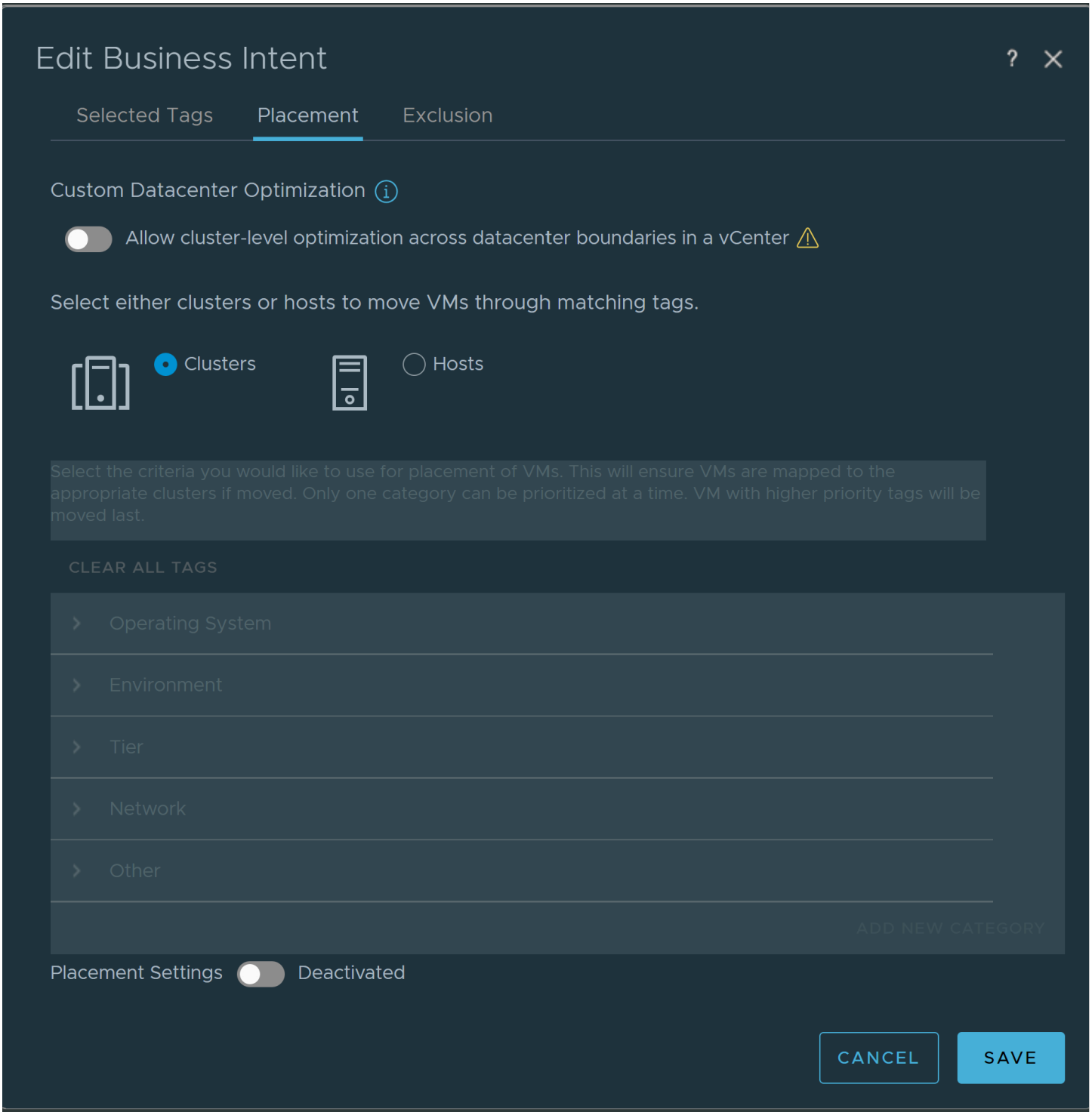
Refer to Workload Automation Details.

Business Intent Workspace

You can use vCenter tagging to tag VMs, hosts, and/or clusters with specific tags. VCF Operations can be configured to leverage tags to define business-related placement constraints. Using tags, you can place VMs on hosts/clusters with matching tags, or exclude VMs and clusters from participating in Workload Placement.

Where You Find Business Intent

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Optimize**. Click on a data center and you will see four tabs - **Reclaim**, **Rightsize**, **Workload Placement,** and **Resource Effeciency**. On the **Workload Placement** tab, select a data center or custom data center from the top row, and click **Edit** in the Business Intent window.



How the Edit Business Intent UI Works

The Edit Business Intent dialog box has three tabs. The **Placement** tab is where you select the tags which VCF Operations uses for Workload Placement. The **Exclusion** tab is where you specify tags to exclude VMs and Clusters from participating in the Workload Placement. For example, you may have VMs running critical workloads in a large environment. These VMs may be running on specific hosts so that they always get the resources they need for running the workloads. You can exclude such VMs from Workload Placement if you do not want the VMs from moving from their original location. The clusters which are excluded using the exclusion tags are not displayed in the Workload Placement page, and are not considered in the Workload Placement calculation.

Note:
Tag exclusion was introduced newly in version 8.16.

The tags that you select in the **Placement** and **Exclusion** tabs are displayed in the **Selected Tags** tab.

You can activate cluster level optimization across data center boundaries in a vCenter if you have created a custom data center across multiple data centers within a single vCenter. You must ensure that the prerequisite networking requirements for movement of VMs are met before running workload optimization on such a custom data center.

To edit Business Intent values, you must have the necessary permissions. When you are logged in with administrative privileges, click Administration on the left menu and go to **Roles** under **Access Control** . Select the name of the role to which you want to provide permissions and then click **Edit** in the permissions section. Select the **Read** and **Write** checkbox under **Administration** > **Configuration** > **WLP Settings**.

Establishing Business Intent

Tags are implemented in vCenter as *key:value* labels that allows operators to add meta-data to vCenter objects. In vCenter terminology, the *key* is the tag category and the *value* is the tag name. Using this construct, the tag OS: Linux can indicate a cluster or VM that is assigned to the category OS with a tag name of Linux. For complete information on vCenter tagging capabilities, refer to the vCenter and Host Management guide.

To specify tags that are either considered for placement, or excluded from placement, first go to the appropriate tab in the Edit Business Intent dialog box.

Select the radio button for the type of object you want to associate with VMs in this business intent session: Clusters or Hosts. Then, click the **Placement Settings** or **VM Exclusion Settings** toggle button.

The system provides several suggested categories. These categories are only suggestions. You must specify the actual categories in vCenter after you expand the section for a suggested category. For example, in section "Tier", you can specify the actual vCenter tag category that represents tier semantics, for instance, "service level".

- Operating System
- Environment
- Tier
- Network
- Other

Any actual categories that you specify must first be created in vCenter.

Then you can associate tagged VMs with clusters or hosts, based on the rules for each type of tagging.

1. Click the chevron to the left of the first suggested category. A **tag category** field appears.
2. Click the drop-down menu indicator and choose a category from the list defined in vCenter.
3. Click the drop-down menu indicator in the Tag Name (Optional) field and choose a tag name from the list defined in vCenter
4. Click **Include Tag**. All VMs with that tag are associated with the category.

Rules for Host-Based Placement

To set host level placement constraints, VCF Operations automatically creates and manages DRS rules. All conflicting user-created DRS rules are deactivated.

These rules include the following:

- Any VM-VM affinity and anti-affinity rules.
- Any VM-Host affinity and anti-affinity rules.

You must check the selection box next to the statement, "I understand that VCF Operations will deactivate all my current and future DRS rules".

View DRS Summary

The View DRS Summary page provides insight and perspective into the actions DRS is taking to balance a cluster. You can view DRS settings for the cluster and cluster balance metrics, and determine if recent vMotions are DRS- or user-initiated.

Where You Find the View DRS Summary Page

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Optimize**. Click on a data center and you will see four tabs - **Reclaim**, **Rightsize**, **Workload Placement,** and **Resource Effeciency**. On the **Workload Placement** tab, select a cluster name in the "Are your clusters meeting your utilization objective?" section. The dimmed View DRS Summary and Set DRS Automation links turn live. Click the link to display the DRS summary information.

Table 1759: DRS Summary Values

Pane/fields		Value
<*cluster name*>		Name of the selected cluster
	Automation Level	Enabled/Disabled. DRS is running or not.
	Migration Threshold	Aggressive/Default/Moderate
	Active Memory Used	False/ *nn*%
Cluster Balance		Shows the variations in the DRS cluster balance metric over time as DRS runs. The graph shows how DRS reacts to and clears any cluster imbalance each time it runs.
	Cluster Imbalance	The range of potential imbalance values, as expressed in vCenter DRS metrics.
	Total Imbalance	The level of imbalance in a cluster, as measured by vCenter DRS metrics.
	Tolerable Threshold	The upper limit of what is tolerable in cluster imbalance. Designated by a green dotted line, this is a vCenter DRS metric.
VM Happiness		A bar graph summarizing the total happy and unhappy VMs in the cluster. For individual VMs, there is a presentation of performance metrics related to its happiness, such as %CPU ready time and memory swapped.
	Happy VMs	Total of happy VMs are shown in green. Click in the green zone to show a list of these VMs in the Happy/Unhappy VMs pane to the right.
	Unhappy VMs	Total of unhappy VMs is shown in red. To show a list of these VMs in the Happy/Unhappy VMs pane to the right, click in the red zone .
Happy/Unhappy VMs		Lists by name all the VMs in the zone you clicked in the VM Happiness pane.
VM Metrics		Shows the trend in VM happiness or unhappiness
Recent vMotions		The number of recent vMotions, plotted against time.
vMotion Details		Shows the number of DRS-initiated and user (non-DRS) initiated vMotions over time. You can choose which type you want to view.
	Date/VM	Date of a given vMotion.
	Source/Destination	Source and destination of moved VMs.
	Type	DRS-initiated or user initiated.

Optimization Schedules

Use the Optimization Schedules page to edit or delete optimization schedules that you set up in the Manage Optimization Schedule Dialog Box at the Workload Optimization main screen.

Where You Find Optimization Schedules

- After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Optimize**. Click on a data center and you will see three tabs - **Reclaim**, **Rightsize**, and **Workload Placement. Optimization Schedules** is available in the **Workload Placement**.
- At the > page, select in the data center whose optimization schedule you want to edit or delete. Then click **SCHEDULE** in the Optimization Recommendation pane.

Table 1760: Optimize Schedules Options

Option	Description
Edit icon	Select a schedule from the list, then click the **Edit** icon. The Manage Optimization Schedules dialog box appears, with the data for the selected schedule filled in.
Delete icon	Select a schedule from the list, then click the **Delete** icon. The selected schedule is deleted and does not run.

See also Example: Run Workload Optimization

Manage Optimization Schedules

Allows you to set up a regular schedule for optimizing a selected container.

Where You Find Manage Optimization Schedules

At the Workload Optimization screen, select **SCHEDULE** from the pane: Optimization Recommendation

Option	Description
Schedule Name	Meaningful name for the schedule
Time Zone	Choose the time zone for the action
Recurrence	Indicate how often you want the optimize action to run. Complex schedules can be defined, for example, select the Monthly option and choose to run the action on Tuesdays and every other Thursday, beginning on the fifth of the month.
Start on:	Day to start the optimization schedule.
Start at:	Time to start the optimization schedule.
Expire after:	Designate a set number of scheduled runs.
Expire on:	Designate an exact date for the actions to end.

See also Example: Schedule a Repeating Optimization Action

Optimize Placement

A two-page dialog box that provides information about optimizing the workload of a selected container. When you run the optimization action, VCF Operations checks which of the VMs can be moved to a different cluster for better optimization of resources, based on the settings you entered in your Operational and Business Intent settings. You can download a report that provides information about the list of VMs that were included in, and excluded from, the move plan. The report provides reasons as to why some VMs were excluded from the plan.

First page: The current workload ("before," for example, CPU 105%) and projected results ("after," for example, storage utilization 45%) for a possible optimizing action.

Second page: The exact moves planned for compute and storage resources.

Note:

It is possible that there is no optimization move plan. Review the report to see why VCF Operations could not provide a move plan.

Where You Find Optimize Placement

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Optimize**. Click on a data center and you see four tabs - **Reclaim**, **Rightsize**, **Workload Placement,** and **Resource Effeciency**. Go to the **Workload Placement** tab. Click **OPTIMIZE NOW** in the **Optimization Recommendation** pane.

Table 1761: Optimize Clusters Options

Option	Description
Compare Cluster Balance	If you are satisfied with the before and after numbers (First page, above), click NEXT.
Review Optimization Moves	If you are satisfied with the moves planned (Second page, above), click BEGIN ACTION. **Note:** Review the optimization plan report before you click BEGIN ACTION.
Download Report	The optimization plan report is in CSV format, and provides the following information: • Summary of the optimization plan. • Summary of the moves that make up the optimization plan. • Issues related to the data center. Resolve these issues before proceeding with the optimization. • Issues and incompatibilities applicable to specific VMs and their configurations. Resolve these issues, if applicable. • Failed move attempts applicable to the specific VMs and their target destinations, as determined from the VM move plan. Resolve these issues and incompatibilities.

See also Example: Run Workload Optimization.

Using Workload Optimization

Use the Workload Optimization UI pages to monitor optimizing moves in a fully automated system. If your system is not fully automated, you can use the UI to conduct research and run actions directly.

VCF Operations monitors virtual objects and collects and analyzes related data that is presented to you in graphical form at the Workload Optimization screen. Depending on what appears on the screen, you might use optimization functions to distribute a workload differently in a data center or custom data center. Or you may decide to perform more research, including checking the Alerts page to determine if any alerts have been generated for objects of interest.

For comprehensive general instructions on responding to alerts and analyzing problems related to objects in your environment, see Configuring Alerts and Using Actions.

The following examples demonstrate the primary ways you can use Workload Optimization to keep your data centers balanced and performing their best.

Example: Run Workload Optimization

As a virtual infrastructure administrator or other IT professional, you use Workload Optimization functions to identify points of resource contention or imbalance. In this example, you manually run an optimization action to consolidate demand.

Ensure that you have all required permissions to access the Workload Optimization UI and manage vCenter objects.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Optimize**. Click on a data center and you see four tabs - **Reclaim**, **Rightsize**, **Workload Placement,** and **Resource Effeciency**. Go to the **Workload Placement** tab.

   The Workload Optimization page appears. Data centers are grouped by Criticality, with the three troubled data centers appearing in a carousel across the top of the page: DC-Bangalore-18, DC-Bangalore-19, DC-Bangalore-20. A Not Optimized badge appears in the lower right corner of each graphic.

2. If no data center is preselected, select DC-Bangalore-18 from the carousel.

   Comprehensive data about the state of the data center follows.

3. Based on the available data, you determine an optimization action is required.

   CPU workloads can be consolidated such that a host in Cluster 3 can be freed up.

Table 1762: Panes and Widgets

Pane	Contents
Workload Optimization	Status shows as Not Optimized. A system message says, "You can consolidate workloads to maximize usage and potentially free up 1 host." The message reflects that you have set policies to emphasize consolidation as a goal in optimization moves. The system is saying you can free up a host through consolidation.
Settings	The current policy is Consolidate. The system advises: Avoid Performance Issues, Consolidate Workloads.
Cluster Workloads	Cluster 1 CPU Workload is 16%. Cluster 2 CPU Workload is 29%. Cluster 3 CPU Workload is 14%. Cluster 4 CPU Workload is 22%.

4. Click **OPTIMIZE NOW** in the Workload Optimization pane.

   The system creates an optimization plan, which depicts BEFORE and (projected) AFTER workload statistics for the optimization action.

5. If you are satisfied with the projected results of the optimization action, click **NEXT**.

   The dialog box updates to show the planned moves.

6. If you need more information about the VMs which are included or excluded in the plan, click **Download Report** to see the optimization plan. You can review the reasons for incompatibilities and why some VMs were excluded from the plan.

7. Optional: If you want to know the total optimization potential of the move, assuming that there were no incompatibilities and all your VMs can be included in the optimization plan, click **Cancel**, and go to the Optimization Potential tab in the Workload Optimization page. Click **Calculate Optimization Potential** to see the total optimization potential of your data center.

8. Review the optimization moves, then click **BEGIN ACTION**.

   The system runs the compute and storage resource moves.

The optimization action moved compute and storage resources from some clusters to other clusters in the data center, and so freed up a host on one cluster.

> **Note:**
> The Workload Optimization page refreshes every five minutes. Depending on when you run an optimization action, the system might not reflect the result for up to five minutes, or longer when longer-running actions extend the processing time.

To confirm that your optimization action was completed, go to the Recent Tasks page by clicking **Administration** in the left pane. In the Recent Tasks page, use the Status function on the menu bar to locate your action by its status. You can also search using a range of filters. For example, first filter on Starting Time and scroll to the time when you began the action, then select the Object Name filter. Finally, enter the name of one of the VMs in the rebalance plan.

Note:
Sometimes an optimizing action may be suggested, for example, to consolidate two hosts, but when you run the optimization, the generated placement plan does not show any potential consolidation. The seeming inconsistency results from the fact that suggested optimization actions are based on current conditions, whereas the placement plan logic includes forecasting. If forecasting predicts that consolidation might incur stress in the future, then consolidation is not suggested.

Example: Schedule a Repeating Optimization Action
As a virtual infrastructure administrator or other IT professional, you determine that compute and storage resources in a given data center are volatile and a regularly scheduled optimization action can address the problem.

Ensure that you have all required permissions to access the Workload Optimization UI and manage vCenter objects.

VCF Operations monitors virtual objects and collects and analyzes related data that is presented to you in graphical form at the Workload Optimization page. Depending on what appears, you may determine that you must schedule optimization functions to distribute a workload more evenly in a data center or custom data center.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Optimize**. Click on a data center and you see four tabs - **Reclaim**, **Rightsize**, **Workload Placement,** and **Resource Effeciency**. Go to the **Workload Placement** tab.
2. From the carousel of data centers across the top of the page, select a data center for which you want to schedule repeated optimization actions.
3. In the Workload Optimization pane, click **SCHEDULE**.
4. Give the schedule a name and choose a time zone.
5. Determine how often you want to repeat the optimization action and click the relevant **radio button** under Recurrence.
   Depending on your selection under Recurrence, additional options appear to the right. In this instance, you choose to repeat the optimization daily.
6. Leave the current date and time.
7. Select the **Repeat every day** radio button.
8. Select the **Expire after** radio button and tick the counter up to 6.
9. Click **Save**.

The optimization action repeats for six days, then stops.

At the Workload Optimization page, the Scheduled button appears in the upper right of the Workload Optimization pane if optimization actions are scheduled for the selected data center. If you want to edit or delete a schedule, click the **Scheduled** button. The Optimization Schedules page appears, where you can perform those actions.

Note:
If you schedule a number of optimization actions close together, and the optimization plans of two or more actions include overlapping functions, that is, they impact the same set of resources, the system shifts the actions into a queue. As a result, some actions may complete later than expected, with longer running actions and other potential system constraints extending the lag time. Optimization actions that do not overlap can run concurrently.

To confirm that your optimization action was finished, go to the Recent Tasks screen. In the Recent Tasks screen, use the Status function on the menu bar to locate your action by its status. You can also search using a range of filters. For example, filter on Event Source and enter the name of the scheduled optimization plan.

Note:
Because real-time data center resource contention is dynamic, the system calculates a new optimization plan each time the scheduled optimization action starts, but before it runs. The system does not run the action if the system determines that the data center container is balanced at this moment. On the Recent Tasks page, the name of the affected data center appears in the Object Name column, and the Message “The optimization of the selected container cannot be improved” appears under Details. Another possibility is that a scheduled optimization plan is attempted, but does not go forward. In this event - which is not the same as a "failed" action - the name of the affected data center also appears in the Object Name column.

Using Memory Tiering for Resource Efficiency

Memory Tiering significantly reduces hardware costs and increases virtual machine (VM) density by allowing clusters to utilize NVMe (Non-Volatile Memory Express) SSDs as a secondary memory tier. By offloading inactive memory pages from expensive DRAM to more cost-effective NVMe devices, you can support more workloads on the same infrastructure without compromising performance.

Using memory tiering, you can see what happens if you add NVMe on clusters, without requiring you to deploy it. It presents the potential cost savings, the possible increase in VM count, the projected memory state with a larger overall footprint, and the expected improvements in CPU utilization. These projections are derived from the current condition of the cluster, including existing CPU and memory usage patterns. The estimated cost savings are based on assumed DRAM and NVMe pricing from common server OEMs. Clusters with active memory usage below 50% and free CPU capacity above 50% are highlighted as strong candidates for memory tiering.

Memory tiering allows you to:

- Simulate Savings: Analyze potential cost reductions and capacity gains before purchasing hardware.
- Increase Efficiency: Support approximately additional VMs by freeing up DRAM.
- Monitor Tiering Health: Track active (DRAM) vs. inactive (NVMe) memory usage and cost savings post-activation.

> **Note:**
>
> Do not enable memory tiering for clusters running latency sensitive workloads or those using large active memory such as telco workloads, in-memory java applications, or high frequency trading.

Working with Memory Tiering

Follow these steps to check the eligibility for memory tiering and activating it.

- Check Eligibility: Click **Operate** from the navigation bar on top, and in the left navigation pane, click **Inventory** and then select the Hosts and Clusters tab and click on a cluster. A cluster is eligible for memory tiering if the active memory is less than 50%. Check the nudge on the cluster summary page and go to the resource efficiency page. Alternatively, navigate to **Capacity** > **Optimize** and click a data center and then the **Resource Effeciency** tab.
- Review Potential Benefits: On the resource efficiency tab, you can see:
  - An introduction to what memory tiering is and why it is needed.
  - The total potential savings across all clusters in the selected data center.
  - The number of additional VMs that can be added by enabling memory tiering, taking the current state into account.
- View Simulation Results (for the selected cluster): For the selected cluster, the resource efficiency tab displays simulation results showing how much you could save if you deployed VMs with the memory tiering feature. The potential savings are calculated as the difference between the cost of adding DRAM and the cost of adding NVMe to double the existing VM count.
- Review Activation Guidance: The resource efficiency tab also shows step-by-step guidance on how to activate the NVMe and enable the memory tiering feature.

Use the comparison data to make data-driven decisions about enabling memory tiering.

You can:

- Evaluate which clusters will benefit most from memory tiering.
- Identify the estimated VM capacity gain.
- Review cost-saving projections based on the current and optimized states.

Where you find Memory Tiering

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Optimize**. Click on an eligible data center and you see four tabs - **Reclaim**, **Rightsize**, **Workload Placement**, and **Resource Effeciency**.

The **Memory Tiering** interface under the **Resource Effeciency** tab allows you to analyze current cluster utilization and simulate the benefits of adding NVMe storage. This page provides an introduction to memory tiering, why it is needed, potential savings across each cluster in the selected data center, and how many additional VMs can be added.

Cluster Selection Pane	Located on the left, this pane lists clusters identified as candidates for memory tiering optimization. Cluster List: Displays available clusters along with their current Active Memory and CPU Utilization percentages. Clusters which have aggregated Active Memory less than 50% or Memory Tiering On are shown here. Clusters are sorted top to bottom based on most affected cluster at the top. Select a specific cluster to populate the utilization data in the main view.
Resource Utilization	The right-hand pane details the current state of the cluster resources. The Cluster Utilization pane visualizes the **Tiering: OFF** and **Tiering: ON** states, highlighting resource bottlenecks caused by inactive memory pages residing in DRAM.
Existing State	Located at the bottom of the dashboard, this displays the current density of Virtual Machines supported by the cluster in its current hardware configuration. It establishes the baseline for comparison and the potential cost savings when you add memory tiering. When you turn on memory tiering, the potential cost savings is displayed.

Memory Tiering Dashboard

More metrics for NVMe usage and performance are collected, and a dashboard is available for monitoring these metrics. The dashboard shows the number of vCenters, clusters, and hosts with memory tiering enabled, as well as the number of VMs across all memory-tiered hosts. It also displays the Tier 0 (DRAM) and Tier 1 (NVMe) total capacity and usage. You can drill down to see details for specific hosts and VMs. For more information on dashboards, see the topic, Configuring Dashboards and Widgets.

Understanding the Comparison View

In the **Memory Tiering** interface under the **Resource Effeciency** tab, left-hand panel displays a list of clusters prioritized by their suitability for memory tiering.

- Sorting Logic: Clusters are sorted by relevance. Ideal candidates typically have active memory usage below 50% and CPU utilization below 50%.
- Selection: Click on a cluster to view its specific simulation data.

The comparison view on the right-hand side of the **Cluster Utilization** pane provides a visual comparison.

Without Memory Tiering	• Shows how much additional DRAM would be required to support new workloads. • Displays the estimated cost for expanding memory using DRAM only.
With Memory Tiering	• Illustrates how memory can be expanded using a mix of DRAM (Tier-0) and NVMe (Tier-1) memory. • Displays cost and capacity benefits when NVMe is used for inactive memory. • Provides a net potential savings estimate based on assumptions. After reviewing the **Net Potential Savings**, you can click the **HOW TO ACTIVATE** link for instructions on how to enable memory tiering on the cluster. When consumed memory reaches almost 80%, the system starts moving inactive memory to NVMe.

How Potential Savings is Calculated

The potential savings displayed in the simulation results show how much you could save by enabling the memory tiering feature if you need to double your existing VM count but were low on memory.

The calculation for potential savings is based on a one-to-one mapping between DRAM and NVMe. The simulation result are derived by comparing the cost of adding DRAM versus the cost of adding NVMe to support doubling the VM count.

Key assumptions for the calculation:

- **DRAM Price:** Assumed to be $10 per 1 GB in the market
- **NVMe Price:** Assumed to be almost $1 per 1 GB

Net Savings is calculated by taking the [Average Capacity per host in GB x $10 - [Minimum Discrete NVMe Disk Capacity in GB x $1]]

Example of the calculation:

- If the current memory used on the cluster were 4600GB, and an additional 4600GB needed to be added:

  - Adding DRAM would cost almost $46,000
  - Adding an NVMe would cost only $7,200

  In this example, there were 9 hosts and we were adding 800GB (discrete NVMe size) on each host, even though 4600GB were required for a 1:1 ratio of DRAM:NVMe.
- The difference between these two numbers ($46,000 and $7,200) is the potential saving. In the example given, this difference is almost $38,000

The calculations for the possible VM count and the NVMe needed are also based on the current usage of CPU, active memory, and inactive memory.

The **How is the NVMe storage distributed among all hosts?** link shows the calculated NVMe capacity and the cost incurred for purchasing the NVMe. The calculation is performed by:

- Taking the average capacity of all hosts in the cluster.
- Selecting the next higher number of available NVMe disk in the market and calculating the cost.

Since the cost saving calculation is based on a one-to-one mapping between DRAM and NVMe, the usable capacity of NVMes is lower than the total available NVMe capacity.

Activating Memory Tiering

Once you have validated the savings via simulation, follow the three-step activation workflow. Click the **HOW TO ACTIVATE** link in the to view the specific requirements.

Step 1: Prepare Hosts	• Hardware Requirement: Purchase and physically install compatible NVMe devices on all hosts within the selected cluster. **Note:** This requires physical access to the data center.
Step 2: Enable Memory Tiering	• Configuration: You can enable tiering at the cluster, host, or VM level. • Methods: Use the vSphere Client, ESXCLI commands, or PowerCLI scripts. • Downtime: Enabling this feature typically requires a host power cycle or maintenance mode.
Step 3: Monitor Performance	• Post-Activation: After enabling, use the Aria Operations dashboards to monitor system performance and memory usage statistics.

Monitoring Memory Tiering Performance

After enabling memory tiering, you can monitor ongoing performance metrics to ensure expected results dashboard updates to show real-time utilization of the memory tiers.

Detailed Memory Breakup

Click on **Detailed Memory Breakup** to view a granular distribution of memory usage.

- Consolidated Memory**:** Displays total available memory (DRAM + NVMe).
- Tier Breakdown:
  - Tier-0 (DRAM): Handles Active Memory. This is the fastest tier used for frequently accessed data.
  - Tier-1 (NVMe): Handles Inactive Memory. Infrequently used pages are moved here to free up expensive DRAM.
- Efficiency Metrics: View exactly how much DRAM has been freed and the current consumption of the NVMe tier.

Monitoring Locations	• Cluster resource pages display real-time memory tiering utilization and performance. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory**. You can see information about memory in the **Summary** tab. • The Launchpad in the VCF Operations home page and predefined dashboards provide high-level summaries and drill-down views. Use these dashboards to continuously evaluate the effectiveness of memory tiering and identify further optimization opportunities.
Available Metrics	The following metrics provide memory tiering information: • Tier-0 and Tier-1 memory usage

- Active vs. inactive memory consumption
- VM performance impact and efficiency trends

How to Plan for Capacity Changes

Planning for capacity changes in VCF Operations involves simulating hypothetical changes to the environment, such as adding new VMs or hosts, to understand their potential impact on resource utilization. This helps in proactive planning.

What-If Analysis: Modeling Workload, Capacity, or Migration Planning

Using the What-If tool, you can plan for an increase or decrease in workload or capacity requirements in your virtual infrastructure. To evaluate the demand and supply for capacity on your resources, and to assess the potential risk to your current capacity, you can create scenarios for adding and removing workloads. You can also determine how much capacity you require to make a migration work. You can run one scenario or group scenarios and run them cumulatively.

Why Create a Scenario

A scenario is a detailed estimation of the resources you must have available in your environment to incorporate upcoming changes. You define scenarios that can potentially add resources to actual data centers. VCF Operations models the scenario and calculates whether your desired workload can fit in the targeted data center. You can save multiple scenarios for comparison or review.

Planning for New Infrastructure

Capacity planning tools now support the simulation of new infrastructure setups, even before these assets are deployed. This helps you plan for additional computational power. You can define a new cluster and VCF Operations will calculate if your workload will fit the newly configured cluster.

Committed Scenarios

When you are sure that you need to reserve capacity, you can commit the scenario to have set VCF Operations aside resources for new, upcoming, or planned workloads. A committed scenario is a supposition about how the capacity and load change on your objects when you change the conditions in your virtual infrastructure environment. You do not have to implement the changes that your committed scenario represents. By committing a scenario, you can determine your capacity requirements before you implement the actual changes.

Why Create a Committed Scenario	In organizations which have separate capacity management and operations teams, committing a scenario helps stakeholders understand the current capacity and upcoming capacity requirements across the board. With committed scenarios, capacity is reserved and this prevents the operations team from performing adhoc resource increase on workloads, while the capacity manager is engaged in resource planning of new projects. Committed Scenario also helps the team responsible for infrastructure expansion, as it provides actionable insights into future scenarios. In the event capacity becomes limited, it could be accounted for in the expansion.

Where You Find What-If Analysis and Committed Scenarios

What-If Analysis	After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. In the **What-If Analysis** tab, click **NEW**. You see a list of seven What-If Analysis tiles.
Committed Scenarios	After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. Then go to the **Committed Scenarios** tab. In the Committed Scenarios page, click the **ADD** button to directly create a committed scenario without first creating a What-If scenario.

The overview tab of the What-If Analysis page has seven tiles. Each tile lets you run What-If scenarios to optimize capacity based on the following areas:

- Workload Planning: Traditional
- Workload Planning: Hyperconverged
- Infrastructure Planning: Traditional
- Infrastructure Planning: Hyperconverged
- Migration Planning: VMware Cloud
- Migration Planning: Public Cloud
- Datacenter Comparison: Private Cloud

How What-If Analysis and Committed Scenarios Work

You can run What-If scenarios to see how much capacity will remain after you add or remove VMs or hosts and add hyperconverged infrastructure (HCI) nodes. Migration planning shows you the capacity and cost information after migrating to cloud based infrastructure.

Scenarios that you save for later are displayed as a list when you browse to the **What-if Analysis** and **Committed Scenarios** pages. You can run, edit or delete the saved scenarios from the ellipsis next to the scenario name. When you run a scenario, whose start date occurs in the past, you get a dialog box asking you if you would like to run the scenario with the current date. If you choose **No**, the scenario is not run. If you choose **Yes**, VCF Operations runs the scenario with the current date as the start date. The end date is not affected.

Use the advanced filter and search box to look for saved scenarios by scenario name, scenario type, datacenter, and cluster. You can select more than one compatible scenarios and run them together. For example, you can create a scenario to remove hosts using the **Physical Infrastructure Planning** tile, because your organization has hardware that will soon become obsolete. You can create another scenario to add hosts to your physical infrastructure to account for new hardware that will replace the obsolete ones. You can run both these scenarios together to see the capacity after removing old hardware and adding new hardware.

You can only combine scenarios that pertain to the same object. Use the filters to narrow down the scenarios in the What-If tab under **Capacity** > **Plan** and filter based on scenario name, type, data center, or cluster. Then, select the scenarios and click **Run** under the ellipsis next to the **NEW** button.

You can select the following combinations of scenarios and run them together:

Workload Planning and Physical Infrastructure Planning	• Add VMs • Remove VMs • Add Hosts • Remove Hosts

The Results Page

The Results page displays the results of running one or more saved scenarios. For Workload and Infrastructure Planning What-If scenarios, the summary page displays the allocation and demand values. You can export the results of the what-if saved scenarios in CSV or PDF format. When you export a saved scenario, the report contains a link to the report details.

If you do not see the allocation values, make sure that the overcommit ratios are activated in the policy. For more information, see the Policy Allocation Model Element topic in the Configuring guide.

To add or remove saved scenarios and run them again cumulatively, select a scenario from the What-If tab under **Capacity** > **Plan** and click **Edit** under the ellipsis. To commit a scenario and reserve capacity, click the **COMMIT SCENARIO** button in the results page, after you run the scenario. The Create Committed Scenario fly-out opens from the right-hand side of the page. Add a name to the scenario you want to commit. Provide and implementation date, and optionally, an end date and click **SAVE**.

What-If Analysis - Workload Planning: Traditional

You define scenarios that can potentially add workloads to actual data centers or you can plan for new clusters. VCF Operations models the scenario and calculates whether your desired workload can fit in the targeted data center, custom data center or new cluster. You can also define scenarios that can potentially remove workloads from data centers. VCF Operations calculates the time remaining and capacity remaining on the cluster when workloads are removed from the cluster.

Where You Find What-If Analysis - Workload Planning: Traditional

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. In the **What-If Analysis** tab, click **NEW**. Click **Add VMs** or **Remove VMs** in the Workload Planning: Traditional tile.

How What-If Analysis - Workload Planning: Traditional Works

Capacity Optimization allows you to forecast successfully the impact of adding a workload to an application. By trying various scenarios, you can arrive at an optimum configuration. You have the following options about the location of the new workload:

- You can select the exact data center or custom data center where you want to locate the new workload.
- You can pick a specific cluster where the workload is to reside.
- You can define a new cluster with new workloads and use the new cluster as a destination for the workloads. This helps you plan for new servers and clusters. VCF Operations can simulate the impact of the capacity on the new infrastructure, without actual deployment. This helps you manage growth and prevents resource bottlenecks, ensuring a smooth expansion. The output of the What-If scenario will show a breakdown of costs and timelines.

In selecting the profile of your workload, you have two options:

- Configure the workload manually by specifying vCPUs, memory, storage, and expected use percentage. You have the further option to click Advanced Configuration and specify more precise characteristics for your workload.
- Use an existing VM or VMs as templates, importing all the attributes of the selected VMs to your workload scenario. The system lets you to specify how many copies of each selected VM you want to add to the proposed workload.

When you have set the profile for the new workload, enter the start and end date for the period when you want the workload to be active. The default is: starting today and ending one year from today. The system can project scenarios ending up to one year from the current date.

At this point, you can save the scenario to edit or run later on. A list of saved scenarios is available on the What-If Analysis main page. Otherwise, run the scenario to get the VCF Operations analysis and assessment of your plan.

The system lets you know immediately if the proposed workload fits or does not fit in the suggested location. If it fits, the results list the prime target cluster and any additional possible locations. The system also projects time remaining before the workload runs out of resources. If you select scenario details, the system displays a graphic depiction of resource use. For each attribute value - vCPU, memory, and storage - the amount by which the workload increases the percentage of total application capacity used is shown against a time line. The graph shows the existing percentage used in blue and the total of existing usage and added usage as a percentage of total capacity in green.

If the proposed workload does not fit, the system announces the outcome and provides the following information:

- How much the added workload reduces the time remaining for the target cluster, for example, from one year to zero.
- The discrepancy between the space available in the target cluster and what the proposed workload requires, for example, 100 GB of memory.
- The cost of the workload on the VMware Hybrid Cloud and on the public cloud.
- The server list configuration if it overfits or underfits.

About Clouds

When you run a scenario in What-If Analysis, you get a recommendation based on cost relative to workload placement on different clouds. This cost-based recommendation varies for different clouds.

Private Cloud and VMware Cloud on AWS costs are computed based on resource usage levels.

Public clouds, AWS, IBM Cloud, Google Cloud, Microsoft Azure, and user-defined cloud costs are dependent on the selected configuration, that is, for the allocated resources. These public cloud instances are selected based on the close proximity rule, with simulated resource allocation values and in some scenarios, the exact configuration match available in the cloud instance list is not available. Due to this issue, these public cloud costs can be inherently higher in comparison.

How What-If Analysis - Remove Workload Works

This feature of Capacity Optimization lets you forecast successfully the impact of removing a workload. By trying various scenarios, you can arrive at an optimum configuration. Once you select the Workload Planning screen, you can select VMs from the concrete cluster data center or from the customer data center from which you want to remove the existing workload.

While removing workloads, you have two options to define the workload:

- Select existing VMs and use their projected utilization to evaluate the impact of removing workloads.
- Configure the workload manually by specifying the vCPUs, memory, storage, and expected use percentage.

Enter the start and end date for the period during which you want the workload to be removed. By default, the start date is today and the end date is one year from today. The end date is left empty by default. The system can project scenarios ending up to one year from the current date.

At this point, you can save the scenario to edit or run later on. A list of saved scenarios is available on the What-If Analysis main page. Otherwise, run the scenario to get theVCF Operations analysis and assessment of your plan.

Table 1763: What-If Analysis Workload Page Options

Option	Description
Add/Remove VMs	Click **Add VMs** or **Remove VMs** to create a scenario for adding or removing workload. When clicked, the command displays the Add Workload or Remove Workload screen.
Scenario Name	In the heading of the Saved Scenarios table. Selecting the check box next to the name selects all scenarios in the list and turns on the dimmed **Delete** button.
Scenario type	Name of the scenario type. Values are Add Workload, Remove Workload, Add Capacity, Remove Capacity, and Migrate.
<*scenario_name*>	Name of a saved scenario. Selecting the check box next to a name turns on the dimmed **Run Scenario**,**Edit**, and **Delete** buttons.
All Filters	Use the filter to search for a specific scenario by name or type.
Show Columns	Click the small button on the lower left to display the Show Columns dialog box. You can select up to four columns to display in the table: Scenario Name, Scenario Type, Date Created, and Scenario Start and End Date.

Add or Remove VMs: Traditional

As part of the What-If workload planning for traditional infrastructure, Workload Planning: Traditional is the tile you use to fill in the details of your virtual machines. You select where to add or remove the workload, configure it yourself or use an existing VM as a template, and establish a time frame. You also have an advanced configuration option that lets you define your configuration more precisely.

Where You Can Add or Remove VMS

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. In the **What-If Analysis** tab, click **NEW**. At the What-If Analysis screen, click **Add VMS** or **Remove VMS** in the Workload Planning: Traditional tile.

Table 1764: Workload Planning: Traditional Add VMs Options

Option	Description
Scenario Name	Name of your scenario
Location	Where do you want to add the workload? Select from three options: • A list of existing data centers, • A list of existing cluster or • A new cluster. You must provide the server type details by selecting from the list in the dialog box that opens.
Application Profile/**Configure**	Allows you to configure the virtual compute resource, including vCPU, memory, and storage.
Application Profile/ **Import from existing VM**	Displays the Select VMs dialog box where you can select one or more existing VMs to use as templates for your workload. Once you have made your selections, you return to this screen to enter the quantity of each chosen VM you want to incorporate as templates into your workload.
Import from Custom Profile	Click the **SELECT CUSTOM PROFILES** button to select one or more Custom Profiles to perform comparative analysis of capacity and cost across different datacenters and clusters in your private cloud. In the Select Custom Profiles dialog box which opens, select from a list of custom profiles and click the **>** icon to add the custom profile to the **SELECTED** list. Click **OK** when done.
Choose Your Workload: • **CPU** • **Memory** • **Disk space**	With the Configure radio button selected, you can size your workload by defining values for vCPU, memory, and disk space. These are your allocation values.
Expected Utilization	Set the projected percentage of total workload capacity you expect to average. Click **Advanced Configuration** to set the percentage of expected utilization for CPU, Memory, and Disk individually and to select thin or thick provisioning. VCF Operations calculates demand based on these values.
Annual Projected Growth	Set the percentage by which you expect your capacity go grow, annually. Click **Advanced Configuration** to set the percentage growth of CPU, Memory, and Disk individually. For example, if the utilization is 100 at the start date, and you set the annual growth % to 10%, then at the end of the year the utilization will grow to 110. The Annual Projected Growth can be set to 0% if no growth is expected.
Number of VMs (optional)/**Quantity**	You can optionally select how many VMs to spread the workload across.
Implementation date/End Date (optional)	Select from pop-up calendars the start and end date for the workload. The end date cannot be later than one year from the current date.
Run Scenario	Click to run the scenario. The system calculates whether it fits into the location you selected.
Save	**Save** the scenario.
Cancel	**Cancel** the scenario.

Table 1765: Workload Planning: Traditional Remove VMs Options

Option	Description
Scenario Name	Name of your scenario.
Location	From where do you want to remove the workload? Select from the list of existing data centers. You can optionally choose the exact cluster from where you want to remove the workload.

Option	Description
Application Profile/**Configure**	Allows you to configure the virtual compute resource, including vCPU, memory, and storage. After you have configured the scenario, enter the quantity of custom VMs that you want to remove.
Application Profile/**Import Existing VMs**	Displays the Select VMs dialog box where you can choose one or more existing VMs. Once you have made your selections, you return to this screen to enter the quantity of each chosen VM you want to remove from your workload. **Note:** The recommended limit is 100 VMs as a maximum for workload removal.
Application Profile / **Custom**: Choose your workload • **CPU** • **Memory** • **Disk space**	With the Configure radio button selected, you can size your workload by defining values for vCPU, memory, and disk space.
Implementation date/End Date (optional)	Select from pop-up calendars the start and end date for the workload. The end date cannot be later than one year from the current date. You can also leave the end date blank.
Run Scenario	Click to run the scenario. The system calculates the impact on the cluster (time remaining and capacity remaining) when removing the workload.
Save	**Save** the scenario.
Cancel	**Cancel** the scenario.

Select VMs

Use the **Select VMs** dialog box to choose the VMs whose attributes you want to copy or remove for your Workload Planning: Traditional or Workload Planning: Hyperconverged what-if scenarios.

Where You Find Select VMs

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. In the **What-If Analysis** tab, click **Add VMS** or **Remove VMS** in the Workload Planning: Traditional or Workload Planning: Hyperconverged pane. When you have entered a **Scenario Name** and **Location**, click the **Import from existing VM/Existing VMs** radio button, then click **Select VMS**. On the left is a selection box that allows you optionally to choose all VMs. To add a VM to the selected list on the right, double-click on the VM name. Following are the rest of your options:

Select VMs

Option	Description
All Filters	Use the advanced filter and search option to look for VMs. Click the drop-down icon to narrow your search based on the available options.
Select (nn).	Select the VMs listed on the current page, from which to import, or remove characteristics.
Select all (nn) VMS	Click to select all the VMs across all the pages, based on the filters you have set. The number of VMs that you can select by clicking this option is limited to 500 VMs.
Selected	List of VMs you selected from RESULTS.
OK	When you have selected the VMs you want, click **OK** to return to the Add Workload or Remove Workload screen, where your selected VMs are listed.

Under Application Profile, in the Selected VMs table, enter the number of copies of each VM you selected to add or remove in the Quantity column.

Scenario Summary Page for Workload Planning - Traditional

The scenario summary page for Workload Planning - Traditional displays the workload values after you add or remove VMs in a data center and cluster you specify.

Summary Section

The summary section displays the inputs that you provided for the Add VMS or Remove VMS scenario. You can change the workload location by selecting a different vCenter and cluster combination. Alternatively, you can edit or save the scenario. You can run saved scenarios later, and commutatively.

Private Cloud and Public Cloud sections for Add VMS

The Private Cloud: Datacenter, Public Cloud, and Private Cloud sections display information which help you understand where your workload would fit, the associated costs, and the time remaining based on peak CPU, Memory, and Disk Space for demand and allocation model after you add VMs. You can choose to commit the Private Cloud: Datacenter results by clicking the **COMMIT SCENARIO** button. To see more details, click **VIEW DETAILS**.

You can see the cost per month for different private and public clouds. Click **Learn More** for a more detailed comparison based on your workload.

Scenario Results Section for Remove VMS

This section displays a graphical view of the projected CPU, Memory, and Disk Space values for the demand and allocation models after the VMs are removed. You can understand how much time you have before the demand runs out.

What-If Analysis - Workload Planning: Hyperconverged and VMC on AWS

You can perform Hyperconverged Infrastructure workload planning by adding or removing VMs to vSAN activated clusters and running What-If scenarios. VCF Operations shows you if the proposed workload fits or does not fit in the suggested location. If it fits, the results list the prime target cluster and any additional possible locations. The system also projects time remaining before the workload runs out of resources. .

Where You Find What-If Analysis - Workload Planning: Hyperconverged

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. In the **What-If Analysis** tab, click **NEW**. In the Workload Planning: Hyperconverged tile, click **Add VMS** or **Remove VMS**.

How What-If Analysis - Workload Planning: Hyperconverged Works

You define scenarios that can potentially add or remove workloads to vSAN environment. The workload scenarios are based on VMs associated with specific storage policy related factors (such as FTT, RAID).

> **Note:**
>
> When a workload is added based on imported VMs, and the VM is currently in a vSAN-activated cluster, the vSAN policy settings are not applied and the current VM disk space is taken as is.

Capacity and Cost Planning Support for Virtual Machines - VMC Datacenter

You can now perform capacity planning and cost calculations for a virtual machine (VM) in hyper-converged environment where the VM is part of the Vmware Cloud on Amazon Web Services (VMC) cluster. VCF Operations provides accurate capacity recommendations and cost calculations when you add or remove VMs in hyper-converged environment from VMC data centers.

The cost calculation is based either on bills collected by VMC adapter or based on reference.

Add or Remove VMs: Hyperconverged

As part of the What-If workload planning for hyperconverged infrastructure, Workload Planning: Hyperconverged is the pane you use to fill in the details of your virtual machines. You select where to add or remove the workload, configure it yourself or use an existing VM as a template, and establish a time frame. The advanced configuration option lets you define your configuration more precisely.

Where You Find Workload Planning

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. In the **What-If Analysis** tab, click **NEW**. Click **Add VMS** or **Remove VMS** in the **Workload Planning: Hyperconverged** pane.

Table 1766: Workload Planning: Hyperconverged Add Options

Option	Description
Scenario Name	Name of your scenario
Location	Where do you want to add the virtual machines? Select from the list of existing data centers. You can optionally select the exact cluster where you want the virtual machine to reside.
Application Profile/**Configure**	Allows you to configure the virtual compute resource, including vCPU, Memory, and Disk Space.
Application Profile/**Import Import from existing VM**	Displays the Select VMs dialog box where you can select one or more existing VMs to use as templates for your workload. Once you have made your selections, you return to this screen to enter the quantity of each selected VM you want to incorporate as templates into your workload.
Select your workload: • **CPU** • **Memory** • **Disk space**	With the **Configure** radio button selected, you can size your workload by defining values for vCPU, Memory, and Disk Space. These are your allocation values.
Expected Utilization	Set the projected percentage of total workload capacity you expect to average. Click **Advanced Configuration** to set the percentage of expected utilization for CPU, Memory, and Disk individually and to select thin or thick provisioning. VCF Operations calculates demand based on these values.
Annual Projected Growth	Set the percentage by which you expect your capacity to grow, annually. Click **Advanced Configuration** to set the percentage growth of CPU, Memory, and Disk individually. For example, if the utilization is 100 at the start date, and you set the annual growth % to 10%, then at the end of the year the utilization will grow to 110. The Annual Projected Growth can be set to 0% if no growth is expected.
Number of VMs (optional)/**Quantity**	You can optionally select how many VMs to spread the workload across.
Additional vSAN configuration	Configure additional vSAN details such as swap space, host failures to tolerate, fault tolerance method, and Dedup.
Implementation date/End Date (optional)	Select from pop-up calendars the start and end date for the workload. The end date cannot be later than one year from the current date.
Run Scenario	Click to run the scenario. The system calculates whether it fits into the location you selected.
Save	**Save** the scenario.
Cancel	**Cancel** the scenario.

Table 1767: Workload Planning: Hyperconverged Remove Options

Option	Description
Scenario Name	Name of your scenario.
Location	From where do you want to remove the VMs? Select from the list of existing data centers. You can optionally select the exact cluster from where you want to remove the workload.
Application Profile/**Configure**	Allows you to configure the virtual compute resource, including vCPU, Memory, and Disk Space. After you have configured the scenario, enter the quantity of custom VMs that you want to remove.
Application Profile/**Import Existing VMs**	Displays the **Select VMs** dialog box where you can select one or more existing VMs. Once you have made your selections, you return to this screen to enter the quantity of each selected VM you want to remove from your workload. **Note:** The recommended limit is 100 VMs as a maximum for workload removal.
Import from Custom Profile	Click the **SELECT CUSTOM PROFILES** button to select one or more Custom Profiles to perform comparative analysis of capacity and cost across different datacenters and clusters in your private cloud. In the Select Custom Profiles dialog box which opens, select from a list of custom profiles and click the the **>** icon to add the custom profile to the **SELECTED** list. Click **OK** when done.
Application Profile / **Custom**: Choose your workload • **CPU** • **Memory** • **Disk space**	With the **Configure** radio button selected, you can size your workload by defining values for vCPU, Memory, and Disk Space.
Expected Utilization	Set the projected percentage of total workload capacity you expect to average. Click **Advanced Configuration** to set the percentage of expected utilization for CPU, Memory, and Disk individually and to select thin or thick provisioning.
Number of VMs (optional)/**Quantity**	You can optionally select how many VMs to spread the workload across.
Additional vSAN configuration	Configure additional vSAN details such as swap space, host failures to tolerate, fault tolerance method, and Dedup.
Start Date/End Date	Select from pop-up calendars the start and end date for the workload. The end date cannot be later than one year from the current date. You can also leave the end date blank.
Run Scenario	Click to run the scenario. The system calculates the impact on the cluster (time remaining and capacity remaining) when removing the workload.
Save	**Save** the scenario.
Cancel	**Cancel** the scenario.

Results: Add or Delete VMs to Hyperconverged Infrastructure

The scenario results are displayed when you run the scenario. In Private Cloud Data Center you can view the recommendation which provides details about the number of VMs to be added or removed from the VMware Cloud. You can also view whether the workload fits in your cloud environment and the cost increase or cost saving depending on whether you are adding or removing a VM from the VMware cloud. The Public Cloud tile displays the cost increase or savings across public clouds like Google Cloud, VMware Cloud on AWS, Amazon Web Services, IBM Cloud and others.

Scenario Summary Page for Workload Planning: Hyperconverged

The scenario summary page for Workload Planning - Hyperconverged displays the workload values after you add or remove VMs to vSAN activated clusters.

Summary Section

The summary section displays the inputs that you provided for the Add VMS or Remove VMS scenario. You can change the workload location by selecting a different vSAN activated clusters. Alternatively, you can edit or save the scenario. You can run saved scenarios later, and commutatively.

Private Cloud and Public Cloud sections for Add VMS

The Private Cloud: Datacenter, Public Cloud, and Private Cloud sections display information which help you understand where your workload will fit, the associated costs, and the time remaining based on peak CPU, Memory and Disk Space for demand and allocation model after you add VMs. You can choose to commit the Private Cloud: Datacenter results by clicking the **COMMIT SCENARIO** button. To see more details, click **VIEW DETAILS**.

You can see the cost per month for different private and public clouds. Click **Learn More** for a more detailed comparison based on your workload.

Scenario Results Section for Remove VMS

This section displays a graphical view of the projected CPU, Memory, and Disk Space values for the demand and allocation models after the VMs are removed. You can understand how much time you have before the demand runs out.

What-If Analysis - Infrastructure Planning: Traditional

You define scenarios that can potentially add capacity to actual data centers or remove capacity from actual data centers. VCF Operations models the scenario and calculates whether your desired workload can fit in the targeted data center or custom data center.

Where You Find Infrastructure Planning: Traditional

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. In the **What-If Analysis** tab, click **NEW**. Click **Add Hosts** or **Remove Hosts** in the Infrastructure Planning: Traditional tile.

How the What-If Analysis for Infrastructure Planning: Traditional Works

Infrastructure Planning for traditional environments lets you forecast successfully the impact of adding capacity to your environment or removing capacity from your environment. By trying various scenarios, you can arrive at an optimum configuration. Once you select the Infrastructure Planning: Traditional tile, you can choose where you want to locate the additional capacity or from where you can remove the existing capacity.

In selecting the profile while removing capacity, you can select a profile only from server types that exist in your cluster.

In selecting the profile while adding capacity, you have two options:

- Select a server type from a list of commercially available servers. You can select from a list of 1) server types already in your cluster or 2) all server types approved for purchase.
- Configure a custom server manually by specifying CPU attributes, memory, and cost.

When you have set the profile for the new server, enter the number of servers to purchase or remove and the start and end date for the period when you want the scenario to be active. The number of servers that you plan to remove is limited by the number of selected server types available in the selected cluster. The system can project scenarios ending up to one year from the current date. By default, the starting date is today and the ending date is one year from today.

At this point, you can save the scenario to edit or run later on. A list of saved scenarios is available on the What-If Analysis main page. Otherwise, run the scenario to get the VCF Operations analysis and assessment of your plan.

The system displays immediately the impact on cluster size of the additional or lesser amount of CPU and memory, and shows the total cost of adding or removing the specified capacity. The system also shows whether adding new capacity or removing capacity extends or shrinks the time remaining before CPU or memory runs out.

As well, the system displays a graphic depiction of resource use. For each attribute value - CPU and memory - the amount by which the workload increases or decreases the percentage of total capacity used is shown against a time line.

Add or Remove Hosts

As part of the What-If analysis for physical infrastructure planning for traditional environments, Infrastructure Planning: Traditional pane is what you use to fill in the details of your What-If scenario. You select where to add or remove hosts, use an existing server type, or configure it yourself (when you add capacity), and establish a time frame.

Where You Find Physical Infrastructure

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. In the **What-If Analysis** tab, click **NEW**. Click **Add Hosts** or **Remove Hosts** in the Infrastructure Planning: Traditional tile.

Table 1768: Add Hosts Options

Option	Description
Scenario Name	Name of your scenario
Location	Where do you want to add capacity? Select from the list of existing data centers, then select the cluster where you want one or more servers to reside.
Server Details	Clicking **Select Server** displays the Select Server Type dialog box, where you can select a commercial brand server or configure a custom server. Number of Servers to add: increment the Quantity counter up to the number of servers you want.
Implementation date/End Date (optional)	Select from pop-up calendars the start and end date for the What-If scenario.
Run Scenario	Click to run the scenario. The system calculates the cost of the scenario and determines any new time remaining number.
Save	**Save** the scenario.
Cancel	**Cancel** the scenario.

The system displays immediately the impact on cluster size of the additional CPU and memory, and shows the total cost of adding the specified capacity. The system also shows in graphical form whether adding the new capacity extends the time remaining before CPU or memory runs out.

Table 1769: Remove Hosts Options

Option	Description
Scenario Name	Name of your scenario
Location	From where do you want to remove capacity? Select from the list of existing data centers, then select the cluster from where you want to remove one or more servers.
Server Details	Clicking **Select Server** displays the Select Server Type dialog box, where you can select only the server types that exist in your selected cluster. The number of servers that you plan to remove is limited by the number of selected server types available in the selected cluster.
Start Date/End Date	Select from pop-up calendars the start and end date for the What-If scenario. You can select to keep the end date blank.
Run Scenario	Click to run the scenario. The system determines any new time remaining number.
Save	**Save** the scenario.
Cancel	**Cancel** the scenario.

The system displays the time remaining and the impact on CPU and memory with reduced capacity. The system also shows in graphical form whether removing capacity decreases the time remaining before CPU or memory runs out.

You can also see that the cost is based on the original purchase cost.

Scenario Summary Page for Infrastructure Planning: Traditional

The scenario summary page for Infrastructure Planning - Traditional displays the capacity values after you add or remove hosts to a server in a data center and cluster you specify.

Summary Section

The summary section displays the inputs that you provided for the Add Hosts or Remove Hosts scenario. You can change the capacity location by selecting a different vCenter and cluster combination, or server. Alternatively, you can edit or save the scenario. You can run saved scenarios later, and commutatively.

Scenario Results Section

This section displays a graphical view of the projected CPU, Memory, and Disk Space capacity values for the demand and allocation models after the hosts are added or removed. You can understand how much time you have before capacity runs out and the costs savings. Click the **COMMIT SCENARIO** button to reserve this capacity.

What-If-Analysis - Infrastructure Planning: Hyperconverged

You can perform infrastructure planning by adding or removing Hyperconverged Infrastructure (HCI) nodes in vSAN activated clusters and running What-If scenarios. VCF Operations displays the cost, time remaining, and capacity remaining for CPU, memory, and disk space in the scenario results.

Where You Find What-If Analysis - Hyperconverged Infrastructure

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. In the **What-If Analysis** tab, click **NEW**. Select the Infrastructure Planning: Hyperconverged tile. To run a What-If scenario click **Add HCI Nodes** or **Remove HCI Nodes**.

How What-If Analysis - Hyperconverged Infrastructure Works

You can add hyperconverged infrastructure to your vSAN activated environment and evaluate the increase in HCI capacity and cost. You can add up to 64 hosts per vSAN cluster. This number accounts for existing hosts in the cluster. VCF Operations only lists vSAN and vXRail clusters in the location property. You can select existing server types from these locations and change the number of instances of these servers to add to your scenario.

Add or Remove HCI Nodes

As part of the what-if analysis for physical infrastructure planning for hyperconverged environments, the Infrastructure Planning: Hyperconverged tile is what you use to fill in the details of your what-if scenario. When you add an HCI node, you can select an existing server type from your vSAN activated data center and change the number of instances of this server to calculate storage, compute capacity, time remaining, and cost. You can run the Remove HCI Nodes scenario to see the capacity changes after you remove HCI nodes from your data center.

Where You Find Workload Planning

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. In the **What-If Analysis** tab, click **NEW**. Click **Add HCI Nodes** or **Remove HCI Nodes** in the **Infrastructure Planning: Hyperconverged** pane.

Table 1770: Add HCI Nodes Options

Option	Description
Scenario Name	Name of your scenario.
Location	Where do you want to add the HCI node? Select from the list of existing data centers. You must also choose the exact cluster where you want the HCI node to reside.
Server Details	Allows you to select an existing server type to calculate capacity, time, and storage remaining based on the number of instances of the server.
Number of servers to add	How many instances of the server do you want to add? **Note:** Only 60 new hosts can be added to the specified vSAN cluster as the maximum allowed is 64 hosts.
Start Date/End Date	Select from pop-up calendars the start and end date for the workload. The end date cannot be later than one year from the current date.
Run Scenario	Click to run the scenario. The system calculates whether it fits into the location you selected.
Save	**Save** the scenario.
Cancel	**Cancel** the scenario.

Table 1771: Remove HCI Nodes Options

Option	Description
Scenario Name	Name of your scenario.
Location	From where do you want to remove capacity? Select from the list of existing data centers, then select the cluster from where you want to remove the server(s).
Server Details	Clicking **Select Server** displays the Select Server Type dialog box, where you can choose only the server types that exist in your selected cluster. The number of servers that you plan to remove is limited by the number of selected server types available in the selected cluster.
Implementation date/End Date (optional)	Select from pop-up calendars the start and end date for the what-if scenario. You can choose to keep the end date blank.
Run Scenario	Click to run the scenario. The system determines any new time remaining number.
Save	**Save** the scenario.
Cancel	**Cancel** the scenario.

Scenario Summary Page for Infrastructure Planning: Hyperconverged

The scenario summary page for Infrastructure Planning - Hyperconverged displays the capacity values after you add or remove Hyperconverged Infrastructure (HCI) nodes in vSAN activated clusters that you specify.

Summary Section

The summary section displays the inputs that you provided for the Add HCI Nodes or Remove HCI Nodes scenario. You can change the vSAN activated cluster location by selecting a different vCenter and cluster combination, or server. Alternatively, you can edit or save the scenario. You can run saved scenarios later, and commutatively.

Scenario Results Section

This section displays a graphical view of the projected CPU, Memory, and Disk Space capacity values for the demand and allocation models after the HCI nodes are added or removed. You can understand how much time you have before capacity runs out and the costs savings. Click the **COMMIT SCENARIO** button to reserve this capacity.

What-If-Analysis - Migration Planning: VMware Cloud

What-If-Analysis - Migration Planning, allows you to evaluate your plan for migrating or moving workloads across different VMware clouds. You can compare capacity and cost of workload across Oracle Cloud VMware Solution (OCVS), VMware Cloud on Dell EMC (VMCD), VMware Cloud on Amazon Web Services (AWS), Azure VMware Solution (AVS), and Google Cloud VMware Engine (GCVE). VCF Operations evaluates the migration plan, calculates the cost and capacity requirements, and provides cost estimate for the selected VMC workload.

Where You Find What-If Analysis - Migration Planning

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. In the **What-If Analysis** tab, click **NEW**. In the Migration Planning: VMware Cloud tile, click **Plan Migration**.

How What-If Analysis - Migration Planning Works

The What-If-Analysis feature of Capacity Optimization lets you forecast successfully the impact of migrating a workload to the VMware Cloud instance such as Oracle Cloud VMware Solution (OCVS), VMware Cloud on Dell EMC (VMCD), VMware Cloud on Amazon Web Services (AWS), Azure VMware Solution (AVS), and Google Cloud VMware Engine (GCVE). Once you select the Migration Planning screen, choose whether you want to run the scenario against a VMware Cloud on AWS or other type of cloud accounts. For the VMware Cloud, select the region where you want to migrate the workload.

When you have set the profile for the migrating workload, run the scenario to get the analysis and assessment of your plan. You can select one VMware cloud at a time and get the estimate for migration planning cost. Alternatively, you can save the scenario to edit or run later on. A list of saved scenarios is available in the Saved Scenarios tab on the What-If analysis page.

If you selected VMware Cloud on AWS for your scenario, the results list the VMware Cloud on AWS Assessment, with details of the VMware configuration. The result also displays the resource-use-level cost and the monthly purchase cost for an on-demand subscription. In addition, the result displays the resource-use-level cost and monthly purchase cost for one-year and three-year subscriptions.

About Clouds

The system might provide a recommendation based on the cost of placing the workload on different VMware clouds. This cost-based recommendation varies for different clouds.

For VMware Cloud on AWS, the system displays the resource-use-level cost and the monthly purchase cost for an on-demand subscription, plus those same costs for one-year and three-year subscriptions.

VMware cloud costs are based on the selected configuration, that is, the allocated resources.

Migration Planning: VMware Cloud

As part of the What-If Analysis function, Migrate is the form you use to fill in the details of your what-if scenario. You choose where to migrate the workload, then select the region.

Where You Find Migration Planning

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. In the **What-If Analysis** tab, click **NEW**. Click **Plan Migration** in the **Migration Planning: VMware Cloud** tile.

When you run a scenario for What If: Migration for VMware Cloud, VCF Operations might suggest the right cloud Instance suitable for the Workload Configuration selected by you. VCF Operations also calculates the cost for that VMware Cloud's instance and displays the same.

Table 1772: Migrate Options

Option	Description
SCENARIO NAME	Name of your scenario
SELECT CLOUDS	Where do you want to migrate the workload? Options: • **VMware Cloud on AWS** • **Azure VMware Solution (AVS)** • **Google Cloud VMware Engine (GCVE)** • **Oracle Cloud VMware Solution (OCVS)** • **VMware Cloud on Dell EMC (VMCD)** **Note:** You can select regions for the above cloud options.
CLUSTER SETTINGS	Specify the following cluster details: • Enter the **Instance Type**. • Enter the **Slack Space** in percentage. • Enter the **Steady state CPU headroom** in percentage.
APPLICATION PROFILE/Configure	Using the Application Profile you can configure the virtual compute resources, like vCPU, memory, and storage.
Select Your Workload: • **CPU** • **Memory** • **Disk Space**	With the Configure radio button selected, you can size your migrating workload by defining values for vCPU, memory, and storage.
Expected Utilization	Specify the expected utilization or Click **ADVANCED CONFIGURATION** and specify the values for the following: • **CPU** • **Memory** • **Disk Space** • **Disk Space Provisioning** - Select either Thin or Thick.
Annual Projected Growth	Specify the annual growth rate so that the system adjust the scenario calculations or click **ADVANCED CONFIGURATION** and specify the values for the following: • **CPU** • **Memory** • **Disk Space**
Number of VMs (OPTIONAL)	You can optionally choose how many VMs to spread the workload across.
Additional vSAN configuration	Select **Account for Swap Space** to reserve swap space for any unreserved virtual machine memory. • Select the **Host failures to tolerate value** from the drop-down list. • Select the **Fault Tolerance Method**, the options are RAID -1 and RAID-5. • Select the **Dedup** value from the drop down list.
APPLICATION PROFILE/Import from existing VM	Displays the **Select VMs** button. When selected, displays the Select VMs workspace, where you can choose one or more existing VMs to use as templates for your workload. You can filter VMs by name, tags, vCenter, or custom group. Once you have made your selections, you return to this screen to enter the quantity of each chosen VM you want to incorporate as templates into your workload.
Import from Custom Profile	Click the **SELECT CUSTOM PROFILES** button to select one or more Custom Profiles to perform comparative analysis of capacity and cost across different datacenters and clusters in your private cloud. In the Select Custom Profiles dialog box which opens, select from a list of custom profiles and click the **>** icon to add the custom profile to the **SELECTED** list. Click **OK** when done.

Option	Description
RUN SCENARIO	Click to run the scenario. The system calculates whether it fits into the location you chose.
SAVE	**SAVE** the scenario.
CANCEL	**CANCEL** the scenario.

VMware Cloud on AWS Assessment - Results

The scenario results are displayed when you run the scenario. You can view the recommendation which provides details about the number of hosts required for the VMware Cloud. You can also view the Total Cost associated with the recommended VMware Cloud for 3 years subscription and the Total Capacity Usage details for CPU, memory, and disk space.

For VMware Cloud on AWS Assessment, you can edit the following options.

- **Edit Configuration** - you can edit the change in Reserved Capacity CPU, Reserved Capacity Memory, Fault Tolerance, and RAID Level values and save the values to the original configuration.
- **Change Plan** - you can use the **Choose Plan** option to change your subscription plan, the available options are one-year plan, three-year plan, or Pay-As-You-Go.
- **Edit Discount** - you can use the edit discount option to specify the discount value, the total cost for the subscription is equal to the actual utilization cost minus the discount percentage.

What-If-Analysis - Migration Planning: Public Cloud

You define scenarios that can potentially migrate workloads to a public cloud. Use this scenario to determine where to move the workloads. VCF Operations models the scenario and calculates the cost and capacity to fit your desired workload.

Where You Find Migration Planning: Public Cloud

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. In the **What-If Analysis** tab, click **NEW**. Click the **PLAN MIGRATION** button in the **Migration Planning: Public Cloud** card.

How What-If Analysis - Migration Planning Works

This feature of Capacity Optimization lets you forecast successfully the impact of migrating a workload to a public cloud instance such as Amazon Web Services, IBM Cloud, Microsoft Azure, or Google Cloud. Select the region where you want to migrate the workload. If the public clouds listed out of the box do not suit your needs, you can also define your own public cloud and upload a rate card.

In defining the profile of your workload, you have two options:

- Configure the workload manually by specifying vCPUs, memory, storage, and expected use percentage.
- Use an existing VM or VMs as templates, importing all the attributes of the selected VMs to your workload scenario. The system allows you to specify how many copies of each selected VM you want to add to the proposed workload.

When you have set the profile for the migrating workload, run the scenario to get the VCF Operations analysis and assessment of your plan. You can also select up to three public clouds to compare results. Alternatively, you can save the scenario to edit or run later on. A list of saved scenarios is available in the **Saved Scenarios** tab on the What-If analysis page.

For a public cloud target, the system lets you know immediately if the workload proposed for migration fits or does not fit in the suggested location. For example, if you selected AWS and the workload fits, the results list the Amazon Web Services Assessment, with details of the VMware Configuration and the AWS Equivalent. If the proposed workload does not fit, an error message appears: "Unable to identify a matching configuration instance in target location."

About Clouds

The system might provide a recommendation based on the cost of placing the workload on different clouds. This cost-based recommendation varies for different clouds. You can modify the costs for public clouds by uploading a new rate card.

Public cloud costs are based on the selected configuration, that is, the allocated resources.

The public instance is selected based on the close proximity rule, with simulated resource allocation values. In some scenarios, an exact configuration match is not available in the list. Due to this lack of availability, the public cost can be inherently higher in comparison.

Migration Planning: Public Cloud

As part of the What-If Analysis function, Migrate is the form you use to fill in the details of your what-if scenario. You choose where to migrate the workload, then select the region.

Where You Find Migration Planning

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. At the What-If Analysis screen, click **PLAN MIGRATION** in the Migration Planning: Public Cloud card.

When you run a scenario for What If: Migration for Public Clouds (Not VMC), VCF Operations on selected by you. VCF Operations also calculates the cost for that Public Cloud's instance and displays the same.

Table 1773: Migrate Options

Option	Description
SCENARIO NAME	Name of your scenario
SELECT CLOUDS	Where do you want to migrate the workload? Options: • Aamazon Web Services • IBM Cloud • Microsoft Azure • Google Cloud **Note:** The cloud providers added in the Add Cloud Provider page are also included in the list. You can select a maximum of three public clouds at a time for comparison. Hold the **Shift** key to select more than one public cloud provider.
ADD CLOUD PROVIDERS	You can add or edit the cloud providers and also edit the rate card of each individual cloud provider.
APPLICATION PROFILE/Configure	Using the Application Profile you can configure the virtual compute resources, like vCPU, memory, and storage.
Select Your Workload: • **CPU** • **Memory** • **Disc Space**	With the Configure radio button selected, you can size your migrating workload by defining values for vCPU, memory, and storage.
APPLICATION PROFILE/Import from existing VM	Displays the Select VMs button. When selected, displays the Select VMs workspace, where you can choose one or more existing VMs to use as templates for your workload. You can filter VMs by name, tags, vCenter, or custom group. Once you have made your selections, you return to this screen to enter the quantity of each chosen VM you want to incorporate as templates into your workload.

Option	Description
Import from Custom Profile	Click the **SELECT CUSTOM PROFILES** button to select one or more Custom Profiles to perform comparative analysis of capacity and cost across different datacenters and clusters in your private cloud. In the Select Custom Profiles dialog box which opens, select from a list of custom profiles and click the **>** icon to add the custom profile to the **SELECTED** list. Click **OK** when done.
Number of VMs (OPTIONAL)/**Quantity**	You can optionally choose how many VMs to spread the workload across.
RUN SCENARIO	Click to run the scenario. The system calculates whether it fits into the location you chose.
SAVE	**SAVE** the scenario.
CANCEL	**CANCEL** the scenario.

What-If Analysis - Data Center Comparison

You can select virtual machines to determine which of the preferred data centers (along with a specific choice of cluster or default cheapest cluster) are best fit from both cost effectiveness and capacity requirements perspective. The comparison helps you to find the right data center to place the workload from a cost and capacity perspective.

Where You Find What-If Analysis – Data Center Comparison

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. In the **What-If Analysis** tab, click **NEW**. Click **COMPARE DATACENTERS** in the Datacenter Comparison: Private Cloud tile.

How What-If Analysis - data center Comparison Works

This feature of Capacity Optimization lets you compare cost across data centers within the private cloud environment. After you select the Datacenter Comparison screen, choose one or more data centers to compare the cost and run the scenario. VCF Operations suggests which data center is most cost effective for the selected workload.

In defining the profile of your workload, you have two options:

- Configure the workload manually by specifying CPU, memory, disk space, expected utilization, and annual projected growth.
- Use an existing VM or VMs as templates, importing all the attributes of the selected VMs to your workload scenario. The system allows you to specify how many copies of each selected VM you want to add to the proposed workload.

When you have set the profile for comparing the workload, run the scenario to get the VCF Operations analysis and assessment of your plan. You can select up to three data centers to compare results. Alternatively, you can save the scenario to edit or run later. A list of saved scenarios is available in the Saved Scenarios tab on the What-If analysis page.

Cost varies from one data center to another depending on cost settings, which include cost drivers such as servers, facility, power, labor, license, network, and storage.

The data center comparison feature solves this problem by allowing you to select a data center which suits your requirement, is least expensive, and has adequate capacity.

Datacenter Comparison

As part of the What-If Analysis function, Compare Datacenters is the form you use to fill in the details of your What-If scenario. Use this scenario to compare cost across data centers within the private cloud environment.

Where You Find Compare Datacenters

After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity** > **Plan**. In the **What-If Analysis** tab, click **NEW**. Click **Compare Datacenters** in the tile titled Datacenter Comparison.

Table 1774: Compare Datacenter Options

Option	Description
Scenario Name	Name of your scenario.
Select Datacenters	Select the data centers for which you want to compare the costs. Use the advanced filter search to find a datacenter.
Application Profile/Configure	Using the Application Profile, you can configure the virtual compute resources, like CPU, memory, disk space, expected utilization, and annual projected growth.
Select Your Workload: • **CPU** • **Memory** • **Disk Space** • **Expected Utilization** • **Annual Projected Growth**	With the Configure radio button selected, you can size your workload by defining values for CPU, memory, disk space, expected utilization, and annual projected growth.
Application Profile/Import from existing VM	Displays the Select VMs button. When selected, displays the Select VMs workspace, where you can choose one or more existing VMs to use as templates for your workload. You can filter VMs by name, tags, vCenter, or custom group. Once you have made your selections, you return to this screen to enter the quantity of each chosen VM you want to incorporate as templates into your workload.
Import from Custom Profile	Click the **SELECT CUSTOM PROFILES** button to select one or more Custom Profiles to perform comparative analysis of capacity and cost across different datacenters and clusters in your private cloud. In the Select Custom Profiles dialog box which opens, select from a list of custom profiles and click the **>** icon to add the custom profile to the **SELECTED** list. Click **OK** when done.
Number of VMs (OPTIONAL)/**Quantity**	You can optionally choose how many VMs to spread the workload across.
Implementation date/End Date (optional)	You can specify the Start Date and End Date to compute the data center infrastructure cost for a specific time period.
Run Scenario	Click to run the scenario. The system calculates the cost of migration and checks whether the selected workload fits into the location you have chosen.
Save	**Save** the scenario.
Cancel	**Cancel** the scenario.

Retain Historical Data of VMs Migrated Using VMware Hybrid Cloud Extension

In VCF Operations you can use VMware Hybrid Cloud Extension (HCX) to perform application migration, workload rebalancing, and business continuity across data centers and clouds. You can also migrate workloads from on-premises data centers to VMware Cloud.

Earlier when you performed HCX bulk migration to migrate workloads from one datacenter to another datacenter, or from one datacenter to VMware Cloud. VCF Operations failed to retain the historic metrics.

Now VCF Operations has implemented a solution which is triggered during HCX migration. The event helps VCF Operations to collect details and manage the target vCenter to identify the workload migration.

After mapping the right attributes of VMs in the source datacenter with the VMs in the destination datacenter, you can verify if VCF Operations is able to retain all the historical metrics. The HCX migration types supported in VCF Operations are:

- Bulk Migration
- vMotion Based Migration (hot & cold)
- Replication Assisted Migration

HCX vMotion

VCF Operations performs the following actions during HCX vMotion.

- Get the target `VCID`, `VM-VC-MOID` from event using the resource key.
- Get the source `VCID`, `VM-VC-MOID` from event using the resource key.
- Maps the correct target `VCID`, `VM-VC-MOID` to the source VM in VCF Operations .

vMotion Generic Scenario

VCF Operations performs the following actions during vMotion.

- Get the target `VCID`, `VM-VC-MOID` from event using the resource key.
- Detects the discovered target VM in VCF Operations , based on the `VCID`, `VM-VC-MOID` attribute.
- For the detected target VMs, get VMs with **VM Entity Instance UUID**, and map the `VCID`, `VM-VC-MOID` for these VMs.
- Search `VCID`, `VM-VC-MOID` in the event message to find the actual vMotion VM.
- Set the correct target `VCID`, `VM-VC-MOID` to the right VM in on source .

Note:

To know more about HCX Migration, see VMware HCX Product Documentation.